

Two- ρ terms from the three- ρ terms proportional to $\log(v/\Lambda)$

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1 What is done here

The input consists of all three- ρ terms of the NLO cross section that are proportional to $\log(v/\Lambda)$:

1. the remainders left after reordering the four- ρ terms, i.e. the three- ρ terms with two and with three Wilson lines of the notes (the remainder with four Wilson lines vanishes, as shown there);
2. the genuine three- ρ terms of Groups I, II and III (file `Evolution_three_rho.tex`). Groups I and III come with their complex conjugate, Group II does not.

In every term the three charges appear as an ordered product of operators. We write each product as its fully symmetrized part plus commutators, Eqs. (3.4)–(3.7) below. Each commutator removes one ρ and produces a δ -function. Those terms are the two- ρ terms we are after. They are derived term by term in Secs. 4–7 and collected in Sec. 8.

Nothing else is changed: the kernels are those of the notes, evaluated at the points set equal by the δ -functions. The two exceptions are a repeated colour index in Row II of Group I and the transformation of that row's kernel to coordinate space (Sec. 5.1 and App. A). The algebra was generated and checked by computer (App. B).

2 Conventions

Common prefactor. Every term carries $\log(\vee/\Lambda)$ and a factor $g^4/(16\pi^5) = \frac{g^4}{8\pi^3} \cdot \frac{1}{2\pi^2}$. We define

$$\mathcal{N} \equiv \frac{1}{(2\pi)^3} \frac{g^4}{16\pi^5} \frac{1}{k^+} \log \frac{\vee}{\Lambda}, \quad (2.1)$$

so that $\frac{1}{(2\pi)^3} \frac{ig^4}{16\pi^5} \frac{1}{k^+} \log \frac{\vee}{\Lambda} = i\mathcal{N}$, $\frac{1}{(2\pi)^3} \frac{ig^4}{8\pi^5} \frac{1}{k^+} \log \frac{\vee}{\Lambda} = 2i\mathcal{N}$ and $\frac{1}{(2\pi)^3} \frac{ig^4}{32\pi^5} \frac{1}{k^+} \log \frac{\vee}{\Lambda} = \frac{i}{2}\mathcal{N}$. All transverse positions except the two measured-gluon positions in the phase are integrated; we write $\int$ for all these integrals.

Kernels. With $\mathcal{K}(X) \equiv X/X^2$ (a two-vector),

$$\frac{(a-b) \cdot (c-d)}{(a-b)^2(c-d)^2} = \mathcal{K}(a-b) \cdot \mathcal{K}(c-d), \quad \mathcal{K}(X) \cdot \mathcal{K}(X) = \frac{1}{X^2}, \quad \mathcal{K}(-X) = -\mathcal{K}(X). \quad (2.2)$$

Each source has a kernel that depends on the three charge positions, e.g. $K_1(x', x, y)$. A δ -function sets two of these equal, e.g. $K_1(x', x', y)$ means K_1 at $x = x'$.

Charges. The valence charges obey

$$[\rho^a(x), \rho^b(y)] = if^{abc} \rho^c(x) \delta^{(2)}(x-y). \quad (2.3)$$

We always integrate the δ -function over the position of the *right* operator. In $[\rho^a(x), \rho^b(y)]$ the point y is replaced by x everywhere: in the kernel, in the Wilson lines and in the remaining charge.

Colour. $(T^a)_{bc} = -if^{abc}$, $f^{acd} f^{bcd} = N_c \delta^{ab}$. For adjoint Wilson lines we use

$$(F) \quad f^{acd} f^{bcd} = N_c \delta^{ab}, \quad (2.4)$$

$$(U1) \quad U^{ab}(x) U^{ac}(x) = \delta^{bc}, \quad (2.5)$$

$$(U2) \quad f^{abc} U^{ad}(x) U^{be}(x) = f^{deh} U^{ch}(x). \quad (2.6)$$

(U2) follows from $U^{aa'} U^{bb'} U^{cc'} f^{a'b'c'} = f^{abc}$ after multiplying by $U^{cc''}$ and using (U1) (written here for the transposed matrix). As written, (U1) and (U2) hold for any adjoint Wilson line: the contracted indices sit in the same slot of both U 's. Your notes use the convention

$$U^{\dagger bc} = U^{bc} = U^{cb}, \quad (2.7)$$

and we follow it. With Eq. (2.7), (U1) and (U2) can also be used when the contracted indices sit in different slots, e.g. $U^{ab}(x) U^{ca}(x) = \delta^{bc}$. Such steps are marked (U1*) and (U2*). They are exact only for a symmetric U ; for a general adjoint matrix $U^{ab} U^{ca} = (U^2)^{cb}$. Finally, f^{abc} contracted with an object symmetric in a, b vanishes.

3 The reordering identity

Let A, B, C be three charges, e.g. $A = \rho^d(x')$, $B = \rho^a(x)$, $C = \rho^c(y)$. Their fully symmetrized product is

$$S(ABC) \equiv \frac{1}{6} (ABC + ACB + BAC + BCA + CAB + CBA). \quad (3.1)$$

Step 1: every ordering in terms of ABC . Moving operators one step at a time,

$$\begin{aligned} ACB &= ABC - A[B, C], \\ BAC &= ABC - [A, B]C, \\ BCA &= BAC - B[A, C] = ABC - [A, B]C - B[A, C], \\ CAB &= ACB - [A, C]B = ABC - A[B, C] - [A, C]B, \\ CBA &= BCA - [B, C]A = ABC - [A, B]C - B[A, C] - [B, C]A. \end{aligned} \quad (3.2)$$

Adding the six orderings,

$$6S(ABC) = 6ABC - 3[A, B]C - 2A[B, C] - 2B[A, C] - [A, C]B - [B, C]A. \quad (3.3)$$

Step 2: symmetrize the products of two. Each commutator is linear in ρ (Eq. (2.3)). For a commutator X and a charge Y , $XY = \frac{1}{2}\{X, Y\} + \frac{1}{2}[X, Y]$ and $YX = \frac{1}{2}\{X, Y\} - \frac{1}{2}[X, Y]$. Here $[X, Y]$ is a double commutator, i.e. a single ρ . Inserting this in Eq. (3.3), the anticommutators come with weights 3 for $\{[A, B], C\}$, $2 + 1 = 3$ for $\{[A, C], B\}$ and $2 + 1 = 3$ for $\{[B, C], A\}$. Hence

$$ABC = S(ABC) + \frac{1}{4} \left(\{[A, B], C\} + \{[A, C], B\} + \{[B, C], A\} \right) + R_1 \quad (3.4)$$

with the one- ρ remainder

$$R_1 = \frac{1}{12} \left(3[[A, B], C] + 2[A, [B, C]] + 2[B, [A, C]] + [[A, C], B] + [[B, C], A] \right). \quad (3.5)$$

Step 3: drop the one- ρ terms. Each term of R_1 is a double commutator, $\propto ff\rho\delta\delta$. It is linear in ρ and multiplies c-numbers (kernels and Wilson lines). Its expectation value vanishes for a colour-neutral projectile, $\langle\rho^a(x)\rangle = 0$. We drop it.

Anticommutator forms. Several rows of the notes contain $\{\rho, \rho\}$. Applying Eq. (3.4) to ABC and ACB and adding, the terms $\{[B, C], A\}$ and $\{[C, B], A\}$ cancel, and $S(ACB) = S(ABC)$:

$$\boxed{A\{B, C\} = 2S(ABC) + \frac{1}{2}\{[A, B], C\} + \frac{1}{2}\{[A, C], B\} + (\text{one } \rho)} \quad (3.6)$$

When the single charge stands on the right, use $\{B, C\}A = A\{B, C\} - [A, \{B, C\}]$ and $[A, \{B, C\}] = \{[A, B], C\} + \{B, [A, C]\}$:

$$\boxed{\{B, C\}A = 2S(ABC) - \frac{1}{2}\{[A, B], C\} - \frac{1}{2}\{[A, C], B\} + (\text{one } \rho)} \quad (3.7)$$

One piece. For $X = \rho^x(p_X)$, $Y = \rho^y(p_Y)$, $Z = \rho^z(p_Z)$, Eq. (2.3) gives

$$\{[X, Y], Z\} = if^{xyh}\{\rho^h(p_X), \rho^z(p_Z)\}\delta^{(2)}(p_X - p_Y), \quad (3.8)$$

and $p_Y \rightarrow p_X$ everywhere. The two- ρ terms are written with the anticommutator $\{\rho, \rho\}$. A different ordering of two charges differs from it by one ρ , which we drop.

The symmetric three- ρ part. For every term, the part $S(\dots)$ is the same expression with $\rho\rho\rho$ replaced by $S(\rho\rho\rho)$ (and by $2S(\rho\rho\rho)$ for the anticommutator forms). These parts behave as products of commuting charges. We do not list them again, and we do not check their cancellation in this note.

Complex conjugate. All prefactors below are $\pm i$, $\pm i/2$ or $\pm 2i$ times $\mathcal{N}$. The commutator supplies one more i , so every two- ρ term has a *real* coefficient. The kernels, f , U and δ are real, and $\{\rho, \rho\}$ is Hermitian. The complex conjugate of a two- ρ term therefore only turns $e^{-ik\cdot(P'-P)}$ into $e^{ik\cdot(P'-P)}$, where P' and P are the measured-gluon positions in the conjugate amplitude and the amplitude. Renaming $P \leftrightarrow P'$ gives back the original phase:

$$\text{c.c.} = (\text{same term with } P \leftrightarrow P'). \quad (3.9)$$

This holds for Groups I and III, with $(P', P) = (w', w)$, except (w', z) and (w', x) in Row V of Group I and (y', w) in Rows III, V and VI of Group III. Group II and the four- ρ remainders get no c.c.

4 Three- ρ terms from reordering the four- ρ terms

4.1 Two Wilson lines, first term

In the notes this term reads $-\frac{i}{(2\pi)^3} \frac{g^4}{8\pi^3} \frac{\log(\sqrt{v}/\Lambda)}{2\pi^2} \frac{1}{k^+} \int(\dots)$, i.e. its prefactor is $-i\mathcal{N}$. It is the complete remainder of reordering the four- ρ terms, so no c.c. is added.

The three- ρ term.

$$\begin{aligned} \left. \frac{d^3N}{d^3k} \right|_{A1} = & -i\mathcal{N} \int e^{-ik\cdot(w'-w)} K_1(x', x, y) \left\{ \right. \\ & + f^{abc} U^{de}(w') U^{eb}(w) \rho^d(x') \rho^a(x) \rho^c(y) \\ & \left. - f^{abc} U^{eb}(w) U^{ed}(x') \rho^d(x') \rho^a(x) \rho^c(y) \right\} \end{aligned} \quad (4.1)$$

with $K_1(x', x, y) = \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \mathcal{K}(x - z) \cdot \mathcal{K}(y - z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.4) with A, B, C the charges at x', x, y . The sub-terms are numbered 1–2 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{abc} U^{de}(w') U^{eb}(w) \rho^d(x') \rho^a(x) \rho^c(y)$.

$$\begin{aligned} [A, B] : [\rho^d(x'), \rho^a(x)] &= if^{dah} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_1(x', x', y) f^{abc} U^{de}(w') U^{eb}(w) f^{dah} \{\rho^h(x'), \rho^c(y)\} \\ &= \frac{1}{4} i K_1(x', x', y) f^{abc} f^{dae} U^{dh}(w') U^{hb}(w) \{\rho^e(x'), \rho^c(y)\} \\ [A, C] : [\rho^d(x'), \rho^c(y)] &= if^{dch} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \end{aligned} \quad (4.2)$$

$$\begin{aligned}\frac{1}{4}\{[A, C], B\} &= \frac{1}{4}i K_1(x', x, x') f^{abc} U^{de}(w') U^{eb}(w) f^{dch} \{\rho^h(x'), \rho^a(x)\} \\ &= \frac{1}{4}i K_1(x', x, x') f^{abc} f^{dce} U^{dh}(w') U^{hb}(w) \{\rho^e(x'), \rho^a(x)\}\end{aligned}\quad (4.3)$$

$$\begin{aligned}[B, C] : [\rho^a(x), \rho^c(y)] &= i f^{ach} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{4}\{[B, C], A\} &= \frac{1}{4}i K_1(x', x, x) f^{abc} U^{de}(w') U^{eb}(w) f^{ach} \{\rho^h(x), \rho^d(x')\} \\ &= -\frac{1}{4}i N_c K_1(x', x, x) U^{ab}(w') U^{bc}(w) \{\rho^c(x), \rho^a(x')\}\end{aligned}\quad (F) \quad (4.4)$$

Sub-term 2: $-f^{abc} U^{eb}(w) U^{ed}(x') \rho^d(x') \rho^a(x) \rho^c(y)$.

$$\begin{aligned}[A, B] : [\rho^d(x'), \rho^a(x)] &= i f^{dah} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4}\{[A, B], C\} &= -\frac{1}{4}i K_1(x', x', y) f^{abc} U^{eb}(w) U^{ed}(x') f^{dah} \{\rho^h(x'), \rho^c(y)\} \\ &= -\frac{1}{4}i K_1(x', x', y) f^{abc} f^{dae} U^{hb}(w) U^{hd}(x') \{\rho^e(x'), \rho^c(y)\}\end{aligned}\quad (4.5)$$

$$\begin{aligned}[A, C] : [\rho^d(x'), \rho^c(y)] &= i f^{dch} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4}\{[A, C], B\} &= -\frac{1}{4}i K_1(x', x, x') f^{abc} U^{eb}(w) U^{ed}(x') f^{dch} \{\rho^h(x'), \rho^a(x)\} \\ &= -\frac{1}{4}i K_1(x', x, x') f^{abc} f^{dce} U^{hb}(w) U^{hd}(x') \{\rho^e(x'), \rho^a(x)\}\end{aligned}\quad (4.6)$$

$$\begin{aligned}[B, C] : [\rho^a(x), \rho^c(y)] &= i f^{ach} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{4}\{[B, C], A\} &= -\frac{1}{4}i K_1(x', x, x) f^{abc} U^{eb}(w) U^{ed}(x') f^{ach} \{\rho^h(x), \rho^d(x')\} \\ &= \frac{1}{4}i N_c K_1(x', x, x) U^{ab}(w) U^{ac}(x') \{\rho^b(x), \rho^c(x')\}\end{aligned}\quad (F) \quad (4.7)$$

Two- ρ terms of this source. Multiplying by the prefactor $-i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}\left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{A1} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\ &\quad + \frac{1}{4} K_1(x', x', y) f^{abc} f^{dae} U^{dh}(w') U^{hb}(w) \{\rho^e(x'), \rho^c(y)\} \\ &\quad + \frac{1}{4} K_1(x', x, x') f^{abc} f^{dce} U^{dh}(w') U^{hb}(w) \{\rho^e(x'), \rho^a(x)\} \\ &\quad - \frac{1}{4} N_c K_1(x', x, x) U^{ab}(w') U^{bc}(w) \{\rho^c(x), \rho^a(x')\} \\ &\quad - \frac{1}{4} K_1(x', x', y) f^{abc} f^{dae} U^{hb}(w) U^{hd}(x') \{\rho^e(x'), \rho^c(y)\} \\ &\quad - \frac{1}{4} K_1(x', x, x') f^{abc} f^{dce} U^{hb}(w) U^{hd}(x') \{\rho^e(x'), \rho^a(x)\} \\ &\quad \left. + \frac{1}{4} N_c K_1(x', x, x) U^{ab}(w) U^{ac}(x') \{\rho^b(x), \rho^c(x')\} \right\}\end{aligned}\quad (4.8)$$

The kernel at the coincident points:

$$\begin{aligned}K_1(x', x', y) &= \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \\ K_1(x', x, x') &= \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \\ K_1(x', x, x) &= \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \frac{1}{(x - z)^2}\end{aligned}$$

4.2 Two Wilson lines, second term

Same prefactor $-i\mathcal{N}$ and no c.c.

The three- ρ term.

$$\begin{aligned}\left. \frac{d^3 N}{d^3 k} \right|_{A2} &= -i\mathcal{N} \int e^{-ik \cdot (w' - w)} K_2(x', x, y) \left\{ \right. \\ &\quad + f^{abc} U^{eb}(w') U^{ed}(y) \rho^c(x') \rho^a(x) \rho^d(y) \\ &\quad \left. - f^{abc} U^{be}(w') U^{ed}(w) \rho^c(x') \rho^a(x) \rho^d(y) \right\}\end{aligned}\quad (4.9)$$

with $K_2(x', x, y) = \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.4) with A, B, C the charges at x', x, y . The sub-terms are numbered 1–2 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{abc} U^{eb}(w') U^{ed}(y) \rho^c(x') \rho^a(x) \rho^d(y)$.

$$\begin{aligned} [A, B] : [\rho^c(x'), \rho^a(x)] &= i f^{cah} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_2(x', x', y) f^{abc} U^{eb}(w') U^{ed}(y) f^{cah} \{\rho^h(x'), \rho^d(y)\} \\ &= \frac{1}{4} i N_c K_2(x', x', y) U^{ab}(w') U^{ac}(y) \{\rho^b(x'), \rho^c(y)\} \quad (\text{F}) \end{aligned} \quad (4.10)$$

$$\begin{aligned} [A, C] : [\rho^c(x'), \rho^d(y)] &= i f^{cdh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_2(x', x, x') f^{abc} U^{eb}(w') U^{ed}(x') f^{cdh} \{\rho^h(x'), \rho^a(x)\} \\ &= \frac{1}{4} i K_2(x', x, x') f^{abc} f^{cde} U^{hb}(w') U^{hd}(x') \{\rho^e(x'), \rho^a(x)\} \end{aligned} \quad (4.11)$$

$$\begin{aligned} [B, C] : [\rho^a(x), \rho^d(y)] &= i f^{adh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_2(x', x, x) f^{abc} U^{eb}(w') U^{ed}(x) f^{adh} \{\rho^h(x), \rho^c(x')\} \\ &= \frac{1}{4} i K_2(x', x, x) f^{abc} f^{ade} U^{hb}(w') U^{hd}(x) \{\rho^e(x), \rho^c(x')\} \end{aligned} \quad (4.12)$$

Sub-term 2: $- f^{abc} U^{be}(w') U^{ed}(w) \rho^c(x') \rho^a(x) \rho^d(y)$.

$$\begin{aligned} [A, B] : [\rho^c(x'), \rho^a(x)] &= i f^{cah} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_2(x', x', y) f^{abc} U^{be}(w') U^{ed}(w) f^{cah} \{\rho^h(x'), \rho^d(y)\} \\ &= -\frac{1}{4} i N_c K_2(x', x', y) U^{ab}(w') U^{bc}(w) \{\rho^a(x'), \rho^c(y)\} \quad (\text{F}) \end{aligned} \quad (4.13)$$

$$\begin{aligned} [A, C] : [\rho^c(x'), \rho^d(y)] &= i f^{cdh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_2(x', x, x') f^{abc} U^{be}(w') U^{ed}(w) f^{cdh} \{\rho^h(x'), \rho^a(x)\} \\ &= -\frac{1}{4} i K_2(x', x, x') f^{abc} f^{cde} U^{bh}(w') U^{hd}(w) \{\rho^e(x'), \rho^a(x)\} \end{aligned} \quad (4.14)$$

$$\begin{aligned} [B, C] : [\rho^a(x), \rho^d(y)] &= i f^{adh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_2(x', x, x) f^{abc} U^{be}(w') U^{ed}(w) f^{adh} \{\rho^h(x), \rho^c(x')\} \\ &= -\frac{1}{4} i K_2(x', x, x) f^{abc} f^{ade} U^{bh}(w') U^{hd}(w) \{\rho^e(x), \rho^c(x')\} \end{aligned} \quad (4.15)$$

Two- ρ terms of this source. Multiplying by the prefactor $-i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} \left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{A_2} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\ &\quad + \frac{1}{4} N_c K_2(x', x', y) U^{ab}(w') U^{ac}(y) \{\rho^b(x'), \rho^c(y)\} \\ &\quad + \frac{1}{4} K_2(x', x, x') f^{abc} f^{cde} U^{hb}(w') U^{hd}(x') \{\rho^e(x'), \rho^a(x)\} \\ &\quad + \frac{1}{4} K_2(x', x, x) f^{abc} f^{ade} U^{hb}(w') U^{hd}(x) \{\rho^e(x), \rho^c(x')\} \\ &\quad - \frac{1}{4} N_c K_2(x', x', y) U^{ab}(w') U^{bc}(w) \{\rho^a(x'), \rho^c(y)\} \\ &\quad - \frac{1}{4} K_2(x', x, x') f^{abc} f^{cde} U^{bh}(w') U^{hd}(w) \{\rho^e(x'), \rho^a(x)\} \\ &\quad \left. - \frac{1}{4} K_2(x', x, x) f^{abc} f^{ade} U^{bh}(w') U^{hd}(w) \{\rho^e(x), \rho^c(x')\} \right\} \end{aligned} \quad (4.16)$$

The kernel at the coincident points:

$$\begin{aligned} K_2(x', x', y) &= \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \frac{1}{(x' - z)^2} \\ K_2(x', x, x') &= \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \\ K_2(x', x, x) &= \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \end{aligned}$$

4.3 Three Wilson lines, kernel K_1

In the notes the three-Wilson-line remainder has prefactor $+\frac{i}{(2\pi)^3} \frac{g^4}{8\pi^3} \frac{\log(\sqrt{\Lambda})}{2\pi^2} \frac{1}{k^+} = +i\mathcal{N}$. Its first and fourth terms share the kernel K_1 and are treated here; no c.c. is added.

The three- ρ term.

$$\begin{aligned} \left. \frac{d^3 N}{d^3 k} \right|_{B_1} &= i\mathcal{N} \int e^{-ik \cdot (w' - w)} K_1(x', x, y) \left\{ \right. \\ &\quad - f^{abc} U^{db}(w') U^{ec}(y) U^{af}(x) \rho^d(x') \rho^f(x) \rho^e(y) \\ &\quad \left. + f^{abc} U^{ec}(y) U^{bd}(x') U^{af}(x) \rho^d(x') \rho^f(x) \rho^e(y) \right\} \end{aligned} \quad (4.17)$$

with $K_1(x', x, y) = \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \mathcal{K}(x - z) \cdot \mathcal{K}(y - z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.4) with A, B, C the charges at x', x, y . The sub-terms are numbered 1–2 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $-f^{abc} U^{db}(w') U^{ec}(y) U^{af}(x) \rho^d(x') \rho^f(x) \rho^e(y)$.

$$\begin{aligned} [A, B] : [\rho^d(x'), \rho^f(x)] &= i f^{dfh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_1(x', x', y) f^{abc} U^{db}(w') U^{ec}(y) U^{af}(x') f^{dfh} \{\rho^h(x'), \rho^e(y)\} \\ &= -\frac{1}{4} i K_1(x', x', y) f^{abc} f^{deh} U^{db}(w') U^{lc}(y) U^{ae}(x') \{\rho^h(x'), \rho^l(y)\} \end{aligned} \quad (4.18)$$

$$\begin{aligned} [A, C] : [\rho^d(x'), \rho^e(y)] &= i f^{deh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_1(x', x, x') f^{abc} U^{db}(w') U^{ec}(x') U^{af}(x) f^{deh} \{\rho^h(x'), \rho^f(x)\} \\ &= -\frac{1}{4} i K_1(x', x, x') f^{abc} f^{deh} U^{db}(w') U^{ec}(x') U^{al}(x) \{\rho^h(x'), \rho^l(x)\} \end{aligned} \quad (4.19)$$

$$\begin{aligned} [B, C] : [\rho^f(x), \rho^e(y)] &= i f^{feh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_1(x', x, x) f^{abc} U^{db}(w') U^{ec}(x) U^{af}(x) f^{feh} \{\rho^h(x), \rho^d(x')\} \\ &= \frac{1}{4} i N_c K_1(x', x, x) U^{ab}(w') U^{bc}(x) \{\rho^c(x), \rho^a(x')\} \quad (\text{U2}^*), \quad (\text{F}) \end{aligned} \quad (4.20)$$

Sub-term 2: $+f^{abc} U^{ec}(y) U^{bd}(x') U^{af}(x) \rho^d(x') \rho^f(x) \rho^e(y)$.

$$\begin{aligned} [A, B] : [\rho^d(x'), \rho^f(x)] &= i f^{dfh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_1(x', x', y) f^{abc} U^{ec}(y) U^{bd}(x') U^{af}(x') f^{dfh} \{\rho^h(x'), \rho^e(y)\} \\ &= -\frac{1}{4} i N_c K_1(x', x', y) U^{ab}(y) U^{bc}(x') \{\rho^c(x'), \rho^a(y)\} \quad (\text{U2}), \quad (\text{F}) \end{aligned} \quad (4.21)$$

$$\begin{aligned} [A, C] : [\rho^d(x'), \rho^e(y)] &= i f^{deh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_1(x', x, x') f^{abc} U^{ec}(x') U^{bd}(x') U^{af}(x) f^{deh} \{\rho^h(x'), \rho^f(x)\} \\ &= \frac{1}{4} i N_c K_1(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \quad (\text{U2}^*), \quad (\text{F}) \end{aligned} \quad (4.22)$$

$$\begin{aligned} [B, C] : [\rho^f(x), \rho^e(y)] &= i f^{feh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_1(x', x, x) f^{abc} U^{ec}(x) U^{bd}(x') U^{af}(x) f^{feh} \{\rho^h(x), \rho^d(x')\} \\ &= -\frac{1}{4} i N_c K_1(x', x, x) U^{ab}(x') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \quad (\text{U2}^*), \quad (\text{F}) \end{aligned} \quad (4.23)$$

Two- ρ terms of this source. Multiplying by the prefactor $i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} \left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{\text{B1}} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\ &\quad + \frac{1}{4} K_1(x', x', y) f^{abc} f^{deh} U^{db}(w') U^{lc}(y) U^{ae}(x') \{\rho^h(x'), \rho^l(y)\} \\ &\quad + \frac{1}{4} K_1(x', x, x') f^{abc} f^{deh} U^{db}(w') U^{ec}(x') U^{al}(x) \{\rho^h(x'), \rho^l(x)\} \\ &\quad - \frac{1}{4} N_c K_1(x', x, x) U^{ab}(w') U^{bc}(x) \{\rho^c(x), \rho^a(x')\} \\ &\quad + \frac{1}{4} N_c K_1(x', x', y) U^{ab}(y) U^{bc}(x') \{\rho^c(x'), \rho^a(y)\} \\ &\quad - \frac{1}{4} N_c K_1(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \\ &\quad \left. + \frac{1}{4} N_c K_1(x', x, x) U^{ab}(x') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \right\} \end{aligned} \quad (4.24)$$

The kernel at the coincident points:

$$\begin{aligned} K_1(x', x', y) &= \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \\ K_1(x', x, x') &= \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \\ K_1(x', x, x) &= \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \frac{1}{(x - z)^2} \end{aligned}$$

4.4 Three Wilson lines, kernel K_3

The second and third terms of the three-Wilson-line remainder share the kernel K_3 ; prefactor $+i\mathcal{N}$, no c.c.

The three- ρ term.

$$\left. \frac{d^3 N}{d^3 k} \right|_{\text{B2}} = i\mathcal{N} \int e^{-ik \cdot (w' - w)} K_3(x', x, y) \left\{ \right.$$

$$\begin{aligned}
& + f^{abc} U^{ec}(x') U^{db}(w) U^{af}(y) \rho^e(x') \rho^d(x) \rho^f(y) \\
& - f^{abc} U^{ec}(x') U^{bd}(x) U^{af}(y) \rho^e(x') \rho^d(x) \rho^f(y) \}
\end{aligned} \tag{4.25}$$

with $K_3(x', x, y) = \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(y - z) \cdot \mathcal{K}(x' - z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.4) with A, B, C the charges at x', x, y . The sub-terms are numbered 1–2 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{abc} U^{ec}(x') U^{db}(w) U^{af}(y) \rho^e(x') \rho^d(x) \rho^f(y)$.

$$\begin{aligned}
[A, B] : [\rho^e(x'), \rho^d(x)] &= i f^{edh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{ [A, B], C \} &= \frac{1}{4} i K_3(x', x', y) f^{abc} U^{ec}(x') U^{db}(w) U^{af}(y) f^{edh} \{ \rho^h(x'), \rho^f(y) \} \\
&= \frac{1}{4} i K_3(x', x', y) f^{abc} f^{deh} U^{dc}(x') U^{eb}(w) U^{al}(y) \{ \rho^h(x'), \rho^l(y) \}
\end{aligned} \tag{4.26}$$

$$\begin{aligned}
[A, C] : [\rho^e(x'), \rho^f(y)] &= i f^{efh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{ [A, C], B \} &= \frac{1}{4} i K_3(x', x, x') f^{abc} U^{ec}(x') U^{db}(w) U^{af}(x') f^{efh} \{ \rho^h(x'), \rho^d(x) \} \\
&= \frac{1}{4} i N_c K_3(x', x, x') U^{ab}(w) U^{bc}(x') \{ \rho^c(x'), \rho^a(x) \} \quad (\text{U2}^*), \quad (\text{F})
\end{aligned} \tag{4.27}$$

$$\begin{aligned}
[B, C] : [\rho^d(x), \rho^f(y)] &= i f^{dfh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
\frac{1}{4} \{ [B, C], A \} &= \frac{1}{4} i K_3(x', x, x) f^{abc} U^{ec}(x') U^{db}(w) U^{af}(x) f^{dfh} \{ \rho^h(x), \rho^e(x') \} \\
&= \frac{1}{4} i K_3(x', x, x) f^{abc} f^{deh} U^{lc}(x') U^{db}(w) U^{ae}(x) \{ \rho^h(x), \rho^l(x') \}
\end{aligned} \tag{4.28}$$

Sub-term 2: $- f^{abc} U^{ec}(x') U^{bd}(x) U^{af}(y) \rho^e(x') \rho^d(x) \rho^f(y)$.

$$\begin{aligned}
[A, B] : [\rho^e(x'), \rho^d(x)] &= i f^{edh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{ [A, B], C \} &= -\frac{1}{4} i K_3(x', x', y) f^{abc} U^{ec}(x') U^{bd}(x') U^{af}(y) f^{edh} \{ \rho^h(x'), \rho^f(y) \} \\
&= -\frac{1}{4} i N_c K_3(x', x', y) U^{ab}(y) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \quad (\text{U2}^*), \quad (\text{F})
\end{aligned} \tag{4.29}$$

$$\begin{aligned}
[A, C] : [\rho^e(x'), \rho^f(y)] &= i f^{efh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{ [A, C], B \} &= -\frac{1}{4} i K_3(x', x, x') f^{abc} U^{ec}(x') U^{bd}(x) U^{af}(x') f^{efh} \{ \rho^h(x'), \rho^d(x) \} \\
&= -\frac{1}{4} i N_c K_3(x', x, x') U^{ab}(x) U^{ac}(x') \{ \rho^c(x'), \rho^b(x) \} \quad (\text{U2}^*), \quad (\text{F})
\end{aligned} \tag{4.30}$$

$$\begin{aligned}
[B, C] : [\rho^d(x), \rho^f(y)] &= i f^{dfh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
\frac{1}{4} \{ [B, C], A \} &= -\frac{1}{4} i K_3(x', x, x) f^{abc} U^{ec}(x') U^{bd}(x) U^{af}(x) f^{dfh} \{ \rho^h(x), \rho^e(x') \} \\
&= \frac{1}{4} i N_c K_3(x', x, x) U^{ab}(x') U^{bc}(x) \{ \rho^c(x), \rho^a(x') \} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{4.31}$$

Two- ρ terms of this source. Multiplying by the prefactor $i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}
\left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{\text{B2}} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \{ \\
&- \frac{1}{4} K_3(x', x', y) f^{abc} f^{deh} U^{dc}(x') U^{eb}(w) U^{al}(y) \{ \rho^h(x'), \rho^l(y) \} \\
&- \frac{1}{4} N_c K_3(x', x, x') U^{ab}(w) U^{bc}(x') \{ \rho^c(x'), \rho^a(x) \} \\
&- \frac{1}{4} K_3(x', x, x) f^{abc} f^{deh} U^{lc}(x') U^{db}(w) U^{ae}(x) \{ \rho^h(x), \rho^l(x') \} \\
&- \frac{1}{4} N_c K_3(x', x', y) U^{ab}(y) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \\
&+ \frac{1}{4} N_c K_3(x', x, x') U^{ab}(x) U^{ac}(x') \{ \rho^c(x'), \rho^b(x) \} \\
&- \frac{1}{4} N_c K_3(x', x, x) U^{ab}(x') U^{bc}(x) \{ \rho^c(x), \rho^a(x') \} \}
\end{aligned} \tag{4.32}$$

The kernel at the coincident points:

$$\begin{aligned}
K_3(x', x', y) &= \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(y - z) \cdot \mathcal{K}(x' - z) \\
K_3(x', x, x') &= \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \frac{1}{(x' - z)^2} \\
K_3(x', x, x) &= \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z)
\end{aligned}$$

5 Group I (with c.c.)

5.1 Group I, Row II (two-charge part of $\mathbb{A}^{(3)}$)

Input. In the notes,

$$\begin{aligned} \frac{d^3 N_2}{d^3 k} &= -\frac{1}{(2\pi)^3} \frac{g^4 f^{abc}}{64\pi^6} \frac{1}{k^+} \int_{w,w'} e^{-ik \cdot (w'-w)} \int d^2 \mathbf{p} d^2 \mathbf{k} \int_{x,y} e^{i\mathbf{k} \cdot (w-y)} e^{i\mathbf{p} \cdot (y-x)} \frac{(x'-w')^i}{(x'-w')^2 \mathbf{p}^2 \mathbf{k}^2} B^i(\mathbf{k}, \mathbf{p}) \log \frac{\vee}{\Lambda} \\ &\quad \times \left[U^{ab'}(x') \rho^{b'}(x') - U^{ab'}(w') \rho^{b'}(x') \right] \left[U^{ab}(w) \{\rho^b(x), \rho^c(y)\} - U^{bd}(x) U^{ce}(y) \{\rho^d(x), \rho^e(y)\} \right] + \text{c.c.}, \\ B^i(\mathbf{k}, \mathbf{p}) &\equiv -\frac{\mathbf{k}^2 \mathbf{p}^i - (\mathbf{k} \cdot \mathbf{p}) \mathbf{k}^i}{(\mathbf{k} - \mathbf{p})^2} - 2\mathbf{k}^i - \frac{\mathbf{p}^2 \mathbf{k}^i + \mathbf{k}^2 \mathbf{p}^i}{(\mathbf{k} - \mathbf{p})^2}. \end{aligned}$$

Two remarks before reordering.

1. *Colour.* As written, the indices a and b each appear three times in the first term of the second bracket: in f^{abc} , $U^{ab'}(x')$, $U^{ab}(w)$ and $\rho^b(x)$. This term is $U^{ab}(w) \mathbb{A}^{(3)b}$ with $\mathbb{A}^{(3)b} \propto f^{bcd} \{\rho^c(x), \rho^d(y)\}$, so we read it as $U^{ah}(w) f^{hbc} \{\rho^b(x), \rho^c(y)\}$. The second term, $f^{abc} U^{bd}(x) U^{ce}(y) \{\rho^d(x), \rho^e(y)\} = \bar{\mathbb{A}}^{(3)a}$, is unchanged.
2. *Kernel.* The Fourier transform is done in App. A. With $\int d^2 \mathbf{p} d^2 \mathbf{k} e^{i\mathbf{k} \cdot (w-y)} e^{i\mathbf{p} \cdot (y-x)} B^i / (\mathbf{p}^2 \mathbf{k}^2) = 2\pi i \mathfrak{F}^i(w; x, y)$, and since both colour structures are odd under $x \leftrightarrow y$, only the part of $\mathfrak{F}^i$ odd under $x \leftrightarrow y$ contributes. It is

$$\mathfrak{F}^i(w; x, y) = \mathcal{K}^i(x - w) \int_{z'} [K(w, y, z') - K(x, y, z')] - \mathcal{K}^i(y - w) \int_{z'} [K(w, x, z') - K(x, y, z')],$$

with $K(a, b, z') = \mathcal{K}(a - z') \cdot \mathcal{K}(b - z')$; equivalently $\mathfrak{F}^i(w; x, y) = -\pi [\mathcal{K}^i(x - w) \log \frac{(w-y)^2}{(x-y)^2} - \mathcal{K}^i(y - w) \log \frac{(w-x)^2}{(x-y)^2}]$.

The prefactor becomes $-\frac{g^4}{64\pi^6} \cdot 2\pi i = -\frac{i}{2} \cdot \frac{g^4}{16\pi^5}$, i.e. $-\frac{i}{2} \mathcal{N}$.

The three- ρ term.

$$\begin{aligned} \frac{d^3 N}{d^3 k} \Big|_{\text{G.I.II}} &= -\frac{1}{2} i \mathcal{N} \int e^{-ik \cdot (w'-w)} G_{\text{I.II}}(x', x, y) \left\{ \right. \\ &\quad + U^{ab'}(x') f^{hbc} U^{ah}(w) \rho^{b'}(x') \{\rho^b(x), \rho^c(y)\} \\ &\quad - U^{ab'}(x') f^{abc} U^{bd}(x) U^{ce}(y) \rho^{b'}(x') \{\rho^d(x), \rho^e(y)\} \\ &\quad - U^{ab'}(w') f^{hbc} U^{ah}(w) \rho^{b'}(x') \{\rho^b(x), \rho^c(y)\} \\ &\quad \left. + U^{ab'}(w') f^{abc} U^{bd}(x) U^{ce}(y) \rho^{b'}(x') \{\rho^d(x), \rho^e(y)\} \right\} + \text{c.c.} \end{aligned} \quad (5.1)$$

with $G_{\text{I.II}}(x', x, y) = \mathcal{K}^i(x' - w') \mathfrak{F}^i(w; x, y)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.6) with A, B, C the charges at x', x, y . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ U^{ab'}(x') f^{hbc} U^{ah}(w) \rho^{b'}(x') \{\rho^b(x), \rho^c(y)\}$.

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^b(x)] &= i f^{b'bl} \rho^l(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{2} \{[A, B], C\} &= \frac{1}{2} i G_{\text{I.II}}(x', x', y) U^{ab'}(x') f^{hbc} U^{ah}(w) f^{b'bl} \{\rho^l(x'), \rho^c(y)\} \\ &= \frac{1}{2} i G_{\text{I.II}}(x', x', y) f^{abc} f^{dbe} U^{hd}(x') U^{ha}(w) \{\rho^e(x'), \rho^c(y)\} \end{aligned} \quad (5.2)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^c(y)] &= i f^{b'cl} \rho^l(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{[A, C], B\} &= \frac{1}{2} i G_{\text{I.II}}(x', x, x') U^{ab'}(x') f^{hbc} U^{ah}(w) f^{b'cl} \{\rho^l(x'), \rho^b(x)\} \\ &= \frac{1}{2} i G_{\text{I.II}}(x', x, x') f^{abc} f^{dce} U^{hd}(x') U^{ha}(w) \{\rho^e(x'), \rho^b(x)\} \end{aligned} \quad (5.3)$$

Sub-term 2: $- U^{ab'}(x') f^{abc} U^{bd}(x) U^{ce}(y) \rho^{b'}(x') \{\rho^d(x), \rho^e(y)\}$.

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^d(x)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{2} \{[A, B], C\} &= -\frac{1}{2} i G_{\text{I.II}}(x', x', y) U^{ab'}(x') f^{abc} U^{bd}(x') U^{ce}(y) f^{b'dh} \{\rho^h(x'), \rho^e(y)\} \\ &= -\frac{1}{2} i N_c G_{\text{I.II}}(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U2}), \quad (\text{F}) \end{aligned} \quad (5.4)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^e(y)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{[A, C], B\} &= -\frac{1}{2} i G_{\text{I.II}}(x', x, x') U^{ab'}(x') f^{abc} U^{bd}(x) U^{ce}(x') f^{b'eh} \{\rho^h(x'), \rho^d(x)\} \\ &= \frac{1}{2} i N_c G_{\text{I.II}}(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \quad (\text{U2}), \quad (\text{F}) \end{aligned} \quad (5.5)$$

Sub-term 3: $-U^{ab'}(w') f^{hbc} U^{ah}(w) \rho^b(x') \{\rho^b(x), \rho^c(y)\}.$

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^b(x)] &= i f^{b'bl} \rho^l(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{2} \{[A, B], C\} &= -\frac{1}{2} i G_{\text{I,II}}(x', x', y) U^{ab'}(w') f^{hbc} U^{ah}(w) f^{b'bl} \{\rho^l(x'), \rho^c(y)\} \\ &= -\frac{1}{2} i G_{\text{I,II}}(x', x', y) f^{abc} f^{dbe} U^{hd}(w') U^{ha}(w) \{\rho^e(x'), \rho^c(y)\} \end{aligned} \quad (5.6)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^c(y)] &= i f^{b'cl} \rho^l(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{[A, C], B\} &= -\frac{1}{2} i G_{\text{I,II}}(x', x, x') U^{ab'}(w') f^{hbc} U^{ah}(w) f^{b'cl} \{\rho^l(x'), \rho^b(x)\} \\ &= -\frac{1}{2} i G_{\text{I,II}}(x', x, x') f^{abc} f^{dce} U^{hd}(w') U^{ha}(w) \{\rho^e(x'), \rho^b(x)\} \end{aligned} \quad (5.7)$$

Sub-term 4: $+U^{ab'}(w') f^{abc} U^{bd}(x) U^{ce}(y) \rho^{b'}(x') \{\rho^d(x), \rho^e(y)\}.$

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^d(x)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{2} \{[A, B], C\} &= \frac{1}{2} i G_{\text{I,II}}(x', x', y) U^{ab'}(w') f^{abc} U^{bd}(x') U^{ce}(y) f^{b'dh} \{\rho^h(x'), \rho^e(y)\} \\ &= \frac{1}{2} i G_{\text{I,II}}(x', x', y) f^{abc} f^{deh} U^{ad}(w') U^{be}(x') U^{cl}(y) \{\rho^h(x'), \rho^l(y)\} \end{aligned} \quad (5.8)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^e(y)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{[A, C], B\} &= \frac{1}{2} i G_{\text{I,II}}(x', x, x') U^{ab'}(w') f^{abc} U^{bd}(x) U^{ce}(x') f^{b'eh} \{\rho^h(x'), \rho^d(x)\} \\ &= \frac{1}{2} i G_{\text{I,II}}(x', x, x') f^{abc} f^{deh} U^{ad}(w') U^{bl}(x) U^{ce}(x') \{\rho^h(x'), \rho^l(x)\} \end{aligned} \quad (5.9)$$

Two- ρ terms of this source. Multiplying by the prefactor $-\frac{1}{2}i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} \left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{\text{GI,II}} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\ &+ \frac{1}{4} G_{\text{I,II}}(x', x', y) f^{abc} f^{dbe} U^{hd}(x') U^{ha}(w) \{\rho^e(x'), \rho^c(y)\} \\ &+ \frac{1}{4} G_{\text{I,II}}(x', x, x') f^{abc} f^{dce} U^{hd}(x') U^{ha}(w) \{\rho^e(x'), \rho^b(x)\} \\ &- \frac{1}{4} N_c G_{\text{I,II}}(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\ &+ \frac{1}{4} N_c G_{\text{I,II}}(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \\ &- \frac{1}{4} G_{\text{I,II}}(x', x', y) f^{abc} f^{dbe} U^{hd}(w') U^{ha}(w) \{\rho^e(x'), \rho^c(y)\} \\ &- \frac{1}{4} G_{\text{I,II}}(x', x, x') f^{abc} f^{dce} U^{hd}(w') U^{ha}(w) \{\rho^e(x'), \rho^b(x)\} \\ &+ \frac{1}{4} G_{\text{I,II}}(x', x', y) f^{abc} f^{deh} U^{ad}(w') U^{be}(x') U^{cl}(y) \{\rho^h(x'), \rho^l(y)\} \\ &+ \frac{1}{4} G_{\text{I,II}}(x', x, x') f^{abc} f^{deh} U^{ad}(w') U^{bl}(x) U^{ce}(x') \{\rho^h(x'), \rho^l(x)\} \left. \right\} + (w' \leftrightarrow w) \end{aligned} \quad (5.10)$$

The kernel at the coincident points:

$$\begin{aligned} G_{\text{I,II}}(x', x', y) &= \mathcal{K}^i(x' - w') \mathfrak{F}^i(w; x', y) \\ G_{\text{I,II}}(x', x, x') &= \mathcal{K}^i(x' - w') \mathfrak{F}^i(w; x, x') \end{aligned}$$

5.2 Group I, Row III ($\bar{\mathbb{B}}_3^\dagger \mathbb{A}$)

Input. Part 1 of the notes has three terms with the same colour structure. The first two (red) are equal and opposite and cancel. The third has prefactor $\frac{ig^4 f^{abd}}{16\pi^5} = i\mathcal{N} f^{abd}$ and kernel $K^{(1)} = \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(y - z)$. Part 2 has five terms, again with the same colour structure and prefactor $\pm \frac{ig^4 f^{abd}}{8\pi^5} = \pm 2i\mathcal{N} f^{abd}$; its terms 1 and 5 are equal, and so are terms 2 and 4. Together,

$$\begin{aligned} \text{Row III} &= i\mathcal{N} f^{abd} \int K_{\text{I,III}}(x', x, y) [\dots][\dots] + \text{c.c.}, \quad K_{\text{I,III}} = K^{(1)} + 2\mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \Phi(x), \\ \Phi(x) &\equiv \frac{1}{(z - w)^2} \left[\frac{2(z - w) \cdot (x - z)(x - w)^2 - 2(z - w) \cdot (x - w)(x - z)^2}{[(x - w)^2 - (x - z)^2]^2} - 1 \right]. \end{aligned}$$

The three- ρ term.

$$\begin{aligned} \left. \frac{d^3 N}{d^3 k} \right|_{\text{GI,III}} &= i\mathcal{N} \int e^{-ik \cdot (w' - w)} K_{\text{I,III}}(x', x, y) \left\{ \right. \\ &+ f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(z) \rho^{b'}(x') \rho^e(x) \rho^c(y) \end{aligned}$$

$$\begin{aligned}
& - f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(y) \rho^{b'}(x') \rho^e(x) \rho^c(y) \\
& - f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(z) \rho^{b'}(x') \rho^e(x) \rho^c(y) \\
& + f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(y) \rho^{b'}(x') \rho^e(x) \rho^c(y) \} + \text{c.c.}
\end{aligned} \tag{5.11}$$

with $K_{\text{I.III}}(x', x, y) = [\mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \Phi(x)]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.4) with A, B, C the charges at x', x, y . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(z) \rho^{b'}(x') \rho^e(x) \rho^c(y)$.

$$\begin{aligned}
[A, B] : [\rho^{b'}(x'), \rho^e(x)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\text{I.III}}(x', x', y) f^{abd} U^{ab'}(x') U^{de}(x') U^{bc}(z) f^{b'eh} \{\rho^h(x'), \rho^c(y)\} \\
&= -\frac{1}{4} i N_c K_{\text{I.III}}(x', x', y) U^{ab}(z) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{5.12}$$

$$\begin{aligned}
[A, C] : [\rho^{b'}(x'), \rho^c(y)] &= i f^{b'ch} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\text{I.III}}(x', x, x') f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(z) f^{b'ch} \{\rho^h(x'), \rho^e(x)\} \\
&= \frac{1}{4} i K_{\text{I.III}}(x', x, x') f^{abc} f^{deh} U^{ad}(x') U^{cl}(x) U^{be}(z) \{\rho^h(x'), \rho^l(x)\}
\end{aligned} \tag{5.13}$$

$$\begin{aligned}
[B, C] : [\rho^e(x), \rho^c(y)] &= i f^{ech} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
\frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\text{I.III}}(x', x, x) f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(z) f^{ech} \{\rho^h(x), \rho^{b'}(x')\} \\
&= \frac{1}{4} i K_{\text{I.III}}(x', x, x) f^{abc} f^{deh} U^{al}(x') U^{cd}(x) U^{be}(z) \{\rho^h(x), \rho^l(x')\}
\end{aligned} \tag{5.14}$$

Sub-term 2: $- f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(y) \rho^{b'}(x') \rho^e(x) \rho^c(y)$.

$$\begin{aligned}
[A, B] : [\rho^{b'}(x'), \rho^e(x)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\text{I.III}}(x', x', y) f^{abd} U^{ab'}(x') U^{de}(x') U^{bc}(y) f^{b'eh} \{\rho^h(x'), \rho^c(y)\} \\
&= \frac{1}{4} i N_c K_{\text{I.III}}(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{5.15}$$

$$\begin{aligned}
[A, C] : [\rho^{b'}(x'), \rho^c(y)] &= i f^{b'ch} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\text{I.III}}(x', x, x') f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(x') f^{b'ch} \{\rho^h(x'), \rho^e(x)\} \\
&= -\frac{1}{4} i N_c K_{\text{I.III}}(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{5.16}$$

$$\begin{aligned}
[B, C] : [\rho^e(x), \rho^c(y)] &= i f^{ech} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
\frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\text{I.III}}(x', x, x) f^{abd} U^{ab'}(x') U^{de}(x) U^{bc}(x) f^{ech} \{\rho^h(x), \rho^{b'}(x')\} \\
&= \frac{1}{4} i N_c K_{\text{I.III}}(x', x, x) U^{ab}(x') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{5.17}$$

Sub-term 3: $- f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(z) \rho^{b'}(x') \rho^e(x) \rho^c(y)$.

$$\begin{aligned}
[A, B] : [\rho^{b'}(x'), \rho^e(x)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\text{I.III}}(x', x', y) f^{abd} U^{ab'}(w') U^{de}(x') U^{bc}(z) f^{b'eh} \{\rho^h(x'), \rho^c(y)\} \\
&= -\frac{1}{4} i K_{\text{I.III}}(x', x', y) f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(z) \{\rho^h(x'), \rho^l(y)\}
\end{aligned} \tag{5.18}$$

$$\begin{aligned}
[A, C] : [\rho^{b'}(x'), \rho^c(y)] &= i f^{b'ch} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\text{I.III}}(x', x, x') f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(z) f^{b'ch} \{\rho^h(x'), \rho^e(x)\} \\
&= -\frac{1}{4} i K_{\text{I.III}}(x', x, x') f^{abc} f^{deh} U^{ad}(w') U^{cl}(x) U^{be}(z) \{\rho^h(x'), \rho^l(x)\}
\end{aligned} \tag{5.19}$$

$$\begin{aligned}
[B, C] : [\rho^e(x), \rho^c(y)] &= i f^{ech} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
\frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\text{I.III}}(x', x, x) f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(z) f^{ech} \{\rho^h(x), \rho^{b'}(x')\} \\
&= -\frac{1}{4} i K_{\text{I.III}}(x', x, x) f^{abc} f^{deh} U^{al}(w') U^{cd}(x) U^{be}(z) \{\rho^h(x), \rho^l(x')\}
\end{aligned} \tag{5.20}$$

Sub-term 4: $+ f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(y) \rho^{b'}(x') \rho^e(x) \rho^c(y)$.

$$\begin{aligned}
[A, B] : [\rho^{b'}(x'), \rho^e(x)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\text{I.III}}(x', x', y) f^{abd} U^{ab'}(w') U^{de}(x') U^{bc}(y) f^{b'eh} \{\rho^h(x'), \rho^c(y)\}
\end{aligned}$$

$$= \frac{1}{4} i K_{\text{I.III}}(x', x', y) f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(y) \{\rho^h(x'), \rho^l(y)\} \quad (5.21)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^c(y)] &= i f^{b'ch} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\text{I.III}}(x', x, x') f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(x') f^{b'ch} \{\rho^h(x'), \rho^e(x)\} \\ &= \frac{1}{4} i K_{\text{I.III}}(x', x, x') f^{abc} f^{deh} U^{ad}(w') U^{cl}(x) U^{be}(x') \{\rho^h(x'), \rho^l(x)\} \end{aligned} \quad (5.22)$$

$$\begin{aligned} [B, C] : [\rho^e(x), \rho^c(y)] &= i f^{ech} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\text{I.III}}(x', x, x) f^{abd} U^{ab'}(w') U^{de}(x) U^{bc}(x) f^{ech} \{\rho^h(x), \rho^{b'}(x')\} \\ &= -\frac{1}{4} i N_c K_{\text{I.III}}(x', x, x) U^{ab}(w') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \quad (\text{U2}), \quad (\text{F}) \end{aligned} \quad (5.23)$$

Two- ρ terms of this source. Multiplying by the prefactor $i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} \frac{d^3 N^{(2\rho)}}{d^3 k} \Big|_{\text{GI.III}} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\ &+ \frac{1}{4} N_c K_{\text{I.III}}(x', x', y) U^{ab}(z) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\ &- \frac{1}{4} K_{\text{I.III}}(x', x, x') f^{abc} f^{deh} U^{ad}(x') U^{cl}(x) U^{be}(z) \{\rho^h(x'), \rho^l(x)\} \\ &- \frac{1}{4} K_{\text{I.III}}(x', x, x) f^{abc} f^{deh} U^{al}(x') U^{cd}(x) U^{be}(z) \{\rho^h(x), \rho^l(x')\} \\ &- \frac{1}{4} N_c K_{\text{I.III}}(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\ &+ \frac{1}{4} N_c K_{\text{I.III}}(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \\ &- \frac{1}{4} N_c K_{\text{I.III}}(x', x, x) U^{ab}(x') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \\ &+ \frac{1}{4} K_{\text{I.III}}(x', x', y) f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(z) \{\rho^h(x'), \rho^l(y)\} \\ &+ \frac{1}{4} K_{\text{I.III}}(x', x, x') f^{abc} f^{deh} U^{ad}(w') U^{cl}(x) U^{be}(z) \{\rho^h(x'), \rho^l(x)\} \\ &+ \frac{1}{4} K_{\text{I.III}}(x', x, x) f^{abc} f^{deh} U^{al}(w') U^{cd}(x) U^{be}(z) \{\rho^h(x), \rho^l(x')\} \\ &- \frac{1}{4} K_{\text{I.III}}(x', x', y) f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(y) \{\rho^h(x'), \rho^l(y)\} \\ &- \frac{1}{4} K_{\text{I.III}}(x', x, x') f^{abc} f^{deh} U^{ad}(w') U^{cl}(x) U^{be}(x') \{\rho^h(x'), \rho^l(x)\} \\ &\left. + \frac{1}{4} N_c K_{\text{I.III}}(x', x, x) U^{ab}(w') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \right\} + (w' \leftrightarrow w) \end{aligned} \quad (5.24)$$

The kernel at the coincident points:

$$\begin{aligned} K_{\text{I.III}}(x', x', y) &= [\mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \Phi(x')] \\ K_{\text{I.III}}(x', x, x') &= [\mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - z) \Phi(x)] \\ K_{\text{I.III}}(x', x, x) &= [\mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \frac{1}{(x - z)^2} + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(x - z) \Phi(x)] \end{aligned}$$

5.3 Group I, Row IV ($\bar{\mathbb{A}}^\dagger \mathbb{B}_2$)

Input. Prefactor $\frac{1}{(2\pi)^3} \frac{ig^4}{8\pi^5} \frac{1}{k^+} \log \frac{\vee}{\Lambda} = 2i\mathcal{N}$. The kernel is the sum of the two kernels of the notes, $K_{\mathfrak{B}} \equiv \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') [\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z)]$.

The three- ρ term.

$$\begin{aligned} \frac{d^3 N}{d^3 k} \Big|_{\text{GI.IV}} &= 2i\mathcal{N} \int e^{-ik \cdot (w' - w)} K_{\mathfrak{B}}(x', x, y) \left\{ \right. \\ &+ U^{ab'}(x') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) \rho^{b'}(x') \rho^e(y) \rho^{c'}(x) \\ &- U^{ab'}(x') f^{abc} U^{be}(y) U^{cd}(x) \rho^{b'}(x') \rho^e(y) \rho^d(x) \\ &- U^{ab'}(w') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) \rho^{b'}(x') \rho^e(y) \rho^{c'}(x) \\ &\left. + U^{ab'}(w') f^{abc} U^{be}(y) U^{cd}(x) \rho^{b'}(x') \rho^e(y) \rho^d(x) \right\} + \text{c.c.} \end{aligned} \quad (5.25)$$

with $K_{\mathfrak{B}}(x', x, y) = \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') [\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z)]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.4) with A, B, C the charges at x', y, x . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+U^{ab'}(x') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) \rho^{b'}(x') \rho^e(y) \rho^{c'}(x).$

$$\begin{aligned}
[A, B] : [\rho^{b'}(x'), \rho^e(y)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\mathfrak{B}}(x', x, x') U^{ab'}(x') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(x') f^{b'eh} \{\rho^h(x'), \rho^{c'}(x)\} \\
&= \frac{1}{4} i K_{\mathfrak{B}}(x', x, x') f^{abc} f^{deh} U^{da}(w) U^{eb}(z) U^{hl}(x') \{\rho^l(x'), \rho^e(x)\} \quad (\text{U2})
\end{aligned} \tag{5.26}$$

$$\begin{aligned}
[A, C] : [\rho^{b'}(x'), \rho^{c'}(x)] &= i f^{b'c'h} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathfrak{B}}(x', x', y) U^{ab'}(x') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) f^{b'c'h} \{\rho^h(x'), \rho^e(y)\} \\
&= \frac{1}{4} i K_{\mathfrak{B}}(x', x', y) f^{abc} f^{dce} U^{hd}(x') U^{ha}(w) U^{lb}(z) U^{lm}(y) \{\rho^e(x'), \rho^m(y)\}
\end{aligned} \tag{5.27}$$

$$\begin{aligned}
[B, C] : [\rho^e(y), \rho^{c'}(x)] &= i f^{ec'h} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\
\frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathfrak{B}}(x', y, y) U^{ab'}(x') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) f^{ec'h} \{\rho^h(y), \rho^{b'}(x')\} \\
&= \frac{1}{4} i K_{\mathfrak{B}}(x', y, y) f^{abc} f^{dce} U^{hl}(x') U^{ha}(w) U^{mb}(z) U^{md}(y) \{\rho^e(y), \rho^l(x')\}
\end{aligned} \tag{5.28}$$

Sub-term 2: $-U^{ab'}(x') f^{abc} U^{be}(y) U^{cd}(x) \rho^{b'}(x') \rho^e(y) \rho^d(x).$

$$\begin{aligned}
[A, B] : [\rho^{b'}(x'), \rho^e(y)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathfrak{B}}(x', x, x') U^{ab'}(x') f^{abc} U^{be}(x') U^{cd}(x) f^{b'eh} \{\rho^h(x'), \rho^d(x)\} \\
&= -\frac{1}{4} i N_c K_{\mathfrak{B}}(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{5.29}$$

$$\begin{aligned}
[A, C] : [\rho^{b'}(x'), \rho^d(x)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathfrak{B}}(x', x', y) U^{ab'}(x') f^{abc} U^{be}(y) U^{cd}(x') f^{b'dh} \{\rho^h(x'), \rho^e(y)\} \\
&= \frac{1}{4} i N_c K_{\mathfrak{B}}(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{5.30}$$

$$\begin{aligned}
[B, C] : [\rho^e(y), \rho^d(x)] &= i f^{edh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\
\frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathfrak{B}}(x', y, y) U^{ab'}(x') f^{abc} U^{be}(y) U^{cd}(y) f^{edh} \{\rho^h(y), \rho^{b'}(x')\} \\
&= -\frac{1}{4} i N_c K_{\mathfrak{B}}(x', y, y) U^{ab}(x') U^{ac}(y) \{\rho^e(y), \rho^b(x')\} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{5.31}$$

Sub-term 3: $-U^{ab'}(w') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) \rho^{b'}(x') \rho^e(y) \rho^{c'}(x).$

$$\begin{aligned}
[A, B] : [\rho^{b'}(x'), \rho^e(y)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathfrak{B}}(x', x, x') U^{ab'}(w') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(x') f^{b'eh} \{\rho^h(x'), \rho^{c'}(x)\} \\
&= -\frac{1}{4} i K_{\mathfrak{B}}(x', x, x') f^{abc} f^{deh} U^{ld}(w') U^{la}(w) U^{mb}(z) U^{me}(x') \{\rho^h(x'), \rho^e(x)\}
\end{aligned} \tag{5.32}$$

$$\begin{aligned}
[A, C] : [\rho^{b'}(x'), \rho^{c'}(x)] &= i f^{b'c'h} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathfrak{B}}(x', x', y) U^{ab'}(w') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) f^{b'c'h} \{\rho^h(x'), \rho^e(y)\} \\
&= -\frac{1}{4} i K_{\mathfrak{B}}(x', x', y) f^{abc} f^{dce} U^{hd}(w') U^{ha}(w) U^{lb}(z) U^{lm}(y) \{\rho^e(x'), \rho^m(y)\}
\end{aligned} \tag{5.33}$$

$$\begin{aligned}
[B, C] : [\rho^e(y), \rho^{c'}(x)] &= i f^{ec'h} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\
\frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathfrak{B}}(x', y, y) U^{ab'}(w') f^{cdc'} U^{ac}(w) U^{bd}(z) U^{be}(y) f^{ec'h} \{\rho^h(y), \rho^{b'}(x')\} \\
&= -\frac{1}{4} i K_{\mathfrak{B}}(x', y, y) f^{abc} f^{dce} U^{hl}(w') U^{ha}(w) U^{mb}(z) U^{md}(y) \{\rho^e(y), \rho^l(x')\}
\end{aligned} \tag{5.34}$$

Sub-term 4: $+U^{ab'}(w') f^{abc} U^{be}(y) U^{cd}(x) \rho^{b'}(x') \rho^e(y) \rho^d(x).$

$$\begin{aligned}
[A, B] : [\rho^{b'}(x'), \rho^e(y)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\mathfrak{B}}(x', x, x') U^{ab'}(w') f^{abc} U^{be}(x') U^{cd}(x) f^{b'eh} \{\rho^h(x'), \rho^d(x)\} \\
&= \frac{1}{4} i K_{\mathfrak{B}}(x', x, x') f^{abc} f^{deh} U^{ad}(w') U^{be}(x') U^{cl}(x) \{\rho^h(x'), \rho^l(x)\}
\end{aligned} \tag{5.35}$$

$$\begin{aligned}
[A, C] : [\rho^{b'}(x'), \rho^d(x)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathfrak{B}}(x', x', y) U^{ab'}(w') f^{abc} U^{be}(y) U^{cd}(x') f^{b'dh} \{\rho^h(x'), \rho^e(y)\} \\
&= \frac{1}{4} i K_{\mathfrak{B}}(x', x', y) f^{abc} f^{deh} U^{ad}(w') U^{bl}(y) U^{ce}(x') \{\rho^h(x'), \rho^l(y)\}
\end{aligned} \tag{5.36}$$

$$\begin{aligned}
[B, C] : [\rho^e(y), \rho^d(x)] &= i f^{edh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\
\frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathfrak{B}}(x', y, y) U^{ab'}(w') f^{abc} U^{be}(y) U^{cd}(y) f^{edh} \{\rho^h(y), \rho^{b'}(x')\} \\
&= \frac{1}{4} i N_c K_{\mathfrak{B}}(x', y, y) U^{ab}(w') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{5.37}$$

Two- ρ terms of this source. Multiplying by the prefactor $2i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} \frac{d^3 N^{(2\rho)}}{d^3 k} \Big|_{\text{GI.IV}} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\ &\quad - \frac{1}{2} K_{\mathfrak{B}}(x', x, x') f^{abc} f^{deh} U^{da}(w) U^{eb}(z) U^{hl}(x') \{ \rho^l(x'), \rho^c(x) \} \\ &\quad - \frac{1}{2} K_{\mathfrak{B}}(x', x', y) f^{abc} f^{dce} U^{hd}(x') U^{ha}(w) U^{lb}(z) U^{lm}(y) \{ \rho^e(x'), \rho^m(y) \} \\ &\quad - \frac{1}{2} K_{\mathfrak{B}}(x', y, y) f^{abc} f^{dce} U^{hl}(x') U^{ha}(w) U^{mb}(z) U^{md}(y) \{ \rho^e(y), \rho^l(x') \} \\ &\quad + \frac{1}{2} N_c K_{\mathfrak{B}}(x', x, x') U^{ab}(x) U^{ac}(x') \{ \rho^c(x'), \rho^b(x) \} \\ &\quad - \frac{1}{2} N_c K_{\mathfrak{B}}(x', x', y) U^{ab}(y) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \\ &\quad + \frac{1}{2} N_c K_{\mathfrak{B}}(x', y, y) U^{ab}(x') U^{ac}(y) \{ \rho^c(y), \rho^b(x') \} \\ &\quad + \frac{1}{2} K_{\mathfrak{B}}(x', x, x') f^{abc} f^{deh} U^{ld}(w') U^{la}(w) U^{mb}(z) U^{me}(x') \{ \rho^h(x'), \rho^c(x) \} \\ &\quad + \frac{1}{2} K_{\mathfrak{B}}(x', x', y) f^{abc} f^{dce} U^{hd}(w') U^{ha}(w) U^{lb}(z) U^{lm}(y) \{ \rho^e(x'), \rho^m(y) \} \\ &\quad + \frac{1}{2} K_{\mathfrak{B}}(x', y, y) f^{abc} f^{dce} U^{hl}(w') U^{ha}(w) U^{mb}(z) U^{md}(y) \{ \rho^e(y), \rho^l(x') \} \\ &\quad - \frac{1}{2} K_{\mathfrak{B}}(x', x, x') f^{abc} f^{deh} U^{ad}(w') U^{be}(x') U^{cl}(x) \{ \rho^h(x'), \rho^l(x) \} \\ &\quad - \frac{1}{2} K_{\mathfrak{B}}(x', x', y) f^{abc} f^{deh} U^{ad}(w') U^{bl}(y) U^{ce}(x') \{ \rho^h(x'), \rho^l(y) \} \\ &\quad \left. - \frac{1}{2} N_c K_{\mathfrak{B}}(x', y, y) U^{ab}(w') U^{ac}(y) \{ \rho^c(y), \rho^b(x') \} \right\} + (w' \leftrightarrow w) \end{aligned} \quad (5.38)$$

The kernel at the coincident points:

$$\begin{aligned} K_{\mathfrak{B}}(x', x, x') &= \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(x' - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \right] \\ K_{\mathfrak{B}}(x', x', y) &= \mathcal{K}(x' - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \right] \\ K_{\mathfrak{B}}(x', y, y) &= \mathcal{K}(y - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \frac{1}{(y - z)^2} \right] \end{aligned}$$

5.4 Group I, Row V, first term (measured gluon at z)

Input. Prefactor $-\frac{1}{(2\pi)^3} \frac{ig^4}{32\pi^5} \frac{1}{k^+} \log \frac{\vee}{\Lambda} = -\frac{i}{2} \mathcal{N}$. After the w integration with the δ -function of $\mathbb{C}_1$ the measured gluon in the amplitude sits at z : the phase is $e^{-ik \cdot (w' - z)}$. The charges sit at x' , y and y' .

The three- ρ term.

$$\begin{aligned} \frac{d^3 N}{d^3 k} \Big|_{\text{GI.V1}} &= -\frac{1}{2} i \mathcal{N} \int e^{-ik \cdot (w' - z)} K_{V1}(x', y, y') \left\{ \right. \\ &\quad + U^{ab'}(x') f^{adc} U^{db}(x) U^{ce}(z) \rho^{b'}(x') \{ \rho^e(y), \rho^b(y') \} \\ &\quad - U^{ab'}(x') f^{abc} U^{cd}(y) U^{be}(y') \rho^{b'}(x') \{ \rho^d(y), \rho^e(y') \} \\ &\quad - U^{ab'}(w') f^{adc} U^{db}(x) U^{ce}(z) \rho^{b'}(x') \{ \rho^e(y), \rho^b(y') \} \\ &\quad \left. + U^{ab'}(w') f^{abc} U^{cd}(y) U^{be}(y') \rho^{b'}(x') \{ \rho^d(y), \rho^e(y') \} \right\} + \text{c.c.} \end{aligned} \quad (5.39)$$

with $K_{V1}(x', y, y') = \mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \mathcal{K}(z - x) \cdot \mathcal{K}(y' - x)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.6) with A, B, C the charges at x', y, y' . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ U^{ab'}(x') f^{adc} U^{db}(x) U^{ce}(z) \rho^{b'}(x') \{ \rho^e(y), \rho^b(y') \}$.

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^e(y)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{ [A, B], C \} &= \frac{1}{2} i K_{V1}(x', x', y') U^{ab'}(x') f^{adc} U^{db}(x) U^{ce}(z) f^{b'eh} \{ \rho^h(x'), \rho^b(y') \} \\ &= \frac{1}{2} i K_{V1}(x', x', y') f^{abc} f^{deh} U^{ad}(x') U^{bl}(x) U^{ce}(z) \{ \rho^h(x'), \rho^l(y') \} \end{aligned} \quad (5.40)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^b(y')] &= i f^{b'bh} \rho^h(x') \delta^{(2)}(x' - y'), \quad y' \rightarrow x' \\ \frac{1}{2} \{ [A, C], B \} &= \frac{1}{2} i K_{V1}(x', y, x') U^{ab'}(x') f^{adc} U^{db}(x) U^{ce}(z) f^{b'bh} \{ \rho^h(x'), \rho^e(y) \} \\ &= \frac{1}{2} i K_{V1}(x', y, x') f^{abc} f^{deh} U^{ad}(x') U^{be}(x) U^{cl}(z) \{ \rho^h(x'), \rho^l(y) \} \end{aligned} \quad (5.41)$$

Sub-term 2: $-U^{ab'}(x') f^{abc} U^{cd}(y) U^{be}(y') \rho^{b'}(x') \{\rho^d(y), \rho^e(y')\}.$

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^d(y)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{[A, B], C\} &= -\frac{1}{2} i K_{V1}(x', x', y') U^{ab'}(x') f^{abc} U^{cd}(x') U^{be}(y') f^{b'dh} \{\rho^h(x'), \rho^e(y')\} \\ &= \frac{1}{2} i N_c K_{V1}(x', x', y') U^{ab}(y') U^{ac}(x') \{\rho^c(x'), \rho^b(y')\} \quad (\text{U2}), \quad (\text{F}) \end{aligned} \quad (5.42)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^e(y')] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y'), \quad y' \rightarrow x' \\ \frac{1}{2} \{[A, C], B\} &= -\frac{1}{2} i K_{V1}(x', y, x') U^{ab'}(x') f^{abc} U^{cd}(y) U^{be}(x') f^{b'eh} \{\rho^h(x'), \rho^d(y)\} \\ &= -\frac{1}{2} i N_c K_{V1}(x', y, x') U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U2}), \quad (\text{F}) \end{aligned} \quad (5.43)$$

Sub-term 3: $-U^{ab'}(w') f^{adc} U^{db}(x) U^{ce}(z) \rho^{b'}(x') \{\rho^e(y), \rho^b(y')\}.$

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^e(y)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{[A, B], C\} &= -\frac{1}{2} i K_{V1}(x', x', y') U^{ab'}(w') f^{adc} U^{db}(x) U^{ce}(z) f^{b'eh} \{\rho^h(x'), \rho^b(y')\} \\ &= -\frac{1}{2} i K_{V1}(x', x', y') f^{abc} f^{deh} U^{ad}(w') U^{bl}(x) U^{ce}(z) \{\rho^h(x'), \rho^l(y')\} \end{aligned} \quad (5.44)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^b(y')] &= i f^{b'bh} \rho^h(x') \delta^{(2)}(x' - y'), \quad y' \rightarrow x' \\ \frac{1}{2} \{[A, C], B\} &= -\frac{1}{2} i K_{V1}(x', y, x') U^{ab'}(w') f^{adc} U^{db}(x) U^{ce}(z) f^{b'bh} \{\rho^h(x'), \rho^e(y)\} \\ &= -\frac{1}{2} i K_{V1}(x', y, x') f^{abc} f^{deh} U^{ad}(w') U^{be}(x) U^{cl}(z) \{\rho^h(x'), \rho^l(y)\} \end{aligned} \quad (5.45)$$

Sub-term 4: $+U^{ab'}(w') f^{abc} U^{cd}(y) U^{be}(y') \rho^{b'}(x') \{\rho^d(y), \rho^e(y')\}.$

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^d(y)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{[A, B], C\} &= \frac{1}{2} i K_{V1}(x', x', y') U^{ab'}(w') f^{abc} U^{cd}(x') U^{be}(y') f^{b'dh} \{\rho^h(x'), \rho^e(y')\} \\ &= \frac{1}{2} i K_{V1}(x', x', y') f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(y') \{\rho^h(x'), \rho^l(y')\} \end{aligned} \quad (5.46)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^e(y')] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y'), \quad y' \rightarrow x' \\ \frac{1}{2} \{[A, C], B\} &= \frac{1}{2} i K_{V1}(x', y, x') U^{ab'}(w') f^{abc} U^{cd}(y) U^{be}(x') f^{b'eh} \{\rho^h(x'), \rho^d(y)\} \\ &= \frac{1}{2} i K_{V1}(x', y, x') f^{abc} f^{deh} U^{ad}(w') U^{cl}(y) U^{be}(x') \{\rho^h(x'), \rho^l(y)\} \end{aligned} \quad (5.47)$$

Two- ρ terms of this source. Multiplying by the prefactor $-\frac{1}{2}i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} \left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{\text{GI.V1}} &= \mathcal{N} \int e^{-ik \cdot (w' - z)} \left\{ \begin{aligned} &+ \frac{1}{4} K_{V1}(x', x', y') f^{abc} f^{deh} U^{ad}(x') U^{bl}(x) U^{ce}(z) \{\rho^h(x'), \rho^l(y')\} \\ &+ \frac{1}{4} K_{V1}(x', y, x') f^{abc} f^{deh} U^{ad}(x') U^{be}(x) U^{cl}(z) \{\rho^h(x'), \rho^l(y)\} \\ &+ \frac{1}{4} N_c K_{V1}(x', x', y') U^{ab}(y') U^{ac}(x') \{\rho^c(x'), \rho^b(y')\} \\ &- \frac{1}{4} N_c K_{V1}(x', y, x') U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\ &- \frac{1}{4} K_{V1}(x', x', y') f^{abc} f^{deh} U^{ad}(w') U^{bl}(x) U^{ce}(z) \{\rho^h(x'), \rho^l(y')\} \\ &- \frac{1}{4} K_{V1}(x', y, x') f^{abc} f^{deh} U^{ad}(w') U^{be}(x) U^{cl}(z) \{\rho^h(x'), \rho^l(y)\} \\ &+ \frac{1}{4} K_{V1}(x', x', y') f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(y') \{\rho^h(x'), \rho^l(y')\} \\ &+ \frac{1}{4} K_{V1}(x', y, x') f^{abc} f^{deh} U^{ad}(w') U^{cl}(y) U^{be}(x') \{\rho^h(x'), \rho^l(y)\} \end{aligned} \right\} + (w' \leftrightarrow z) \end{aligned} \quad (5.48)$$

The kernel at the coincident points:

$$\begin{aligned} K_{V1}(x', x', y') &= \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - z) \mathcal{K}(z - x) \cdot \mathcal{K}(y' - x) \\ K_{V1}(x', y, x') &= \mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \mathcal{K}(z - x) \cdot \mathcal{K}(x' - x) \end{aligned}$$

5.5 Group I, Row V, second term (measured gluon at x)

Input. Prefactor $-\frac{i}{2}\mathcal{N}$; the measured gluon in the amplitude sits at x (phase $e^{-ik \cdot (w' - x)}$). Same colour structure as the first term.

The three- ρ term.

$$\begin{aligned} \left. \frac{d^3 N}{d^3 k} \right|_{\text{GI.V2}} = & -\frac{1}{2} i \mathcal{N} \int e^{-ik \cdot (w' - x)} K_{V2}(x', y, y') \{ \\ & + U^{ab'}(x') f^{adc} U^{db}(x) U^{ce}(z) \rho^{b'}(x') \{\rho^e(y), \rho^b(y')\} \\ & - U^{ab'}(x') f^{abc} U^{cd}(y) U^{be}(y') \rho^{b'}(x') \{\rho^d(y), \rho^e(y')\} \\ & - U^{ab'}(w') f^{adc} U^{db}(x) U^{ce}(z) \rho^{b'}(x') \{\rho^e(y), \rho^b(y')\} \\ & + U^{ab'}(w') f^{abc} U^{cd}(y) U^{be}(y') \rho^{b'}(x') \{\rho^d(y), \rho^e(y')\} \} + \text{c.c.} \end{aligned} \quad (5.49)$$

with $K_{V2}(x', y, y') = \mathcal{K}(x' - w') \cdot \mathcal{K}(y' - x) \mathcal{K}(z - x) \cdot \mathcal{K}(y - z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.6) with A, B, C the charges at x', y, y' . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ U^{ab'}(x') f^{adc} U^{db}(x) U^{ce}(z) \rho^{b'}(x') \{\rho^e(y), \rho^b(y')\}$.

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^e(y)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{[A, B], C\} &= \frac{1}{2} i K_{V2}(x', x', y') U^{ab'}(x') f^{adc} U^{db}(x) U^{ce}(z) f^{b'eh} \{\rho^h(x'), \rho^b(y')\} \\ &= \frac{1}{2} i K_{V2}(x', x', y') f^{abc} f^{deh} U^{ad}(x') U^{bl}(x) U^{ce}(z) \{\rho^h(x'), \rho^l(y')\} \end{aligned} \quad (5.50)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^b(y')] &= i f^{b'bh} \rho^h(x') \delta^{(2)}(x' - y'), \quad y' \rightarrow x' \\ \frac{1}{2} \{[A, C], B\} &= \frac{1}{2} i K_{V2}(x', y, x') U^{ab'}(x') f^{adc} U^{db}(x) U^{ce}(z) f^{b'bh} \{\rho^h(x'), \rho^e(y)\} \\ &= \frac{1}{2} i K_{V2}(x', y, x') f^{abc} f^{deh} U^{ad}(x') U^{be}(x) U^{cl}(z) \{\rho^h(x'), \rho^l(y)\} \end{aligned} \quad (5.51)$$

Sub-term 2: $- U^{ab'}(x') f^{abc} U^{cd}(y) U^{be}(y') \rho^{b'}(x') \{\rho^d(y), \rho^e(y')\}$.

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^d(y)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{[A, B], C\} &= -\frac{1}{2} i K_{V2}(x', x', y') U^{ab'}(x') f^{abc} U^{cd}(x') U^{be}(y') f^{b'dh} \{\rho^h(x'), \rho^e(y')\} \\ &= \frac{1}{2} i N_c K_{V2}(x', x', y') U^{ab}(y') U^{ac}(x') \{\rho^c(x'), \rho^b(y')\} \quad (\text{U2}), \quad (\text{F}) \end{aligned} \quad (5.52)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^e(y')] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y'), \quad y' \rightarrow x' \\ \frac{1}{2} \{[A, C], B\} &= -\frac{1}{2} i K_{V2}(x', y, x') U^{ab'}(x') f^{abc} U^{cd}(y) U^{be}(x') f^{b'eh} \{\rho^h(x'), \rho^d(y)\} \\ &= -\frac{1}{2} i N_c K_{V2}(x', y, x') U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U2}), \quad (\text{F}) \end{aligned} \quad (5.53)$$

Sub-term 3: $- U^{ab'}(w') f^{adc} U^{db}(x) U^{ce}(z) \rho^{b'}(x') \{\rho^e(y), \rho^b(y')\}$.

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^e(y)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{[A, B], C\} &= -\frac{1}{2} i K_{V2}(x', x', y') U^{ab'}(w') f^{adc} U^{db}(x) U^{ce}(z) f^{b'eh} \{\rho^h(x'), \rho^b(y')\} \\ &= -\frac{1}{2} i K_{V2}(x', x', y') f^{abc} f^{deh} U^{ad}(w') U^{bl}(x) U^{ce}(z) \{\rho^h(x'), \rho^l(y')\} \end{aligned} \quad (5.54)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^b(y')] &= i f^{b'bh} \rho^h(x') \delta^{(2)}(x' - y'), \quad y' \rightarrow x' \\ \frac{1}{2} \{[A, C], B\} &= -\frac{1}{2} i K_{V2}(x', y, x') U^{ab'}(w') f^{adc} U^{db}(x) U^{ce}(z) f^{b'bh} \{\rho^h(x'), \rho^e(y)\} \\ &= -\frac{1}{2} i K_{V2}(x', y, x') f^{abc} f^{deh} U^{ad}(w') U^{be}(x) U^{cl}(z) \{\rho^h(x'), \rho^l(y)\} \end{aligned} \quad (5.55)$$

Sub-term 4: $+ U^{ab'}(w') f^{abc} U^{cd}(y) U^{be}(y') \rho^{b'}(x') \{\rho^d(y), \rho^e(y')\}$.

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^d(y)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{[A, B], C\} &= \frac{1}{2} i K_{V2}(x', x', y') U^{ab'}(w') f^{abc} U^{cd}(x') U^{be}(y') f^{b'dh} \{\rho^h(x'), \rho^e(y')\} \\ &= \frac{1}{2} i K_{V2}(x', x', y') f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(y') \{\rho^h(x'), \rho^l(y')\} \end{aligned} \quad (5.56)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^e(y')] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - y'), \quad y' \rightarrow x' \\ \frac{1}{2} \{[A, C], B\} &= \frac{1}{2} i K_{V2}(x', y, x') U^{ab'}(w') f^{abc} U^{cd}(y) U^{be}(x') f^{b'eh} \{\rho^h(x'), \rho^d(y)\} \\ &= \frac{1}{2} i K_{V2}(x', y, x') f^{abc} f^{deh} U^{ad}(w') U^{cl}(y) U^{be}(x') \{\rho^h(x'), \rho^l(y)\} \end{aligned} \quad (5.57)$$

Two- ρ terms of this source. Multiplying by the prefactor $-\frac{1}{2}i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} \left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{\text{GI.V2}} &= \mathcal{N} \int e^{-ik \cdot (w' - x)} \left\{ \right. \\ &+ \frac{1}{4} K_{V2}(x', x', y') f^{abc} f^{deh} U^{ad}(x') U^{bl}(x) U^{ce}(z) \{\rho^h(x'), \rho^l(y')\} \\ &+ \frac{1}{4} K_{V2}(x', y, x') f^{abc} f^{deh} U^{ad}(x') U^{be}(x) U^{cl}(z) \{\rho^h(x'), \rho^l(y')\} \\ &+ \frac{1}{4} N_c K_{V2}(x', x', y') U^{ab}(y') U^{ac}(x') \{\rho^c(x'), \rho^b(y')\} \\ &- \frac{1}{4} N_c K_{V2}(x', y, x') U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y')\} \\ &- \frac{1}{4} K_{V2}(x', x', y') f^{abc} f^{deh} U^{ad}(w') U^{bl}(x) U^{ce}(z) \{\rho^h(x'), \rho^l(y')\} \\ &- \frac{1}{4} K_{V2}(x', y, x') f^{abc} f^{deh} U^{ad}(w') U^{be}(x) U^{cl}(z) \{\rho^h(x'), \rho^l(y')\} \\ &+ \frac{1}{4} K_{V2}(x', x', y') f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(y') \{\rho^h(x'), \rho^l(y')\} \\ &+ \frac{1}{4} K_{V2}(x', y, x') f^{abc} f^{deh} U^{ad}(w') U^{cl}(y) U^{be}(x') \{\rho^h(x'), \rho^l(y')\} \left. \right\} + (w' \leftrightarrow x) \end{aligned} \quad (5.58)$$

The kernel at the coincident points:

$$\begin{aligned} K_{V2}(x', x', y') &= \mathcal{K}(x' - w') \cdot \mathcal{K}(y' - x) \mathcal{K}(z - x) \cdot \mathcal{K}(x' - z) \\ K_{V2}(x', y, x') &= \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - x) \mathcal{K}(z - x) \cdot \mathcal{K}(y - z) \end{aligned}$$

6 Group II (no c.c.)

6.1 Group II, Row I, first term

Input. The prefactor is $-\frac{1}{(2\pi)^3} \frac{ig^4}{16\pi^5} \frac{1}{k^+} \log \frac{\vee}{\wedge} = -i\mathcal{N}$ and the kernel is $K_{\mathfrak{B}}$, as in Row IV of Group I. The charges appear as $\{\rho(x'), \rho(y)\}\rho(x)$. Group II has no c.c.; the second term of the row is its conjugate.

The three- ρ term.

$$\begin{aligned} \left. \frac{d^3 N}{d^3 k} \right|_{\text{GII.II}} &= -i\mathcal{N} \int e^{-ik \cdot (w' - w)} K_{\mathfrak{B}}(x', x, y) \left\{ \right. \\ &+ U^{ac'}(w') f^{cbd} U^{ac}(w) \{\rho^{c'}(x'), \rho^b(y)\} \rho^d(x) \\ &- U^{ac'}(w') f^{acd} U^{bc}(z) U^{de}(x) \{\rho^{c'}(x'), \rho^b(y)\} \rho^e(x) \\ &- U^{bc'}(z) U^{ad'}(x') U^{c'e'}(y) f^{cbd} U^{ac}(w) \{\rho^{d'}(x'), \rho^{e'}(y)\} \rho^d(x) \\ &+ U^{bc'}(z) U^{ad'}(x') U^{c'e'}(y) f^{acd} U^{bc}(z) U^{de}(x) \{\rho^{d'}(x'), \rho^{e'}(y)\} \rho^e(x) \left. \right\} \end{aligned} \quad (6.1)$$

with $K_{\mathfrak{B}}(x', x, y) = \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.7) with A, B, C the charges at x, x', y . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ U^{ac'}(w') f^{cbd} U^{ac}(w) \{\rho^{c'}(x'), \rho^b(y)\} \rho^d(x)$.

$$\begin{aligned} [A, B] : [\rho^d(x), \rho^{c'}(x')] &= i f^{dc'h} \rho^h(x) \delta^{(2)}(x - x'), \quad x' \rightarrow x \\ -\frac{1}{2} \{[A, B], C\} &= -\frac{1}{2} i K_{\mathfrak{B}}(x, x, y) U^{ac'}(w') f^{cbd} U^{ac}(w) f^{dc'h} \{\rho^h(x), \rho^b(y)\} \\ &= -\frac{1}{2} i K_{\mathfrak{B}}(x, x, y) f^{abc} f^{cde} U^{hd}(w') U^{ha}(w) \{\rho^e(x), \rho^b(y)\} \end{aligned} \quad (6.2)$$

$$\begin{aligned} [A, C] : [\rho^d(x), \rho^b(y)] &= i f^{dbh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ -\frac{1}{2} \{[A, C], B\} &= -\frac{1}{2} i K_{\mathfrak{B}}(x', x, x) U^{ac'}(w') f^{cbd} U^{ac}(w) f^{dbh} \{\rho^h(x), \rho^{c'}(x')\} \\ &= \frac{1}{2} i N_c K_{\mathfrak{B}}(x', x, x) U^{ab}(w') U^{ac}(w) \{\rho^c(x), \rho^b(x')\} \quad (\text{F}) \end{aligned} \quad (6.3)$$

Sub-term 2: $- U^{ac'}(w') f^{acd} U^{bc}(z) U^{de}(x) \{\rho^{c'}(x'), \rho^b(y)\} \rho^e(x)$.

$$\begin{aligned} [A, B] : [\rho^e(x), \rho^{c'}(x')] &= i f^{ec'h} \rho^h(x) \delta^{(2)}(x - x'), \quad x' \rightarrow x \\ -\frac{1}{2} \{[A, B], C\} &= \frac{1}{2} i K_{\mathfrak{B}}(x, x, y) U^{ac'}(w') f^{acd} U^{bc}(z) U^{de}(x) f^{ec'h} \{\rho^h(x), \rho^b(y)\} \end{aligned}$$

$$= \frac{1}{2} i K_{\mathfrak{B}}(x, x, y) f^{abc} f^{deh} U^{ae}(w') U^{lb}(z) U^{cd}(x) \{\rho^h(x), \rho^l(y)\} \quad (6.4)$$

$$\begin{aligned} [A, C] : [\rho^e(x), \rho^b(y)] &= i f^{ebh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ -\frac{1}{2} \{[A, C], B\} &= \frac{1}{2} i K_{\mathfrak{B}}(x', x, x) U^{ac'}(w') f^{acd} U^{bc}(z) U^{de}(x) f^{ebh} \{\rho^h(x), \rho^{e'}(x')\} \\ &= \frac{1}{2} i K_{\mathfrak{B}}(x', x, x) f^{abc} f^{deh} U^{al}(w') U^{eb}(z) U^{cd}(x) \{\rho^h(x), \rho^l(x')\} \end{aligned} \quad (6.5)$$

Sub-term 3: $-U^{bc'}(z) U^{ad'}(x') U^{c'e'}(y) f^{cbd} U^{ac}(w) \{\rho^{d'}(x'), \rho^{e'}(y)\} \rho^d(x).$

$$\begin{aligned} [A, B] : [\rho^d(x), \rho^{d'}(x')] &= i f^{dd'h} \rho^h(x) \delta^{(2)}(x - x'), \quad x' \rightarrow x \\ -\frac{1}{2} \{[A, B], C\} &= \frac{1}{2} i K_{\mathfrak{B}}(x, x, y) U^{bc'}(z) U^{ad'}(x) U^{c'e'}(y) f^{cbd} U^{ac}(w) f^{dd'h} \{\rho^h(x), \rho^{e'}(y)\} \\ &= \frac{1}{2} i K_{\mathfrak{B}}(x, x, y) f^{abc} f^{cde} U^{bh}(z) U^{ld}(x) U^{hm}(y) U^{la}(w) \{\rho^e(x), \rho^m(y)\} \end{aligned} \quad (6.6)$$

$$\begin{aligned} [A, C] : [\rho^d(x), \rho^{e'}(y)] &= i f^{de'h} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ -\frac{1}{2} \{[A, C], B\} &= \frac{1}{2} i K_{\mathfrak{B}}(x', x, x) U^{bc'}(z) U^{ad'}(x') U^{c'e'}(x) f^{cbd} U^{ac}(w) f^{de'h} \{\rho^h(x), \rho^{d'}(x')\} \\ &= \frac{1}{2} i K_{\mathfrak{B}}(x', x, x) f^{abc} f^{cde} U^{bh}(z) U^{lm}(x') U^{hd}(x) U^{la}(w) \{\rho^e(x), \rho^m(x')\} \end{aligned} \quad (6.7)$$

Sub-term 4: $+U^{bc'}(z) U^{ad'}(x') U^{c'e'}(y) f^{acd} U^{bc}(z) U^{de}(x) \{\rho^{d'}(x'), \rho^{e'}(y)\} \rho^e(x).$

$$\begin{aligned} [A, B] : [\rho^e(x), \rho^{d'}(x')] &= i f^{ed'h} \rho^h(x) \delta^{(2)}(x - x'), \quad x' \rightarrow x \\ -\frac{1}{2} \{[A, B], C\} &= -\frac{1}{2} i K_{\mathfrak{B}}(x, x, y) U^{bc'}(z) U^{ad'}(x) U^{c'e'}(y) f^{acd} U^{bc}(z) U^{de}(x) f^{ed'h} \{\rho^h(x), \rho^{e'}(y)\} \\ &= -\frac{1}{2} i N_c K_{\mathfrak{B}}(x, x, y) U^{ab}(y) U^{ac}(x) \{\rho^c(x), \rho^b(y)\} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (6.8)$$

$$\begin{aligned} [A, C] : [\rho^e(x), \rho^{e'}(y)] &= i f^{ee'h} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ -\frac{1}{2} \{[A, C], B\} &= -\frac{1}{2} i K_{\mathfrak{B}}(x', x, x) U^{bc'}(z) U^{ad'}(x') U^{c'e'}(x) f^{acd} U^{bc}(z) U^{de}(x) f^{ee'h} \{\rho^h(x), \rho^{d'}(x')\} \\ &= \frac{1}{2} i N_c K_{\mathfrak{B}}(x', x, x) U^{ab}(x') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (6.9)$$

Two- ρ terms of this source. Multiplying by the prefactor $-i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} \frac{d^3 N^{(2\rho)}}{d^3 k} \Big|_{\text{GH.II}} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\ &\quad -\frac{1}{2} K_{\mathfrak{B}}(x, x, y) f^{abc} f^{cde} U^{hd}(w') U^{ha}(w) \{\rho^e(x), \rho^b(y)\} \\ &\quad +\frac{1}{2} N_c K_{\mathfrak{B}}(x', x, x) U^{ab}(w') U^{ac}(w) \{\rho^c(x), \rho^b(x')\} \\ &\quad +\frac{1}{2} K_{\mathfrak{B}}(x, x, y) f^{abc} f^{deh} U^{ae}(w') U^{lb}(z) U^{cd}(x) \{\rho^h(x), \rho^l(y)\} \\ &\quad +\frac{1}{2} K_{\mathfrak{B}}(x', x, x) f^{abc} f^{deh} U^{al}(w') U^{eb}(z) U^{cd}(x) \{\rho^h(x), \rho^l(x')\} \\ &\quad +\frac{1}{2} K_{\mathfrak{B}}(x, x, y) f^{abc} f^{cde} U^{bh}(z) U^{ld}(x) U^{hm}(y) U^{la}(w) \{\rho^e(x), \rho^m(y)\} \\ &\quad +\frac{1}{2} K_{\mathfrak{B}}(x', x, x) f^{abc} f^{cde} U^{bh}(z) U^{lm}(x') U^{hd}(x) U^{la}(w) \{\rho^e(x), \rho^m(x')\} \\ &\quad -\frac{1}{2} N_c K_{\mathfrak{B}}(x, x, y) U^{ab}(y) U^{ac}(x) \{\rho^c(x), \rho^b(y)\} \\ &\quad \left. +\frac{1}{2} N_c K_{\mathfrak{B}}(x', x, x) U^{ab}(x') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \right\} \end{aligned} \quad (6.10)$$

The kernel at the coincident points:

$$\begin{aligned} K_{\mathfrak{B}}(x, x, y) &= \mathcal{K}(x - w) \cdot \mathcal{K}(x - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right] \\ K_{\mathfrak{B}}(x', x, x) &= \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(x - z) + \frac{1}{2} \frac{1}{(x - z)^2} \right] \end{aligned}$$

6.2 Group II, Row I, second term

Input. Prefactor $+i\mathcal{N}$ and kernel

$$\bar{K}_{\mathfrak{B}} = K_{\mathfrak{B}}|_{w \leftrightarrow w'} = \mathcal{K}(x - w') \cdot \mathcal{K}(x' - w) \left[\mathcal{K}(z - w') \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right].$$

The charges appear as $\rho(x) \{\rho(x'), \rho(y)\}$.

The three- ρ term.

$$\begin{aligned} \frac{d^3 N}{d^3 k} \Big|_{\text{GII.12}} &= i \mathcal{N} \int e^{-ik \cdot (w' - w)} \bar{K}_{\mathfrak{B}}(x', x, y) \left\{ \right. \\ &\quad + f^{c'bd'} U^{ac'}(w') U^{ac}(w) \rho^{d'}(x) \{ \rho^c(x'), \rho^b(y) \} \\ &\quad - f^{c'bd'} U^{ac'}(w') U^{bc}(z) U^{ad}(x') U^{ce}(y) \rho^{d'}(x) \{ \rho^d(x'), \rho^e(y) \} \\ &\quad - f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{ac}(w) \rho^{e'}(x) \{ \rho^c(x'), \rho^b(y) \} \\ &\quad \left. + f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{bc}(z) U^{ad}(x') U^{ce}(y) \rho^{e'}(x) \{ \rho^d(x'), \rho^e(y) \} \right\} \end{aligned} \quad (6.11)$$

$$\text{with } \bar{K}_{\mathfrak{B}}(x', x, y) = \mathcal{K}(x - w') \cdot \mathcal{K}(x' - w) \left[\mathcal{K}(z - w') \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right].$$

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.6) with A, B, C the charges at x, x', y . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{c'bd'} U^{ac'}(w') U^{ac}(w) \rho^{d'}(x) \{ \rho^c(x'), \rho^b(y) \}.$

$$\begin{aligned} [A, B] : [\rho^{d'}(x), \rho^c(x')] &= i f^{d'ch} \rho^h(x) \delta^{(2)}(x - x'), \quad x' \rightarrow x \\ \frac{1}{2} \{ [A, B], C \} &= \frac{1}{2} i \bar{K}_{\mathfrak{B}}(x, x, y) f^{c'bd'} U^{ac'}(w') U^{ac}(w) f^{d'ch} \{ \rho^h(x), \rho^b(y) \} \\ &= \frac{1}{2} i \bar{K}_{\mathfrak{B}}(x, x, y) f^{abc} f^{cde} U^{ha}(w') U^{hd}(w) \{ \rho^e(x), \rho^b(y) \} \end{aligned} \quad (6.12)$$

$$\begin{aligned} [A, C] : [\rho^{d'}(x), \rho^b(y)] &= i f^{d'bh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{2} \{ [A, C], B \} &= \frac{1}{2} i \bar{K}_{\mathfrak{B}}(x', x, x) f^{c'bd'} U^{ac'}(w') U^{ac}(w) f^{d'bh} \{ \rho^h(x), \rho^c(x') \} \\ &= -\frac{1}{2} i N_c \bar{K}_{\mathfrak{B}}(x', x, x) U^{ab}(w') U^{ac}(w) \{ \rho^b(x), \rho^c(x') \} \quad (\text{F}) \end{aligned} \quad (6.13)$$

Sub-term 2: $- f^{c'bd'} U^{ac'}(w') U^{bc}(z) U^{ad}(x') U^{ce}(y) \rho^{d'}(x) \{ \rho^d(x'), \rho^e(y) \}.$

$$\begin{aligned} [A, B] : [\rho^{d'}(x), \rho^d(x')] &= i f^{d'dh} \rho^h(x) \delta^{(2)}(x - x'), \quad x' \rightarrow x \\ \frac{1}{2} \{ [A, B], C \} &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}(x, x, y) f^{c'bd'} U^{ac'}(w') U^{bc}(z) U^{ad}(x') U^{ce}(y) f^{d'dh} \{ \rho^h(x), \rho^e(y) \} \\ &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}(x, x, y) f^{abc} f^{cde} U^{ha}(w') U^{bl}(z) U^{hd}(x) U^{lm}(y) \{ \rho^e(x), \rho^m(y) \} \end{aligned} \quad (6.14)$$

$$\begin{aligned} [A, C] : [\rho^{d'}(x), \rho^e(y)] &= i f^{d'eh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{2} \{ [A, C], B \} &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}(x', x, x) f^{c'bd'} U^{ac'}(w') U^{bc}(z) U^{ad}(x') U^{ce}(x) f^{d'eh} \{ \rho^h(x), \rho^d(x') \} \\ &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}(x', x, x) f^{abc} f^{cde} U^{ha}(w') U^{bl}(z) U^{hm}(x') U^{ld}(x) \{ \rho^e(x), \rho^m(x') \} \end{aligned} \quad (6.15)$$

Sub-term 3: $- f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{ac}(w) \rho^{e'}(x) \{ \rho^c(x'), \rho^b(y) \}.$

$$\begin{aligned} [A, B] : [\rho^{e'}(x), \rho^c(x')] &= i f^{e'ch} \rho^h(x) \delta^{(2)}(x - x'), \quad x' \rightarrow x \\ \frac{1}{2} \{ [A, B], C \} &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}(x, x, y) f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{ac}(w) f^{e'ch} \{ \rho^h(x), \rho^b(y) \} \\ &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}(x, x, y) f^{abc} f^{deh} U^{lb}(z) U^{cd}(x) U^{ae}(w) \{ \rho^h(x), \rho^l(y) \} \end{aligned} \quad (6.16)$$

$$\begin{aligned} [A, C] : [\rho^{e'}(x), \rho^b(y)] &= i f^{e'bh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{2} \{ [A, C], B \} &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}(x', x, x) f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{ac}(w) f^{e'bh} \{ \rho^h(x), \rho^c(x') \} \\ &= -\frac{1}{2} i \bar{K}_{\mathfrak{B}}(x', x, x) f^{abc} f^{deh} U^{eb}(z) U^{cd}(x) U^{al}(w) \{ \rho^h(x), \rho^l(x') \} \end{aligned} \quad (6.17)$$

Sub-term 4: $+ f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{bc}(z) U^{ad}(x') U^{ce}(y) \rho^{e'}(x) \{ \rho^d(x'), \rho^e(y) \}.$

$$\begin{aligned} [A, B] : [\rho^{e'}(x), \rho^d(x')] &= i f^{e'dh} \rho^h(x) \delta^{(2)}(x - x'), \quad x' \rightarrow x \\ \frac{1}{2} \{ [A, B], C \} &= \frac{1}{2} i \bar{K}_{\mathfrak{B}}(x, x, y) f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{bc}(z) U^{ad}(x') U^{ce}(y) f^{e'dh} \{ \rho^h(x), \rho^e(y) \} \\ &= \frac{1}{2} i N_c \bar{K}_{\mathfrak{B}}(x, x, y) U^{ab}(y) U^{ac}(x) \{ \rho^c(x), \rho^b(y) \} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (6.18)$$

$$\begin{aligned} [A, C] : [\rho^{e'}(x), \rho^e(y)] &= i f^{e'eh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{2} \{ [A, C], B \} &= \frac{1}{2} i \bar{K}_{\mathfrak{B}}(x', x, x) f^{ac'd'} U^{bc'}(z) U^{d'e'}(x) U^{bc}(z) U^{ad}(x') U^{ce}(x) f^{e'eh} \{ \rho^h(x), \rho^d(x') \} \\ &= -\frac{1}{2} i N_c \bar{K}_{\mathfrak{B}}(x', x, x) U^{ab}(x') U^{ac}(x) \{ \rho^c(x), \rho^b(x') \} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (6.19)$$

Two- ρ terms of this source. Multiplying by the prefactor $i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} \frac{d^3 N^{(2\rho)}}{d^3 k} \Big|_{\text{GII.I2}} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\ &\quad - \frac{1}{2} \bar{K}_{\mathfrak{B}}(x, x, y) f^{abc} f^{cde} U^{ha}(w') U^{hd}(w) \{ \rho^e(x), \rho^b(y) \} \\ &\quad + \frac{1}{2} N_c \bar{K}_{\mathfrak{B}}(x', x, x) U^{ab}(w') U^{ac}(w) \{ \rho^b(x), \rho^c(x') \} \\ &\quad + \frac{1}{2} \bar{K}_{\mathfrak{B}}(x, x, y) f^{abc} f^{cde} U^{ha}(w') U^{bl}(z) U^{hd}(x) U^{lm}(y) \{ \rho^e(x), \rho^m(y) \} \\ &\quad + \frac{1}{2} \bar{K}_{\mathfrak{B}}(x', x, x) f^{abc} f^{cde} U^{ha}(w') U^{bl}(z) U^{hm}(x') U^{ld}(x) \{ \rho^e(x), \rho^m(x') \} \\ &\quad + \frac{1}{2} \bar{K}_{\mathfrak{B}}(x, x, y) f^{abc} f^{deh} U^{lb}(z) U^{cd}(x) U^{ae}(w) \{ \rho^h(x), \rho^l(y) \} \\ &\quad + \frac{1}{2} \bar{K}_{\mathfrak{B}}(x', x, x) f^{abc} f^{deh} U^{eb}(z) U^{cd}(x) U^{al}(w) \{ \rho^h(x), \rho^l(x') \} \\ &\quad - \frac{1}{2} N_c \bar{K}_{\mathfrak{B}}(x, x, y) U^{ab}(y) U^{ac}(x) \{ \rho^c(x), \rho^b(y) \} \\ &\quad \left. + \frac{1}{2} N_c \bar{K}_{\mathfrak{B}}(x', x, x) U^{ab}(x') U^{ac}(x) \{ \rho^c(x), \rho^b(x') \} \right\} \end{aligned} \quad (6.20)$$

The kernel at the coincident points:

$$\begin{aligned} \bar{K}_{\mathfrak{B}}(x, x, y) &= \mathcal{K}(x - w') \cdot \mathcal{K}(x - w) \left[\mathcal{K}(z - w') \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right] \\ \bar{K}_{\mathfrak{B}}(x', x, x) &= \mathcal{K}(x - w') \cdot \mathcal{K}(x' - w) \left[\mathcal{K}(z - w') \cdot \mathcal{K}(x - z) + \frac{1}{2} \frac{1}{(x - z)^2} \right] \end{aligned}$$

7 Group III (with c.c.)

7.1 Group III, Row I

Input. Prefactor $-\frac{1}{(2\pi)^3} \frac{ig^4}{8\pi^5} \frac{1}{k^+} \log \frac{\vee}{\Lambda} = -2i\mathcal{N}$, kernel $K_{\mathfrak{B}}$. Order of the charges: $\rho(x')\rho(y)\rho(x)$.

The three- ρ term.

$$\begin{aligned} \frac{d^3 N}{d^3 k} \Big|_{\text{GIII.I}} &= -2i\mathcal{N} \int e^{-ik \cdot (w' - w)} K_{\mathfrak{B}}(x', x, y) \left\{ \right. \\ &\quad + U^{ab'}(x') U^{c'd'}(y) U^{c'b}(z) f^{cbd} U^{ac}(w) \rho^{b'}(x') \rho^{d'}(y) \rho^d(x) \\ &\quad - U^{ab'}(x') U^{c'd'}(y) U^{c'b}(z) f^{acd} U^{bc}(z) U^{de}(x) \rho^{b'}(x') \rho^{d'}(y) \rho^e(x) \\ &\quad - U^{bd'}(z) U^{d'e'}(y) U^{ac'}(w') f^{cbd} U^{ac}(w) \rho^{c'}(x') \rho^{e'}(y) \rho^d(x) \\ &\quad \left. + U^{bd'}(z) U^{d'e'}(y) U^{ac'}(w') f^{acd} U^{bc}(z) U^{de}(x) \rho^{c'}(x') \rho^{e'}(y) \rho^e(x) \right\} + \text{c.c.} \end{aligned} \quad (7.1)$$

with $K_{\mathfrak{B}}(x', x, y) = \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.4) with A, B, C the charges at x', y, x . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ U^{ab'}(x') U^{c'd'}(y) U^{c'b}(z) f^{cbd} U^{ac}(w) \rho^{b'}(x') \rho^{d'}(y) \rho^d(x)$.

$$\begin{aligned} [A, B] : [\rho^{b'}(x'), \rho^{d'}(y)] &= i f^{b'd'h} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{ [A, B], C \} &= \frac{1}{4} i K_{\mathfrak{B}}(x', x, x') U^{ab'}(x') U^{c'd'}(x') U^{c'b}(z) f^{cbd} U^{ac}(w) f^{b'd'h} \{ \rho^h(x'), \rho^d(x) \} \\ &= \frac{1}{4} i K_{\mathfrak{B}}(x', x, x') f^{abc} f^{deh} U^{eb}(z) U^{da}(w) U^{hl}(x') \{ \rho^l(x'), \rho^c(x) \} \quad (\text{U2}) \end{aligned} \quad (7.2)$$

$$\begin{aligned} [A, C] : [\rho^{b'}(x'), \rho^d(x)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{ [A, C], B \} &= \frac{1}{4} i K_{\mathfrak{B}}(x', x', y) U^{ab'}(x') U^{c'd'}(y) U^{c'b}(z) f^{cbd} U^{ac}(w) f^{b'dh} \{ \rho^h(x'), \rho^{d'}(y) \} \\ &= \frac{1}{4} i K_{\mathfrak{B}}(x', x', y) f^{abc} f^{dce} U^{hd}(x') U^{lm}(y) U^{lb}(z) U^{ha}(w) \{ \rho^e(x'), \rho^m(y) \} \end{aligned} \quad (7.3)$$

$$\begin{aligned} [B, C] : [\rho^{d'}(y), \rho^d(x)] &= i f^{d'dh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{ [B, C], A \} &= \frac{1}{4} i K_{\mathfrak{B}}(x', y, y) U^{ab'}(x') U^{c'd'}(y) U^{c'b}(z) f^{cbd} U^{ac}(w) f^{d'dh} \{ \rho^h(y), \rho^{b'}(x') \} \\ &= \frac{1}{4} i K_{\mathfrak{B}}(x', y, y) f^{abc} f^{dce} U^{hl}(x') U^{md}(y) U^{mb}(z) U^{ha}(w) \{ \rho^e(y), \rho^l(x') \} \end{aligned} \quad (7.4)$$

Sub-term 2: $-U^{ab'}(x')U^{c'd'}(y)U^{c'b}(z)f^{acd}U^{bc}(z)U^{de}(x)\rho^{b'}(x')\rho^{d'}(y)\rho^e(x)$.

$$\begin{aligned}[A, B] : [\rho^{b'}(x'), \rho^{d'}(y)] &= if^{b'd'h}\rho^h(x')\delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4}\{[A, B], C\} &= -\frac{1}{4}i K_{\mathfrak{B}}(x', x, x')U^{ab'}(x')U^{c'd'}(x')U^{c'b}(z)f^{acd}U^{bc}(z)U^{de}(x)f^{b'd'h}\{\rho^h(x'), \rho^e(x)\} \\ &= -\frac{1}{4}iN_c K_{\mathfrak{B}}(x', x, x')U^{ab}(x)U^{ac}(x')\{\rho^c(x'), \rho^b(x')\} \quad (\text{U1}^*), (\text{U2}), (\text{F})\end{aligned}\quad (7.5)$$

$$\begin{aligned}[A, C] : [\rho^{b'}(x'), \rho^e(x)] &= if^{b'eh}\rho^h(x')\delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4}\{[A, C], B\} &= -\frac{1}{4}i K_{\mathfrak{B}}(x', x', y)U^{ab'}(x')U^{c'd'}(y)U^{c'b}(z)f^{acd}U^{bc}(z)U^{de}(x')f^{b'eh}\{\rho^h(x'), \rho^{d'}(y)\} \\ &= \frac{1}{4}iN_c K_{\mathfrak{B}}(x', x', y)U^{ab}(y)U^{ac}(x')\{\rho^c(x'), \rho^b(y)\} \quad (\text{U1}^*), (\text{U2}), (\text{F})\end{aligned}\quad (7.6)$$

$$\begin{aligned}[B, C] : [\rho^{d'}(y), \rho^e(x)] &= if^{d'eh}\rho^h(y)\delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4}\{[B, C], A\} &= -\frac{1}{4}i K_{\mathfrak{B}}(x', y, y)U^{ab'}(x')U^{c'd'}(y)U^{c'b}(z)f^{acd}U^{bc}(z)U^{de}(y)f^{d'eh}\{\rho^h(y), \rho^{b'}(x')\} \\ &= -\frac{1}{4}iN_c K_{\mathfrak{B}}(x', y, y)U^{ab}(x')U^{ac}(y)\{\rho^c(y), \rho^b(x')\} \quad (\text{U1}^*), (\text{U2}), (\text{F})\end{aligned}\quad (7.7)$$

Sub-term 3: $-U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')f^{cbd}U^{ac}(w)\rho^{c'}(x')\rho^{e'}(y)\rho^d(x)$.

$$\begin{aligned}[A, B] : [\rho^{c'}(x'), \rho^{e'}(y)] &= if^{c'e'h}\rho^h(x')\delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4}\{[A, B], C\} &= -\frac{1}{4}i K_{\mathfrak{B}}(x', x, x')U^{bd'}(z)U^{d'e'}(x')U^{ac'}(w')f^{cbd}U^{ac}(w)f^{c'e'h}\{\rho^h(x'), \rho^d(x)\} \\ &= -\frac{1}{4}i K_{\mathfrak{B}}(x', x, x')f^{abc}f^{deh}U^{bl}(z)U^{le}(x')U^{md}(w')U^{ma}(w)\{\rho^h(x'), \rho^c(x)\}\end{aligned}\quad (7.8)$$

$$\begin{aligned}[A, C] : [\rho^{c'}(x'), \rho^d(x)] &= if^{c'dh}\rho^h(x')\delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4}\{[A, C], B\} &= -\frac{1}{4}i K_{\mathfrak{B}}(x', x', y)U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')f^{cbd}U^{ac}(w)f^{c'dh}\{\rho^h(x'), \rho^{e'}(y)\} \\ &= -\frac{1}{4}i K_{\mathfrak{B}}(x', x', y)f^{abc}f^{dce}U^{bh}(z)U^{hl}(y)U^{md}(w')U^{ma}(w)\{\rho^e(x'), \rho^l(y)\}\end{aligned}\quad (7.9)$$

$$\begin{aligned}[B, C] : [\rho^{e'}(y), \rho^d(x)] &= if^{e'dh}\rho^h(y)\delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4}\{[B, C], A\} &= -\frac{1}{4}i K_{\mathfrak{B}}(x', y, y)U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')f^{cbd}U^{ac}(w)f^{e'dh}\{\rho^h(y), \rho^{c'}(x')\} \\ &= -\frac{1}{4}i K_{\mathfrak{B}}(x', y, y)f^{abc}f^{dce}U^{bh}(z)U^{hd}(y)U^{lm}(w')U^{la}(w)\{\rho^e(y), \rho^m(x')\}\end{aligned}\quad (7.10)$$

Sub-term 4: $+U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')f^{acd}U^{bc}(z)U^{de}(x)\rho^{c'}(x')\rho^{e'}(y)\rho^e(x)$.

$$\begin{aligned}[A, B] : [\rho^{c'}(x'), \rho^{e'}(y)] &= if^{c'e'h}\rho^h(x')\delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4}\{[A, B], C\} &= \frac{1}{4}i K_{\mathfrak{B}}(x', x, x')U^{bd'}(z)U^{d'e'}(x')U^{ac'}(w')f^{acd}U^{bc}(z)U^{de}(x)f^{c'e'h}\{\rho^h(x'), \rho^e(x)\} \\ &= \frac{1}{4}i K_{\mathfrak{B}}(x', x, x')f^{abc}f^{deh}U^{be}(x')U^{ad}(w')U^{cl}(x)\{\rho^h(x'), \rho^l(x)\} \quad (\text{U1})\end{aligned}\quad (7.11)$$

$$\begin{aligned}[A, C] : [\rho^{c'}(x'), \rho^e(x)] &= if^{c'eh}\rho^h(x')\delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4}\{[A, C], B\} &= \frac{1}{4}i K_{\mathfrak{B}}(x', x', y)U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')f^{acd}U^{bc}(z)U^{de}(x')f^{c'eh}\{\rho^h(x'), \rho^{e'}(y)\} \\ &= \frac{1}{4}i K_{\mathfrak{B}}(x', x', y)f^{abc}f^{deh}U^{bl}(y)U^{ad}(w')U^{ce}(x')\{\rho^h(x'), \rho^l(y)\} \quad (\text{U1})\end{aligned}\quad (7.12)$$

$$\begin{aligned}[B, C] : [\rho^{e'}(y), \rho^e(x)] &= if^{e'eh}\rho^h(y)\delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4}\{[B, C], A\} &= \frac{1}{4}i K_{\mathfrak{B}}(x', y, y)U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')f^{acd}U^{bc}(z)U^{de}(y)f^{e'eh}\{\rho^h(y), \rho^{c'}(x')\} \\ &= \frac{1}{4}iN_c K_{\mathfrak{B}}(x', y, y)U^{ab}(w')U^{ac}(y)\{\rho^c(y), \rho^b(x')\} \quad (\text{U1}), (\text{U2}), (\text{F})\end{aligned}\quad (7.13)$$

Two- ρ terms of this source. Multiplying by the prefactor $-2i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GIII.I}} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\ &+ \frac{1}{2} K_{\mathfrak{B}}(x', x, x')f^{abc}f^{deh}U^{eb}(z)U^{da}(w)U^{hl}(x')\{\rho^l(x'), \rho^c(x)\} \\ &+ \frac{1}{2} K_{\mathfrak{B}}(x', x', y)f^{abc}f^{dce}U^{hd}(x')U^{lm}(y)U^{lb}(z)U^{ha}(w)\{\rho^e(x'), \rho^m(y)\} \\ &+ \frac{1}{2} K_{\mathfrak{B}}(x', y, y)f^{abc}f^{dce}U^{hl}(x')U^{md}(y)U^{mb}(z)U^{ha}(w)\{\rho^e(y), \rho^l(x')\} \\ &- \frac{1}{2} N_c K_{\mathfrak{B}}(x', x, x')U^{ab}(x)U^{ac}(x')\{\rho^c(x'), \rho^b(x)\} \\ &+ \frac{1}{2} N_c K_{\mathfrak{B}}(x', x', y)U^{ab}(y)U^{ac}(x')\{\rho^c(x'), \rho^b(y)\} \\ &- \frac{1}{2} N_c K_{\mathfrak{B}}(x', y, y)U^{ab}(x')U^{ac}(y)\{\rho^c(y), \rho^b(x')\} \\ &- \frac{1}{2} K_{\mathfrak{B}}(x', x, x')f^{abc}f^{deh}U^{bl}(z)U^{le}(x')U^{md}(w')U^{ma}(w)\{\rho^h(x'), \rho^c(x)\} \\ &- \frac{1}{2} K_{\mathfrak{B}}(x', x', y)f^{abc}f^{dce}U^{bh}(z)U^{hl}(y)U^{md}(w')U^{ma}(w)\{\rho^e(x'), \rho^l(y)\} \\ &- \frac{1}{2} K_{\mathfrak{B}}(x', y, y)f^{abc}f^{dce}U^{bh}(z)U^{hd}(y)U^{lm}(w')U^{la}(w)\{\rho^e(y), \rho^m(x')\} \\ &\left. \right\}\end{aligned}$$

$$\begin{aligned}
& + \frac{1}{2} K_{\mathfrak{B}}(x', x, x') f^{abc} f^{deh} U^{be}(x') U^{ad}(w') U^{cl}(x) \{\rho^h(x'), \rho^l(x)\} \\
& + \frac{1}{2} K_{\mathfrak{B}}(x', x', y) f^{abc} f^{deh} U^{bl}(y) U^{ad}(w') U^{ce}(x') \{\rho^h(x'), \rho^l(y)\} \\
& + \frac{1}{2} N_c K_{\mathfrak{B}}(x', y, y) U^{ab}(w') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \} + (w' \leftrightarrow w)
\end{aligned} \tag{7.14}$$

The kernel at the coincident points:

$$\begin{aligned}
K_{\mathfrak{B}}(x', x, x') &= \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(x' - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \right] \\
K_{\mathfrak{B}}(x', x', y) &= \mathcal{K}(x' - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \right] \\
K_{\mathfrak{B}}(x', y, y) &= \mathcal{K}(y - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \frac{1}{(y - z)^2} \right]
\end{aligned}$$

7.2 Group III, Row II

Input. Prefactor $-2i\mathcal{N}$, kernel $K_{\mathfrak{B}}$. Order of the charges: $\rho(y)\rho(x')\rho(x)$.

The three- ρ term.

$$\begin{aligned}
\left. \frac{d^3 N}{d^3 k} \right|_{\text{GIII.II}} &= -2i\mathcal{N} \int e^{-ik \cdot (w' - w)} K_{\mathfrak{B}}(x', x, y) \left\{ \right. \\
& + U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{cbd} U^{ac}(w) \rho^{d'}(y) \rho^{b'}(x') \rho^d(x) \\
& - U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{acd} U^{bc}(z) U^{de}(x) \rho^{d'}(y) \rho^{b'}(x') \rho^e(x) \\
& - U^{ab'}(x') f^{cbd} U^{ac}(w) \rho^b(y) \rho^{b'}(x') \rho^d(x) \\
& \left. + U^{ab'}(x') f^{acd} U^{bc}(z) U^{de}(x) \rho^b(y) \rho^{b'}(x') \rho^e(x) \right\} + \text{c.c.}
\end{aligned} \tag{7.15}$$

with $K_{\mathfrak{B}}(x', x, y) = \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.4) with A, B, C the charges at y, x', x . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{cbd} U^{ac}(w) \rho^{d'}(y) \rho^{b'}(x') \rho^d(x)$.

$$\begin{aligned}
[A, B] : [\rho^{d'}(y), \rho^{b'}(x')] &= i f^{d'b'h} \rho^h(y) \delta^{(2)}(y - x'), \quad x' \rightarrow y \\
\frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\mathfrak{B}}(y, x, y) U^{c'd'}(y) U^{ab'}(y) U^{bc'}(z) f^{cbd} U^{ac}(w) f^{d'b'h} \{\rho^h(y), \rho^d(x)\} \\
&= \frac{1}{4} i K_{\mathfrak{B}}(y, x, y) f^{abc} f^{deh} U^{bd}(z) U^{ea}(w) U^{hl}(y) \{\rho^l(y), \rho^e(x)\} \quad (\text{U2})
\end{aligned} \tag{7.16}$$

$$\begin{aligned}
[A, C] : [\rho^{d'}(y), \rho^d(x)] &= i f^{d'dh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\
\frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathfrak{B}}(x', y, y) U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{cbd} U^{ac}(w) f^{d'dh} \{\rho^h(y), \rho^{b'}(x')\} \\
&= \frac{1}{4} i K_{\mathfrak{B}}(x', y, y) f^{abc} f^{dce} U^{hd}(y) U^{lm}(x') U^{bh}(z) U^{la}(w) \{\rho^e(y), \rho^m(x')\}
\end{aligned} \tag{7.17}$$

$$\begin{aligned}
[B, C] : [\rho^{b'}(x'), \rho^d(x)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathfrak{B}}(x', x', y) U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{cbd} U^{ac}(w) f^{b'dh} \{\rho^h(x'), \rho^{d'}(y)\} \\
&= \frac{1}{4} i K_{\mathfrak{B}}(x', x', y) f^{abc} f^{dce} U^{hl}(y) U^{md}(x') U^{bh}(z) U^{ma}(w) \{\rho^e(x'), \rho^l(y)\}
\end{aligned} \tag{7.18}$$

Sub-term 2: $- U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{acd} U^{bc}(z) U^{de}(x) \rho^{d'}(y) \rho^{b'}(x') \rho^e(x)$.

$$\begin{aligned}
[A, B] : [\rho^{d'}(y), \rho^{b'}(x')] &= i f^{d'b'h} \rho^h(y) \delta^{(2)}(y - x'), \quad x' \rightarrow y \\
\frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathfrak{B}}(y, x, y) U^{c'd'}(y) U^{ab'}(y) U^{bc'}(z) f^{acd} U^{bc}(z) U^{de}(x) f^{d'b'h} \{\rho^h(y), \rho^e(x)\} \\
&= \frac{1}{4} i N_c K_{\mathfrak{B}}(y, x, y) U^{ab}(x) U^{ac}(y) \{\rho^c(y), \rho^b(x)\} \quad (\text{U1}), (\text{U2}), (\text{F})
\end{aligned} \tag{7.19}$$

$$\begin{aligned}
[A, C] : [\rho^{d'}(y), \rho^e(x)] &= i f^{d'eh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\
\frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathfrak{B}}(x', y, y) U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{acd} U^{bc}(z) U^{de}(y) f^{d'eh} \{\rho^h(y), \rho^{b'}(x')\} \\
&= -\frac{1}{4} i N_c K_{\mathfrak{B}}(x', y, y) U^{ab}(x') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \quad (\text{U1}), (\text{U2}), (\text{F})
\end{aligned} \tag{7.20}$$

$$\begin{aligned}
[B, C] : [\rho^{b'}(x'), \rho^e(x)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathfrak{B}}(x', x', y) U^{c'd'}(y) U^{ab'}(x') U^{bc'}(z) f^{acd} U^{bc}(z) U^{de}(x') f^{b'eh} \{\rho^h(x'), \rho^{d'}(y)\} \\
&= \frac{1}{4} i N_c K_{\mathfrak{B}}(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U1}), (\text{U2}), (\text{F})
\end{aligned} \tag{7.21}$$

Sub-term 3: $-U^{ab'}(x') f^{cbd} U^{ac}(w) \rho^b(y) \rho^{b'}(x') \rho^d(x)$.

$$\begin{aligned} [A, B] : [\rho^b(y), \rho^{b'}(x')] &= i f^{bb'h} \rho^h(y) \delta^{(2)}(y - x'), \quad x' \rightarrow y \\ \frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathfrak{B}}(y, x, y) U^{ab'}(y) f^{cbd} U^{ac}(w) f^{bb'h} \{\rho^h(y), \rho^d(x)\} \\ &= -\frac{1}{4} i K_{\mathfrak{B}}(y, x, y) f^{abc} f^{bde} U^{hd}(y) U^{ha}(w) \{\rho^e(y), \rho^c(x)\} \end{aligned} \quad (7.22)$$

$$\begin{aligned} [A, C] : [\rho^b(y), \rho^d(x)] &= i f^{bdh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathfrak{B}}(x', y, y) U^{ab'}(x') f^{cbd} U^{ac}(w) f^{bdh} \{\rho^h(y), \rho^{b'}(x')\} \\ &= -\frac{1}{4} i N_c K_{\mathfrak{B}}(x', y, y) U^{ab}(x') U^{ac}(w) \{\rho^c(y), \rho^b(x')\} \quad (\text{F}) \end{aligned} \quad (7.23)$$

$$\begin{aligned} [B, C] : [\rho^{b'}(x'), \rho^d(x)] &= i f^{b'dh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathfrak{B}}(x', x', y) U^{ab'}(x') f^{cbd} U^{ac}(w) f^{b'dh} \{\rho^h(x'), \rho^b(y)\} \\ &= -\frac{1}{4} i K_{\mathfrak{B}}(x', x', y) f^{abc} f^{dce} U^{hd}(x') U^{ha}(w) \{\rho^e(x'), \rho^b(y)\} \end{aligned} \quad (7.24)$$

Sub-term 4: $+U^{ab'}(x') f^{acd} U^{bc}(z) U^{de}(x) \rho^b(y) \rho^{b'}(x') \rho^e(x)$.

$$\begin{aligned} [A, B] : [\rho^b(y), \rho^{b'}(x')] &= i f^{bb'h} \rho^h(y) \delta^{(2)}(y - x'), \quad x' \rightarrow y \\ \frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\mathfrak{B}}(y, x, y) U^{ab'}(y) f^{acd} U^{bc}(z) U^{de}(x) f^{bb'h} \{\rho^h(y), \rho^e(x)\} \\ &= \frac{1}{4} i K_{\mathfrak{B}}(y, x, y) f^{abc} f^{deh} U^{ae}(y) U^{db}(z) U^{cl}(x) \{\rho^h(y), \rho^l(x)\} \end{aligned} \quad (7.25)$$

$$\begin{aligned} [A, C] : [\rho^b(y), \rho^e(x)] &= i f^{beh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathfrak{B}}(x', y, y) U^{ab'}(x') f^{acd} U^{bc}(z) U^{de}(y) f^{beh} \{\rho^h(y), \rho^{b'}(x')\} \\ &= \frac{1}{4} i K_{\mathfrak{B}}(x', y, y) f^{abc} f^{deh} U^{al}(x') U^{db}(z) U^{ce}(y) \{\rho^h(y), \rho^l(x')\} \end{aligned} \quad (7.26)$$

$$\begin{aligned} [B, C] : [\rho^{b'}(x'), \rho^e(x)] &= i f^{b'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathfrak{B}}(x', x', y) U^{ab'}(x') f^{acd} U^{bc}(z) U^{de}(x') f^{b'eh} \{\rho^h(x'), \rho^b(y)\} \\ &= -\frac{1}{4} i N_c K_{\mathfrak{B}}(x', x', y) U^{ab}(z) U^{bc}(x') \{\rho^c(x'), \rho^a(y)\} \quad (\text{U2}), (\text{F}) \end{aligned} \quad (7.27)$$

Two- ρ terms of this source. Multiplying by the prefactor $-2i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} \left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{\text{GIII.II}} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \begin{aligned} &+ \frac{1}{2} K_{\mathfrak{B}}(y, x, y) f^{abc} f^{deh} U^{bd}(z) U^{ea}(w) U^{hl}(y) \{\rho^l(y), \rho^c(x)\} \\ &+ \frac{1}{2} K_{\mathfrak{B}}(x', y, y) f^{abc} f^{dce} U^{hd}(y) U^{lm}(x') U^{bh}(z) U^{la}(w) \{\rho^e(y), \rho^m(x')\} \\ &+ \frac{1}{2} K_{\mathfrak{B}}(x', x', y) f^{abc} f^{dce} U^{hl}(y) U^{md}(x') U^{bh}(z) U^{ma}(w) \{\rho^e(x'), \rho^l(y)\} \\ &+ \frac{1}{2} N_c K_{\mathfrak{B}}(y, x, y) U^{ab}(x) U^{ac}(y) \{\rho^c(y), \rho^b(x)\} \\ &- \frac{1}{2} N_c K_{\mathfrak{B}}(x', y, y) U^{ab}(x') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \\ &+ \frac{1}{2} N_c K_{\mathfrak{B}}(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\ &- \frac{1}{2} K_{\mathfrak{B}}(y, x, y) f^{abc} f^{bde} U^{hd}(y) U^{ha}(w) \{\rho^e(y), \rho^c(x)\} \\ &- \frac{1}{2} N_c K_{\mathfrak{B}}(x', y, y) U^{ab}(x') U^{ac}(w) \{\rho^c(y), \rho^b(x')\} \\ &- \frac{1}{2} K_{\mathfrak{B}}(x', x', y) f^{abc} f^{dce} U^{hd}(x') U^{ha}(w) \{\rho^e(x'), \rho^b(y)\} \\ &+ \frac{1}{2} K_{\mathfrak{B}}(y, x, y) f^{abc} f^{deh} U^{ae}(y) U^{db}(z) U^{cl}(x) \{\rho^h(y), \rho^l(x)\} \\ &+ \frac{1}{2} K_{\mathfrak{B}}(x', y, y) f^{abc} f^{deh} U^{al}(x') U^{db}(z) U^{ce}(y) \{\rho^h(y), \rho^l(x')\} \\ &- \frac{1}{2} N_c K_{\mathfrak{B}}(x', x', y) U^{ab}(z) U^{bc}(x') \{\rho^c(x'), \rho^a(y)\} \end{aligned} \right\} + (w' \leftrightarrow w) \end{aligned} \quad (7.28)$$

The kernel at the coincident points:

$$\begin{aligned} K_{\mathfrak{B}}(y, x, y) &= \mathcal{K}(x - w) \cdot \mathcal{K}(y - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right] \\ K_{\mathfrak{B}}(x', y, y) &= \mathcal{K}(y - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \frac{1}{(y - z)^2} \right] \\ K_{\mathfrak{B}}(x', x', y) &= \mathcal{K}(x' - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \right] \end{aligned}$$

7.3 Group III, Row III, first part

Input. Prefactor $-\frac{1}{(2\pi)^3} \frac{ig^4 f^{c'd'a}}{16\pi^5} \frac{1}{k^+} \log \frac{\vee}{\Lambda} = -i\mathcal{N} f^{c'd'a}$. After the w' integration the measured gluon of the conjugate amplitude sits at y' : the phase is $e^{-ik \cdot (y'-w)}$, and the kernel is $K_{\mathfrak{C}} \equiv \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w)$.

The three- ρ term.

$$\begin{aligned} \left. \frac{d^3 N}{d^3 k} \right|_{\text{GIII.III}} = & -i\mathcal{N} \int e^{-ik \cdot (y'-w)} K_{\mathfrak{C}}(x', x, y) \left\{ \right. \\ & + f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{ac}(w) \rho^{e'}(x') \{ \rho^c(y), \rho^b(x) \} \\ & - f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \{ \rho^d(y), \rho^e(x) \} \\ & - f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ac}(w) \rho^{e'}(x') \{ \rho^c(y), \rho^b(x) \} \\ & \left. + f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \{ \rho^d(y), \rho^e(x) \} \right\} + \text{c.c.} \end{aligned} \quad (7.29)$$

with $K_{\mathfrak{C}}(x', x, y) = \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.6) with A, B, C the charges at x', y, x . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{ac}(w) \rho^{e'}(x') \{ \rho^c(y), \rho^b(x) \}$.

$$\begin{aligned} [A, B] : [\rho^{e'}(x'), \rho^c(y)] &= i f^{e'ch} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{ [A, B], C \} &= \frac{1}{2} i K_{\mathfrak{C}}(x', x, x') f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{ac}(w) f^{e'ch} \{ \rho^h(x'), \rho^b(x) \} \\ &= \frac{1}{2} i K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{da}(y') U^{bl}(z) U^{ce}(w) \{ \rho^h(x'), \rho^l(x) \} \end{aligned} \quad (7.30)$$

$$\begin{aligned} [A, C] : [\rho^{e'}(x'), \rho^b(x)] &= i f^{e'bh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{2} \{ [A, C], B \} &= \frac{1}{2} i K_{\mathfrak{C}}(x', x', y) f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{ac}(w) f^{e'bh} \{ \rho^h(x'), \rho^c(y) \} \\ &= \frac{1}{2} i K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{da}(y') U^{be}(z) U^{cl}(w) \{ \rho^h(x'), \rho^l(y) \} \end{aligned} \quad (7.31)$$

Sub-term 2: $- f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \{ \rho^d(y), \rho^e(x) \}$.

$$\begin{aligned} [A, B] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{ [A, B], C \} &= -\frac{1}{2} i K_{\mathfrak{C}}(x', x, x') f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{bc}(z) U^{ad}(x') U^{ce}(x) f^{e'dh} \{ \rho^h(x'), \rho^e(x) \} \\ &= -\frac{1}{2} i K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{da}(y') U^{ce}(x') U^{bl}(x) \{ \rho^h(x'), \rho^l(x) \} \quad (\text{U1}^*) \end{aligned} \quad (7.32)$$

$$\begin{aligned} [A, C] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{2} \{ [A, C], B \} &= -\frac{1}{2} i K_{\mathfrak{C}}(x', x', y) f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{bc}(z) U^{ad}(y) U^{ce}(x') f^{e'eh} \{ \rho^h(x'), \rho^d(y) \} \\ &= -\frac{1}{2} i K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{da}(y') U^{cl}(y) U^{be}(x') \{ \rho^h(x'), \rho^l(y) \} \quad (\text{U1}^*) \end{aligned} \quad (7.33)$$

Sub-term 3: $- f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ac}(w) \rho^{e'}(x') \{ \rho^c(y), \rho^b(x) \}$.

$$\begin{aligned} [A, B] : [\rho^{e'}(x'), \rho^c(y)] &= i f^{e'ch} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{ [A, B], C \} &= -\frac{1}{2} i K_{\mathfrak{C}}(x', x, x') f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ac}(w) f^{e'ch} \{ \rho^h(x'), \rho^b(x) \} \\ &= -\frac{1}{2} i K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{lb}(z) U^{ad}(x') U^{ce}(w) \{ \rho^h(x'), \rho^l(x) \} \end{aligned} \quad (7.34)$$

$$\begin{aligned} [A, C] : [\rho^{e'}(x'), \rho^b(x)] &= i f^{e'bh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{2} \{ [A, C], B \} &= -\frac{1}{2} i K_{\mathfrak{C}}(x', x', y) f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ac}(w) f^{e'bh} \{ \rho^h(x'), \rho^c(y) \} \\ &= -\frac{1}{2} i K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{eb}(z) U^{ad}(x') U^{cl}(w) \{ \rho^h(x'), \rho^l(y) \} \end{aligned} \quad (7.35)$$

Sub-term 4: $+ f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \{ \rho^d(y), \rho^e(x) \}$.

$$\begin{aligned} [A, B] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{2} \{ [A, B], C \} &= \frac{1}{2} i K_{\mathfrak{C}}(x', x, x') f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(x') U^{ce}(x) f^{e'dh} \{ \rho^h(x'), \rho^e(x) \} \\ &= -\frac{1}{2} i N_c K_{\mathfrak{C}}(x', x, x') U^{ab}(x) U^{ac}(x') \{ \rho^c(x'), \rho^b(x) \} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (7.36)$$

$$[A, C] : [\rho^{e'}(x'), \rho^e(x)] = i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x'$$

$$\begin{aligned}\frac{1}{2}\{[A, C], B\} &= \frac{1}{2}i K_{\mathfrak{C}}(x', x', y) f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(y) U^{ce}(x') f^{e'eh} \{\rho^h(x'), \rho^d(y)\} \\ &= \frac{1}{2}i N_c K_{\mathfrak{C}}(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U1}), (\text{U2}), (\text{F})\end{aligned}\quad (7.37)$$

Two- ρ terms of this source. Multiplying by the prefactor $-i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}\frac{d^3 N^{(2\rho)}}{d^3 k} \Big|_{\text{GIII.III}} &= \mathcal{N} \int e^{-ik \cdot (y' - w)} \left\{ \right. \\ &\quad + \frac{1}{2} K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{da}(y') U^{bl}(z) U^{ce}(w) \{\rho^h(x'), \rho^l(x)\} \\ &\quad + \frac{1}{2} K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{da}(y') U^{be}(z) U^{cl}(w) \{\rho^h(x'), \rho^l(y)\} \\ &\quad - \frac{1}{2} K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{da}(y') U^{ce}(x') U^{bl}(x) \{\rho^h(x'), \rho^l(x)\} \\ &\quad - \frac{1}{2} K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{da}(y') U^{cl}(y) U^{be}(x') \{\rho^h(x'), \rho^l(y)\} \\ &\quad - \frac{1}{2} K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{lb}(z) U^{ad}(x') U^{ce}(w) \{\rho^h(x'), \rho^l(x)\} \\ &\quad - \frac{1}{2} K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{eb}(z) U^{ad}(x') U^{cl}(w) \{\rho^h(x'), \rho^l(y)\} \\ &\quad - \frac{1}{2} N_c K_{\mathfrak{C}}(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \\ &\quad \left. + \frac{1}{2} N_c K_{\mathfrak{C}}(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \right\} + (y' \leftrightarrow w)\end{aligned}\quad (7.38)$$

The kernel at the coincident points:

$$\begin{aligned}K_{\mathfrak{C}}(x', x, x') &= \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x' - w) \\ K_{\mathfrak{C}}(x', x', y) &= \mathcal{K}(y' - z) \cdot \mathcal{K}(x' - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w)\end{aligned}$$

7.4 Group III, Row V

Input. Prefactor $-\frac{1}{(2\pi)^3} \frac{ig^4 f^{c'ca}}{8\pi^5} \frac{1}{k^+} \log \frac{\vee}{\Lambda} = -2i\mathcal{N} f^{c'ca}$, phase $e^{-ik \cdot (y' - w)}$, kernel $K_{\mathfrak{C}}$. Order of the charges: $\rho(x')\rho(x)\rho(y)$.

The three- ρ term.

$$\begin{aligned}\frac{d^3 N}{d^3 k} \Big|_{\text{GIII.V}} &= -2i\mathcal{N} \int e^{-ik \cdot (y' - w)} K_{\mathfrak{C}}(x', x, y) \left\{ \right. \\ &\quad + f^{c'ca} U^{e'e'}(y') U^{ce}(x) U^{ad}(y) \rho^{e'}(x') \rho^e(x) \rho^d(y) \\ &\quad - f^{c'ca} U^{e'e'}(y') U^{ad}(w) U^{ce}(x) \rho^{e'}(x') \rho^e(x) \rho^d(y) \\ &\quad - f^{c'ca} U^{e'e'}(x') U^{ce}(x) U^{ad}(y) \rho^{e'}(x') \rho^e(x) \rho^d(y) \\ &\quad \left. + f^{c'ca} U^{e'e'}(x') U^{ad}(w) U^{ce}(x) \rho^{e'}(x') \rho^e(x) \rho^d(y) \right\} + \text{c.c.}\end{aligned}\quad (7.39)$$

with $K_{\mathfrak{C}}(x', x, y) = \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.4) with A, B, C the charges at x', x, y . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{c'ca} U^{e'e'}(y') U^{ce}(x) U^{ad}(y) \rho^{e'}(x') \rho^e(x) \rho^d(y)$.

$$\begin{aligned}[A, B] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4}\{[A, B], C\} &= \frac{1}{4}i K_{\mathfrak{C}}(x', x', y) f^{c'ca} U^{e'e'}(y') U^{ce}(x') U^{ad}(y) f^{e'eh} \{\rho^h(x'), \rho^d(y)\} \\ &= \frac{1}{4}i K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{da}(y') U^{be}(x') U^{cl}(y) \{\rho^h(x'), \rho^l(y)\}\end{aligned}\quad (7.40)$$

$$\begin{aligned}[A, C] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4}\{[A, C], B\} &= \frac{1}{4}i K_{\mathfrak{C}}(x', x, x') f^{c'ca} U^{e'e'}(y') U^{ce}(x) U^{ad}(x') f^{e'dh} \{\rho^h(x'), \rho^e(x)\} \\ &= \frac{1}{4}i K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{da}(y') U^{bl}(x) U^{ce}(x') \{\rho^h(x'), \rho^l(x)\}\end{aligned}\quad (7.41)$$

$$\begin{aligned}[B, C] : [\rho^e(x), \rho^d(y)] &= i f^{edh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\ \frac{1}{4}\{[B, C], A\} &= \frac{1}{4}i K_{\mathfrak{C}}(x', x, x) f^{c'ca} U^{e'e'}(y') U^{ce}(x) U^{ad}(x) f^{edh} \{\rho^h(x), \rho^{e'}(x')\} \\ &= \frac{1}{4}i N_c K_{\mathfrak{C}}(x', x, x) U^{ab}(y') U^{bc}(x) \{\rho^c(x), \rho^a(x')\} \quad (\text{U2}), (\text{F})\end{aligned}\quad (7.42)$$

Sub-term 2: $-f^{c'ca} U^{e'e'}(y') U^{ad}(w) U^{ce}(x) \rho^{e'}(x') \rho^e(x) \rho^d(y).$

$$\begin{aligned}
[A, B] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathfrak{C}}(x', x', y) f^{c'ca} U^{e'e'}(y') U^{ad}(w) U^{ce}(x') f^{e'eh} \{\rho^h(x'), \rho^d(y)\} \\
&= -\frac{1}{4} i K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{da}(y') U^{cl}(w) U^{be}(x') \{\rho^h(x'), \rho^l(y)\}
\end{aligned} \tag{7.43}$$

$$\begin{aligned}
[A, C] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathfrak{C}}(x', x, x') f^{c'ca} U^{e'e'}(y') U^{ad}(w) U^{ce}(x) f^{e'dh} \{\rho^h(x'), \rho^e(x)\} \\
&= -\frac{1}{4} i K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{da}(y') U^{ce}(w) U^{bl}(x) \{\rho^h(x'), \rho^l(x)\}
\end{aligned} \tag{7.44}$$

$$\begin{aligned}
[B, C] : [\rho^e(x), \rho^d(y)] &= i f^{edh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
\frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathfrak{C}}(x', x, x) f^{c'ca} U^{e'e'}(y') U^{ad}(w) U^{ce}(x) f^{edh} \{\rho^h(x), \rho^{e'}(x')\} \\
&= -\frac{1}{4} i K_{\mathfrak{C}}(x', x, x) f^{abc} f^{deh} U^{la}(y') U^{ce}(w) U^{bd}(x) \{\rho^h(x), \rho^l(x')\}
\end{aligned} \tag{7.45}$$

Sub-term 3: $-f^{c'ca} U^{c'e'}(x') U^{ce}(x) U^{ad}(y) \rho^{e'}(x') \rho^e(x) \rho^d(y).$

$$\begin{aligned}
[A, B] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathfrak{C}}(x', x', y) f^{c'ca} U^{c'e'}(x') U^{ce}(x') U^{ad}(y) f^{e'eh} \{\rho^h(x'), \rho^d(y)\} \\
&= -\frac{1}{4} i N_c K_{\mathfrak{C}}(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{7.46}$$

$$\begin{aligned}
[A, C] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathfrak{C}}(x', x, x') f^{c'ca} U^{c'e'}(x') U^{ce}(x) U^{ad}(x') f^{e'dh} \{\rho^h(x'), \rho^e(x)\} \\
&= -\frac{1}{4} i N_c K_{\mathfrak{C}}(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{7.47}$$

$$\begin{aligned}
[B, C] : [\rho^e(x), \rho^d(y)] &= i f^{edh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
\frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathfrak{C}}(x', x, x) f^{c'ca} U^{c'e'}(x') U^{ce}(x) U^{ad}(x) f^{edh} \{\rho^h(x), \rho^{e'}(x')\} \\
&= -\frac{1}{4} i N_c K_{\mathfrak{C}}(x', x, x) U^{ab}(x') U^{ac}(x) \{\rho^c(x), \rho^b(x')\} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{7.48}$$

Sub-term 4: $+f^{c'ca} U^{c'e'}(x') U^{ad}(w) U^{ce}(x) \rho^{e'}(x') \rho^e(x) \rho^d(y).$

$$\begin{aligned}
[A, B] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\mathfrak{C}}(x', x', y) f^{c'ca} U^{c'e'}(x') U^{ad}(w) U^{ce}(x') f^{e'eh} \{\rho^h(x'), \rho^d(y)\} \\
&= \frac{1}{4} i N_c K_{\mathfrak{C}}(x', x', y) U^{ab}(w) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U2}), \quad (\text{F})
\end{aligned} \tag{7.49}$$

$$\begin{aligned}
[A, C] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathfrak{C}}(x', x, x') f^{c'ca} U^{c'e'}(x') U^{ad}(w) U^{ce}(x) f^{e'dh} \{\rho^h(x'), \rho^e(x)\} \\
&= \frac{1}{4} i K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{ad}(x') U^{ce}(w) U^{bl}(x) \{\rho^h(x'), \rho^l(x)\}
\end{aligned} \tag{7.50}$$

$$\begin{aligned}
[B, C] : [\rho^e(x), \rho^d(y)] &= i f^{edh} \rho^h(x) \delta^{(2)}(x - y), \quad y \rightarrow x \\
\frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathfrak{C}}(x', x, x) f^{c'ca} U^{c'e'}(x') U^{ad}(w) U^{ce}(x) f^{edh} \{\rho^h(x), \rho^{e'}(x')\} \\
&= \frac{1}{4} i K_{\mathfrak{C}}(x', x, x) f^{abc} f^{deh} U^{al}(x') U^{ce}(w) U^{bd}(x) \{\rho^h(x), \rho^l(x')\}
\end{aligned} \tag{7.51}$$

Two- ρ terms of this source. Multiplying by the prefactor $-2i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}
\left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{\text{GHI.V}} &= \mathcal{N} \int e^{-ik \cdot (y' - w)} \{ \\
&\quad + \frac{1}{2} K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{da}(y') U^{be}(x') U^{cl}(y) \{\rho^h(x'), \rho^l(y)\} \\
&\quad + \frac{1}{2} K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{da}(y') U^{bl}(x) U^{ce}(x') \{\rho^h(x'), \rho^l(x)\} \\
&\quad + \frac{1}{2} N_c K_{\mathfrak{C}}(x', x, x) U^{ab}(y') U^{bc}(x) \{\rho^c(x), \rho^a(x')\} \\
&\quad - \frac{1}{2} K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{da}(y') U^{cl}(w) U^{be}(x') \{\rho^h(x'), \rho^l(y)\} \\
&\quad - \frac{1}{2} K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{da}(y') U^{ce}(w) U^{bl}(x) \{\rho^h(x'), \rho^l(x)\} \\
&\quad - \frac{1}{2} K_{\mathfrak{C}}(x', x, x) f^{abc} f^{deh} U^{la}(y') U^{ce}(w) U^{bd}(x) \{\rho^h(x), \rho^l(x')\} \\
&\quad - \frac{1}{2} N_c K_{\mathfrak{C}}(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\
&\quad + \frac{1}{2} N_c K_{\mathfrak{C}}(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \\
&\quad - \frac{1}{2} N_c K_{\mathfrak{C}}(x', x, x) U^{ab}(x') U^{ac}(x) \{\rho^c(x), \rho^b(x')\}
\end{aligned}$$

$$\begin{aligned}
& + \frac{1}{2} N_c K_{\mathfrak{C}}(x', x', y) U^{ab}(w) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \\
& + \frac{1}{2} K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{ad}(x') U^{ce}(w) U^{bl}(x) \{ \rho^h(x'), \rho^l(x) \} \\
& + \frac{1}{2} K_{\mathfrak{C}}(x', x, x) f^{abc} f^{deh} U^{al}(x') U^{ce}(w) U^{bd}(x) \{ \rho^h(x), \rho^l(x') \} \} + (y' \leftrightarrow w)
\end{aligned} \tag{7.52}$$

The kernel at the coincident points:

$$\begin{aligned}
K_{\mathfrak{C}}(x', x', y) &= \mathcal{K}(y' - z) \cdot \mathcal{K}(x' - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \\
K_{\mathfrak{C}}(x', x, x') &= \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x' - w) \\
K_{\mathfrak{C}}(x', x, x) &= \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x - w)
\end{aligned}$$

7.5 Group III, Row VI

Input. Prefactor $-2i\mathcal{N} f^{c'd'a}$, phase $e^{-ik \cdot (y' - w)}$, kernel $K_{\mathfrak{C}}$. Order of the charges: $\rho(x')\rho(y)\rho(x)$.

The three- ρ term.

$$\begin{aligned}
\left. \frac{d^3 N}{d^3 k} \right|_{\text{GIII.VI}} &= -2i\mathcal{N} \int e^{-ik \cdot (y' - w)} K_{\mathfrak{C}}(x', x, y) \{ \\
& + f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \rho^d(y) \rho^e(x) \\
& - f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{ad}(y) \rho^{e'}(x') \rho^d(y) \rho^b(x) \\
& - f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \rho^d(y) \rho^e(x) \\
& + f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ad}(y) \rho^{e'}(x') \rho^d(y) \rho^b(x) \} + \text{c.c.}
\end{aligned} \tag{7.53}$$

with $K_{\mathfrak{C}}(x', x, y) = \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (3.4) with A, B, C the charges at x', y, x . The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two- ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \rho^d(y) \rho^e(x)$.

$$\begin{aligned}
[A, B] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{ [A, B], C \} &= \frac{1}{4} i K_{\mathfrak{C}}(x', x, x') f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{bc}(z) U^{ad}(x') U^{ce}(x) f^{e'dh} \{ \rho^h(x'), \rho^e(x) \} \\
&= \frac{1}{4} i K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{da}(y') U^{ce}(x') U^{bl}(x) \{ \rho^h(x'), \rho^l(x) \} \quad (\text{U1}^*)
\end{aligned} \tag{7.54}$$

$$\begin{aligned}
[A, C] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{ [A, C], B \} &= \frac{1}{4} i K_{\mathfrak{C}}(x', x', y) f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{bc}(z) U^{ad}(y) U^{ce}(x') f^{e'eh} \{ \rho^h(x'), \rho^d(y) \} \\
&= \frac{1}{4} i K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{da}(y') U^{cl}(y) U^{be}(x') \{ \rho^h(x'), \rho^l(y) \} \quad (\text{U1}^*)
\end{aligned} \tag{7.55}$$

$$\begin{aligned}
[B, C] : [\rho^d(y), \rho^e(x)] &= i f^{deh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\
\frac{1}{4} \{ [B, C], A \} &= \frac{1}{4} i K_{\mathfrak{C}}(x', y, y) f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{bc}(z) U^{ad}(y) U^{ce}(y) f^{deh} \{ \rho^h(y), \rho^{e'}(x') \} \\
&= -\frac{1}{4} i N_c K_{\mathfrak{C}}(x', y, y) U^{ab}(y') U^{bc}(y) \{ \rho^c(y), \rho^a(x') \} \quad (\text{U1}^*), (\text{U2}), (\text{F})
\end{aligned} \tag{7.56}$$

Sub-term 2: $- f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{ad}(y) \rho^{e'}(x') \rho^d(y) \rho^b(x)$.

$$\begin{aligned}
[A, B] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\
\frac{1}{4} \{ [A, B], C \} &= -\frac{1}{4} i K_{\mathfrak{C}}(x', x, x') f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{ad}(x') f^{e'dh} \{ \rho^h(x'), \rho^b(x) \} \\
&= -\frac{1}{4} i K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{da}(y') U^{bl}(z) U^{ce}(x') \{ \rho^h(x'), \rho^l(x) \}
\end{aligned} \tag{7.57}$$

$$\begin{aligned}
[A, C] : [\rho^{e'}(x'), \rho^b(x)] &= i f^{e'bh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\
\frac{1}{4} \{ [A, C], B \} &= -\frac{1}{4} i K_{\mathfrak{C}}(x', x', y) f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{ad}(y) f^{e'bh} \{ \rho^h(x'), \rho^d(y) \} \\
&= -\frac{1}{4} i K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{da}(y') U^{be}(z) U^{cl}(y) \{ \rho^h(x'), \rho^l(y) \}
\end{aligned} \tag{7.58}$$

$$\begin{aligned}
[B, C] : [\rho^d(y), \rho^b(x)] &= i f^{dbh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\
\frac{1}{4} \{ [B, C], A \} &= -\frac{1}{4} i K_{\mathfrak{C}}(x', y, y) f^{c'd'a} U^{e'c'}(y') U^{d'b}(z) U^{ad}(y) f^{dbh} \{ \rho^h(y), \rho^{e'}(x') \} \\
&= -\frac{1}{4} i K_{\mathfrak{C}}(x', y, y) f^{abc} f^{deh} U^{la}(y') U^{be}(z) U^{cd}(y) \{ \rho^h(y), \rho^l(x') \}
\end{aligned} \tag{7.59}$$

Sub-term 3: $-f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(y) U^{ce}(x) \rho^{e'}(x') \rho^d(y) \rho^e(x)$.

$$\begin{aligned} [A, B] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= -\frac{1}{4} i K_{\mathfrak{C}}(x', x, x') f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(x') U^{ce}(x) f^{e'dh} \{\rho^h(x'), \rho^e(x)\} \\ &= \frac{1}{4} i N_c K_{\mathfrak{C}}(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (7.60)$$

$$\begin{aligned} [A, C] : [\rho^{e'}(x'), \rho^e(x)] &= i f^{e'eh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= -\frac{1}{4} i K_{\mathfrak{C}}(x', x', y) f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(y) U^{ce}(x') f^{e'eh} \{\rho^h(x'), \rho^d(y)\} \\ &= -\frac{1}{4} i N_c K_{\mathfrak{C}}(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (7.61)$$

$$\begin{aligned} [B, C] : [\rho^d(y), \rho^e(x)] &= i f^{deh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[B, C], A\} &= -\frac{1}{4} i K_{\mathfrak{C}}(x', y, y) f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{bc}(z) U^{ad}(y) U^{ce}(y) f^{deh} \{\rho^h(y), \rho^{e'}(x')\} \\ &= \frac{1}{4} i N_c K_{\mathfrak{C}}(x', y, y) U^{ab}(x') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \quad (\text{U1}), (\text{U2}), (\text{F}) \end{aligned} \quad (7.62)$$

Sub-term 4: $+f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ad}(y) \rho^{e'}(x') \rho^d(y) \rho^b(x)$.

$$\begin{aligned} [A, B] : [\rho^{e'}(x'), \rho^d(y)] &= i f^{e'dh} \rho^h(x') \delta^{(2)}(x' - y), \quad y \rightarrow x' \\ \frac{1}{4} \{[A, B], C\} &= \frac{1}{4} i K_{\mathfrak{C}}(x', x, x') f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ad}(x') f^{e'dh} \{\rho^h(x'), \rho^b(x)\} \\ &= -\frac{1}{4} i N_c K_{\mathfrak{C}}(x', x, x') U^{ab}(z) U^{bc}(x') \{\rho^c(x'), \rho^a(x)\} \quad (\text{U2}), (\text{F}) \end{aligned} \quad (7.63)$$

$$\begin{aligned} [A, C] : [\rho^{e'}(x'), \rho^b(x)] &= i f^{e'bh} \rho^h(x') \delta^{(2)}(x' - x), \quad x \rightarrow x' \\ \frac{1}{4} \{[A, C], B\} &= \frac{1}{4} i K_{\mathfrak{C}}(x', x', y) f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ad}(y) f^{e'bh} \{\rho^h(x'), \rho^d(y)\} \\ &= \frac{1}{4} i K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{eb}(z) U^{ad}(x') U^{cl}(y) \{\rho^h(x'), \rho^l(y)\} \end{aligned} \quad (7.64)$$

$$\begin{aligned} [B, C] : [\rho^d(y), \rho^b(x)] &= i f^{dbh} \rho^h(y) \delta^{(2)}(y - x), \quad x \rightarrow y \\ \frac{1}{4} \{[B, C], A\} &= \frac{1}{4} i K_{\mathfrak{C}}(x', y, y) f^{c'd'a} U^{bd'}(z) U^{c'e'}(x') U^{ad}(y) f^{dbh} \{\rho^h(y), \rho^{e'}(x')\} \\ &= \frac{1}{4} i K_{\mathfrak{C}}(x', y, y) f^{abc} f^{deh} U^{eb}(z) U^{al}(x') U^{cd}(y) \{\rho^h(y), \rho^l(x')\} \end{aligned} \quad (7.65)$$

Two- ρ terms of this source. Multiplying by the prefactor $-2i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} \frac{d^3 N^{(2\rho)}}{d^3 k} \Big|_{\text{GIII.VI}} &= \mathcal{N} \int e^{-ik \cdot (y' - w)} \left\{ \begin{aligned} &+ \frac{1}{2} K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{da}(y') U^{ce}(x') U^{bl}(x) \{\rho^h(x'), \rho^l(x)\} \\ &+ \frac{1}{2} K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{da}(y') U^{cl}(y) U^{be}(x') \{\rho^h(x'), \rho^l(y)\} \\ &- \frac{1}{2} N_c K_{\mathfrak{C}}(x', y, y) U^{ab}(y') U^{bc}(y) \{\rho^c(y), \rho^a(x')\} \\ &- \frac{1}{2} K_{\mathfrak{C}}(x', x, x') f^{abc} f^{deh} U^{da}(y') U^{bl}(z) U^{ce}(x') \{\rho^h(x'), \rho^l(x)\} \\ &- \frac{1}{2} K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{da}(y') U^{be}(z) U^{cl}(y) \{\rho^h(x'), \rho^l(y)\} \\ &- \frac{1}{2} K_{\mathfrak{C}}(x', y, y) f^{abc} f^{deh} U^{la}(y') U^{be}(z) U^{cd}(y) \{\rho^h(y), \rho^l(x')\} \\ &+ \frac{1}{2} N_c K_{\mathfrak{C}}(x', x, x') U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \\ &- \frac{1}{2} N_c K_{\mathfrak{C}}(x', x', y) U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\ &+ \frac{1}{2} N_c K_{\mathfrak{C}}(x', y, y) U^{ab}(x') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \\ &- \frac{1}{2} N_c K_{\mathfrak{C}}(x', x, x') U^{ab}(z) U^{bc}(x') \{\rho^c(x'), \rho^a(x)\} \\ &+ \frac{1}{2} K_{\mathfrak{C}}(x', x', y) f^{abc} f^{deh} U^{eb}(z) U^{ad}(x') U^{cl}(y) \{\rho^h(x'), \rho^l(y)\} \\ &+ \frac{1}{2} K_{\mathfrak{C}}(x', y, y) f^{abc} f^{deh} U^{eb}(z) U^{al}(x') U^{cd}(y) \{\rho^h(y), \rho^l(x')\} \end{aligned} \right\} + (y' \leftrightarrow w) \end{aligned} \quad (7.66)$$

The kernel at the coincident points:

$$\begin{aligned} K_{\mathfrak{C}}(x', x, x') &= \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x' - w) \\ K_{\mathfrak{C}}(x', x', y) &= \mathcal{K}(y' - z) \cdot \mathcal{K}(x' - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \\ K_{\mathfrak{C}}(x', y, y) &= \mathcal{K}(y' - z) \cdot \mathcal{K}(y - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \end{aligned}$$

8 Summary: all two- ρ terms

Below are all two- ρ terms generated by the reordering, with the kernels written out. Every coefficient is real. For Groups I and III the complex conjugate is included as the exchange of the two measured-gluon positions in the phase, shown

at the end of each line. The four- ρ remainders and Group II have no c.c. (Sec. 2). The symmetric three- ρ remainders are not listed (Sec. 3).

Three- ρ terms from reordering the four- ρ terms

$$\begin{aligned}
\left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{A1} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\
&+ \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \\
&\quad \times f^{abc} f^{dae} U^{dh}(w') U^{hb}(w) \{ \rho^e(x'), \rho^c(y) \} \\
&+ \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \\
&\quad \times f^{abc} f^{dce} U^{dh}(w') U^{hb}(w) \{ \rho^e(x'), \rho^a(x) \} \\
&- \frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \frac{1}{(x - z)^2} \\
&\quad \times U^{ab}(w') U^{bc}(w) \{ \rho^c(x), \rho^a(x') \} \\
&- \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \\
&\quad \times f^{abc} f^{dae} U^{hb}(w) U^{hd}(x') \{ \rho^e(x'), \rho^c(y) \} \\
&- \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \\
&\quad \times f^{abc} f^{dce} U^{hb}(w) U^{hd}(x') \{ \rho^e(x'), \rho^a(x) \} \\
&+ \frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \frac{1}{(x - z)^2} \\
&\quad \times U^{ab}(w) U^{ac}(x') \{ \rho^b(x), \rho^c(x') \} \left. \right\} \tag{8.1}
\end{aligned}$$

$$\begin{aligned}
\left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{A2} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\
&+ \frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \frac{1}{(x' - z)^2} \\
&\quad \times U^{ab}(w') U^{ac}(y) \{ \rho^b(x'), \rho^c(y) \} \\
&+ \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \\
&\quad \times f^{abc} f^{cde} U^{hb}(w') U^{hd}(x') \{ \rho^e(x'), \rho^a(x) \} \\
&+ \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \\
&\quad \times f^{abc} f^{ade} U^{hb}(w') U^{hd}(x) \{ \rho^e(x), \rho^c(x') \} \\
&- \frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \frac{1}{(x' - z)^2} \\
&\quad \times U^{ab}(w') U^{bc}(w) \{ \rho^a(x'), \rho^c(y) \} \\
&- \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \\
&\quad \times f^{abc} f^{cde} U^{bh}(w') U^{hd}(w) \{ \rho^e(x'), \rho^a(x) \} \\
&- \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \\
&\quad \times f^{abc} f^{ade} U^{bh}(w') U^{hd}(w) \{ \rho^e(x), \rho^c(x') \} \left. \right\} \tag{8.2}
\end{aligned}$$

$$\begin{aligned}
\left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{B1} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\
&+ \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \\
&\quad \times f^{abc} f^{deh} U^{db}(w') U^{lc}(y) U^{ae}(x') \{ \rho^h(x'), \rho^l(y) \} \\
&+ \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \\
&\quad \times f^{abc} f^{deh} U^{db}(w') U^{ec}(x') U^{al}(x) \{ \rho^h(x'), \rho^l(x) \} \\
&- \frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \frac{1}{(x - z)^2} \\
&\quad \times U^{ab}(w') U^{bc}(x) \{ \rho^c(x), \rho^a(x') \} \\
&+ \frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(y - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \\
&\quad \times U^{ab}(y) U^{bc}(x') \{ \rho^c(x'), \rho^a(y) \} \left. \right\}
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \\
& \quad \times U^{ab}(x) U^{ac}(x') \{ \rho^c(x'), \rho^b(x) \} \\
& + \frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \frac{1}{(x - z)^2} \\
& \quad \times U^{ab}(x') U^{ac}(x) \{ \rho^c(x), \rho^b(x') \} \}
\end{aligned} \tag{8.3}$$

$$\begin{aligned}
\left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{\text{B2}} = \mathcal{N} \int e^{-ik \cdot (w' - w)} \{ & \\
& -\frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(y - z) \cdot \mathcal{K}(x' - z) \\
& \quad \times f^{abc} f^{deh} U^{dc}(x') U^{eb}(w) U^{al}(y) \{ \rho^h(x'), \rho^l(y) \} \\
& -\frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \frac{1}{(x' - z)^2} \\
& \quad \times U^{ab}(w) U^{bc}(x') \{ \rho^c(x'), \rho^a(x) \} \\
& -\frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \\
& \quad \times f^{abc} f^{deh} U^{lc}(x') U^{db}(w) U^{ae}(x) \{ \rho^h(x), \rho^l(x') \} \\
& -\frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(y - z) \cdot \mathcal{K}(x' - z) \\
& \quad \times U^{ab}(y) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \\
& + \frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \frac{1}{(x' - z)^2} \\
& \quad \times U^{ab}(x) U^{ac}(x') \{ \rho^c(x'), \rho^b(x) \} \\
& -\frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \\
& \quad \times U^{ab}(x') U^{bc}(x) \{ \rho^c(x), \rho^a(x') \} \}
\end{aligned} \tag{8.4}$$

Group I (with c.c.)

$$\begin{aligned}
\left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{\text{GI.II}} = \mathcal{N} \int e^{-ik \cdot (w' - w)} \{ & \\
& + \frac{1}{4} \mathcal{K}^i(x' - w') \mathfrak{F}^i(w; x', y) \\
& \quad \times f^{abc} f^{dbe} U^{hd}(x') U^{ha}(w) \{ \rho^e(x'), \rho^c(y) \} \\
& + \frac{1}{4} \mathcal{K}^i(x' - w') \mathfrak{F}^i(w; x, x') \\
& \quad \times f^{abc} f^{dce} U^{hd}(x') U^{ha}(w) \{ \rho^e(x'), \rho^b(x) \} \\
& - \frac{1}{4} N_c \mathcal{K}^i(x' - w') \mathfrak{F}^i(w; x', y) \\
& \quad \times U^{ab}(y) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \\
& + \frac{1}{4} N_c \mathcal{K}^i(x' - w') \mathfrak{F}^i(w; x, x') \\
& \quad \times U^{ab}(x) U^{ac}(x') \{ \rho^c(x'), \rho^b(x) \} \\
& - \frac{1}{4} \mathcal{K}^i(x' - w') \mathfrak{F}^i(w; x', y) \\
& \quad \times f^{abc} f^{dbe} U^{hd}(w') U^{ha}(w) \{ \rho^e(x'), \rho^c(y) \} \\
& - \frac{1}{4} \mathcal{K}^i(x' - w') \mathfrak{F}^i(w; x, x') \\
& \quad \times f^{abc} f^{dce} U^{hd}(w') U^{ha}(w) \{ \rho^e(x'), \rho^b(x) \} \\
& + \frac{1}{4} \mathcal{K}^i(x' - w') \mathfrak{F}^i(w; x', y) \\
& \quad \times f^{abc} f^{deh} U^{ad}(w') U^{be}(x') U^{cl}(y) \{ \rho^h(x'), \rho^l(y) \} \\
& + \frac{1}{4} \mathcal{K}^i(x' - w') \mathfrak{F}^i(w; x, x') \\
& \quad \times f^{abc} f^{deh} U^{ad}(w') U^{bl}(x) U^{ce}(x') \{ \rho^h(x'), \rho^l(x) \} \} + (w' \leftrightarrow w)
\end{aligned} \tag{8.5}$$

$$\begin{aligned}
\left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{\text{GI.III}} = \mathcal{N} \int e^{-ik \cdot (w' - w)} \{ & \\
& + \frac{1}{4} N_c \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \Phi(x') \right] \\
& \quad \times U^{ab}(z) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \}
\end{aligned}$$

$$\begin{aligned}
& -\frac{1}{4} \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - z) \Phi(x) \right] \\
& \quad \times f^{abc} f^{deh} U^{ad}(x') U^{cl}(x) U^{be}(z) \{ \rho^h(x'), \rho^l(x) \} \\
& -\frac{1}{4} \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \frac{1}{(x - z)^2} + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(x - z) \Phi(x) \right] \\
& \quad \times f^{abc} f^{deh} U^{al}(x') U^{cd}(x) U^{be}(z) \{ \rho^h(x), \rho^l(x') \} \\
& -\frac{1}{4} N_c \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \Phi(x') \right] \\
& \quad \times U^{ab}(y) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \\
& +\frac{1}{4} N_c \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - z) \Phi(x) \right] \\
& \quad \times U^{ab}(x) U^{ac}(x') \{ \rho^c(x'), \rho^b(x) \} \\
& -\frac{1}{4} N_c \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \frac{1}{(x - z)^2} + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(x - z) \Phi(x) \right] \\
& \quad \times U^{ab}(x') U^{ac}(x) \{ \rho^c(x), \rho^b(x') \} \\
& +\frac{1}{4} \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \Phi(x') \right] \\
& \quad \times f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(z) \{ \rho^h(x'), \rho^l(y) \} \\
& +\frac{1}{4} \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - z) \Phi(x) \right] \\
& \quad \times f^{abc} f^{deh} U^{ad}(w') U^{cl}(x) U^{be}(z) \{ \rho^h(x'), \rho^l(x) \} \\
& +\frac{1}{4} \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \frac{1}{(x - z)^2} + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(x - z) \Phi(x) \right] \\
& \quad \times f^{abc} f^{deh} U^{al}(w') U^{cd}(x) U^{be}(z) \{ \rho^h(x), \rho^l(x') \} \\
& -\frac{1}{4} \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(x' - w) \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \Phi(x') \right] \\
& \quad \times f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(y) \{ \rho^h(x'), \rho^l(y) \} \\
& -\frac{1}{4} \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - z) \Phi(x) \right] \\
& \quad \times f^{abc} f^{deh} U^{ad}(w') U^{cl}(x) U^{be}(x') \{ \rho^h(x'), \rho^l(x) \} \\
& +\frac{1}{4} N_c \left[\mathcal{K}(x' - w') \cdot \mathcal{K}(x - w) \frac{1}{(x - z)^2} + 2 \mathcal{K}(x' - w') \cdot \mathcal{K}(x - z) \Phi(x) \right] \\
& \quad \times U^{ab}(w') U^{ac}(x) \{ \rho^c(x), \rho^b(x') \} \} + (w' \leftrightarrow w)
\end{aligned} \tag{8.6}$$

$$\begin{aligned}
\frac{d^3 N^{(2\rho)}}{d^3 k} \Big|_{\text{GI.IV}} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\
& -\frac{1}{2} \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(x' - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \right] \\
& \quad \times f^{abc} f^{deh} U^{da}(w) U^{eb}(z) U^{hl}(x') \{ \rho^l(x'), \rho^c(x) \} \\
& -\frac{1}{2} \mathcal{K}(x' - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \right] \\
& \quad \times f^{abc} f^{dce} U^{hd}(x') U^{ha}(w) U^{lb}(z) U^{lm}(y) \{ \rho^e(x'), \rho^m(y) \} \\
& -\frac{1}{2} \mathcal{K}(y - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \frac{1}{(y - z)^2} \right] \\
& \quad \times f^{abc} f^{dce} U^{hl}(x') U^{ha}(w) U^{mb}(z) U^{md}(y) \{ \rho^e(y), \rho^l(x') \} \\
& +\frac{1}{2} N_c \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(x' - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \right] \\
& \quad \times U^{ab}(x) U^{ac}(x') \{ \rho^c(x'), \rho^b(x) \} \\
& -\frac{1}{2} N_c \mathcal{K}(x' - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \right] \\
& \quad \times U^{ab}(y) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \\
& +\frac{1}{2} N_c \mathcal{K}(y - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \frac{1}{(y - z)^2} \right] \\
& \quad \times U^{ab}(x') U^{ac}(y) \{ \rho^c(y), \rho^b(x') \} \\
& +\frac{1}{2} \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(x' - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \right] \\
& \quad \times f^{abc} f^{deh} U^{ld}(w') U^{la}(w) U^{mb}(z) U^{me}(x') \{ \rho^h(x'), \rho^c(x) \}
\end{aligned}$$

$$\begin{aligned}
& + \frac{1}{2} \mathcal{K}(x' - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \right] \\
& \quad \times f^{abc} f^{dce} U^{hd}(w') U^{ha}(w) U^{lb}(z) U^{lm}(y) \{ \rho^e(x'), \rho^m(y) \} \\
& + \frac{1}{2} \mathcal{K}(y - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \frac{1}{(y - z)^2} \right] \\
& \quad \times f^{abc} f^{dce} U^{hl}(w') U^{ha}(w) U^{mb}(z) U^{md}(y) \{ \rho^e(y), \rho^l(x') \} \\
& - \frac{1}{2} \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(x' - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(x' - z) \right] \\
& \quad \times f^{abc} f^{deh} U^{ad}(w') U^{be}(x') U^{cl}(x) \{ \rho^h(x'), \rho^l(x) \} \\
& - \frac{1}{2} \mathcal{K}(x' - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z) \right] \\
& \quad \times f^{abc} f^{deh} U^{ad}(w') U^{bl}(y) U^{ce}(x') \{ \rho^h(x'), \rho^l(y) \} \\
& - \frac{1}{2} N_c \mathcal{K}(y - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \frac{1}{(y - z)^2} \right] \\
& \quad \times U^{ab}(w') U^{ac}(y) \{ \rho^c(y), \rho^b(x') \} \} + (w' \leftrightarrow w)
\end{aligned} \tag{8.7}$$

$$\begin{aligned}
\frac{d^3 N^{(2\rho)}}{d^3 k} \Big|_{\text{GI.V1}} &= \mathcal{N} \int e^{-ik \cdot (w' - z)} \left\{ \right. \\
& + \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - z) \mathcal{K}(z - x) \cdot \mathcal{K}(y' - x) \\
& \quad \times f^{abc} f^{deh} U^{ad}(x') U^{bl}(x) U^{ce}(z) \{ \rho^h(x'), \rho^l(y') \} \\
& + \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \mathcal{K}(z - x) \cdot \mathcal{K}(x' - x) \\
& \quad \times f^{abc} f^{deh} U^{ad}(x') U^{be}(x) U^{cl}(z) \{ \rho^h(x'), \rho^l(y) \} \\
& + \frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - z) \mathcal{K}(z - x) \cdot \mathcal{K}(y' - x) \\
& \quad \times U^{ab}(y') U^{ac}(x') \{ \rho^c(x'), \rho^b(y') \} \\
& - \frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \mathcal{K}(z - x) \cdot \mathcal{K}(x' - x) \\
& \quad \times U^{ab}(y) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \\
& - \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - z) \mathcal{K}(z - x) \cdot \mathcal{K}(y' - x) \\
& \quad \times f^{abc} f^{deh} U^{ad}(w') U^{bl}(x) U^{ce}(z) \{ \rho^h(x'), \rho^l(y') \} \\
& - \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \mathcal{K}(z - x) \cdot \mathcal{K}(x' - x) \\
& \quad \times f^{abc} f^{deh} U^{ad}(w') U^{be}(x) U^{cl}(z) \{ \rho^h(x'), \rho^l(y) \} \\
& + \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - z) \mathcal{K}(z - x) \cdot \mathcal{K}(y' - x) \\
& \quad \times f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(y') \{ \rho^h(x'), \rho^l(y') \} \\
& + \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(y - z) \mathcal{K}(z - x) \cdot \mathcal{K}(x' - x) \\
& \quad \times f^{abc} f^{deh} U^{ad}(w') U^{cl}(y) U^{be}(x') \{ \rho^h(x'), \rho^l(y) \} \} + (w' \leftrightarrow z)
\end{aligned} \tag{8.8}$$

$$\begin{aligned}
\frac{d^3 N^{(2\rho)}}{d^3 k} \Big|_{\text{GI.V2}} &= \mathcal{N} \int e^{-ik \cdot (w' - x)} \left\{ \right. \\
& + \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(y' - x) \mathcal{K}(z - x) \cdot \mathcal{K}(x' - z) \\
& \quad \times f^{abc} f^{deh} U^{ad}(x') U^{bl}(x) U^{ce}(z) \{ \rho^h(x'), \rho^l(y') \} \\
& + \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - x) \mathcal{K}(z - x) \cdot \mathcal{K}(y - z) \\
& \quad \times f^{abc} f^{deh} U^{ad}(x') U^{be}(x) U^{cl}(z) \{ \rho^h(x'), \rho^l(y) \} \\
& + \frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(y' - x) \mathcal{K}(z - x) \cdot \mathcal{K}(x' - z) \\
& \quad \times U^{ab}(y') U^{ac}(x') \{ \rho^c(x'), \rho^b(y') \} \\
& - \frac{1}{4} N_c \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - x) \mathcal{K}(z - x) \cdot \mathcal{K}(y - z) \\
& \quad \times U^{ab}(y) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \\
& - \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(y' - x) \mathcal{K}(z - x) \cdot \mathcal{K}(x' - z) \\
& \quad \times f^{abc} f^{deh} U^{ad}(w') U^{bl}(x) U^{ce}(z) \{ \rho^h(x'), \rho^l(y') \} \\
& - \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - x) \mathcal{K}(z - x) \cdot \mathcal{K}(y - z) \\
& \quad \times f^{abc} f^{deh} U^{ad}(w') U^{be}(x) U^{cl}(z) \{ \rho^h(x'), \rho^l(y) \} \\
& + \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(y' - x) \mathcal{K}(z - x) \cdot \mathcal{K}(x' - z)
\end{aligned}$$

$$\begin{aligned}
& \times f^{abc} f^{deh} U^{ad}(w') U^{ce}(x') U^{bl}(y') \{ \rho^h(x'), \rho^l(y') \} \\
& + \frac{1}{4} \mathcal{K}(x' - w') \cdot \mathcal{K}(x' - x) \mathcal{K}(z - x) \cdot \mathcal{K}(y - z) \\
& \times f^{abc} f^{deh} U^{ad}(w') U^{cl}(y) U^{be}(x') \{ \rho^h(x'), \rho^l(y) \} \} + (w' \leftrightarrow x)
\end{aligned} \tag{8.9}$$

Group II (no c.c.)

$$\begin{aligned}
\frac{d^3 N^{(2\rho)}}{d^3 k} \Big|_{\text{GII.I1}} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\
& - \frac{1}{2} \mathcal{K}(x - w) \cdot \mathcal{K}(x - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right] \\
& \times f^{abc} f^{cde} U^{hd}(w') U^{ha}(w) \{ \rho^e(x), \rho^b(y) \} \\
& + \frac{1}{2} N_c \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(x - z) + \frac{1}{2} \frac{1}{(x - z)^2} \right] \\
& \times U^{ab}(w') U^{ac}(w) \{ \rho^c(x), \rho^b(x') \} \\
& + \frac{1}{2} \mathcal{K}(x - w) \cdot \mathcal{K}(x - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right] \\
& \times f^{abc} f^{deh} U^{ae}(w') U^{lb}(z) U^{cd}(x) \{ \rho^h(x), \rho^l(y) \} \\
& + \frac{1}{2} \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(x - z) + \frac{1}{2} \frac{1}{(x - z)^2} \right] \\
& \times f^{abc} f^{deh} U^{al}(w') U^{eb}(z) U^{cd}(x) \{ \rho^h(x), \rho^l(x') \} \\
& + \frac{1}{2} \mathcal{K}(x - w) \cdot \mathcal{K}(x - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right] \\
& \times f^{abc} f^{cde} U^{bh}(z) U^{ld}(x) U^{hm}(y) U^{la}(w) \{ \rho^e(x), \rho^m(y) \} \\
& + \frac{1}{2} \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(x - z) + \frac{1}{2} \frac{1}{(x - z)^2} \right] \\
& \times f^{abc} f^{cde} U^{bh}(z) U^{lm}(x') U^{hd}(x) U^{la}(w) \{ \rho^e(x), \rho^m(x') \} \\
& - \frac{1}{2} N_c \mathcal{K}(x - w) \cdot \mathcal{K}(x - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right] \\
& \times U^{ab}(y) U^{ac}(x) \{ \rho^c(x), \rho^b(y) \} \\
& + \frac{1}{2} N_c \mathcal{K}(x - w) \cdot \mathcal{K}(x' - w') \left[\mathcal{K}(z - w) \cdot \mathcal{K}(x - z) + \frac{1}{2} \frac{1}{(x - z)^2} \right] \\
& \times U^{ab}(x') U^{ac}(x) \{ \rho^c(x), \rho^b(x') \} \left. \right\}
\end{aligned} \tag{8.10}$$

$$\begin{aligned}
\frac{d^3 N^{(2\rho)}}{d^3 k} \Big|_{\text{GII.I2}} &= \mathcal{N} \int e^{-ik \cdot (w' - w)} \left\{ \right. \\
& - \frac{1}{2} \mathcal{K}(x - w') \cdot \mathcal{K}(x - w) \left[\mathcal{K}(z - w') \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right] \\
& \times f^{abc} f^{cde} U^{ha}(w') U^{hd}(w) \{ \rho^e(x), \rho^b(y) \} \\
& + \frac{1}{2} N_c \mathcal{K}(x - w') \cdot \mathcal{K}(x' - w) \left[\mathcal{K}(z - w') \cdot \mathcal{K}(x - z) + \frac{1}{2} \frac{1}{(x - z)^2} \right] \\
& \times U^{ab}(w') U^{ac}(w) \{ \rho^b(x), \rho^c(x') \} \\
& + \frac{1}{2} \mathcal{K}(x - w') \cdot \mathcal{K}(x - w) \left[\mathcal{K}(z - w') \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right] \\
& \times f^{abc} f^{cde} U^{ha}(w') U^{bl}(z) U^{hd}(x) U^{lm}(y) \{ \rho^e(x), \rho^m(y) \} \\
& + \frac{1}{2} \mathcal{K}(x - w') \cdot \mathcal{K}(x' - w) \left[\mathcal{K}(z - w') \cdot \mathcal{K}(x - z) + \frac{1}{2} \frac{1}{(x - z)^2} \right] \\
& \times f^{abc} f^{cde} U^{ha}(w') U^{bl}(z) U^{hm}(x') U^{ld}(x) \{ \rho^e(x), \rho^m(x') \} \\
& + \frac{1}{2} \mathcal{K}(x - w') \cdot \mathcal{K}(x - w) \left[\mathcal{K}(z - w') \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right] \\
& \times f^{abc} f^{deh} U^{lb}(z) U^{cd}(x) U^{ae}(w) \{ \rho^h(x), \rho^l(y) \} \\
& + \frac{1}{2} \mathcal{K}(x - w') \cdot \mathcal{K}(x' - w) \left[\mathcal{K}(z - w') \cdot \mathcal{K}(x - z) + \frac{1}{2} \frac{1}{(x - z)^2} \right] \\
& \times f^{abc} f^{deh} U^{eb}(z) U^{cd}(x) U^{al}(w) \{ \rho^h(x), \rho^l(x') \} \\
& - \frac{1}{2} N_c \mathcal{K}(x - w') \cdot \mathcal{K}(x - w) \left[\mathcal{K}(z - w') \cdot \mathcal{K}(y - z) + \frac{1}{2} \mathcal{K}(x - z) \cdot \mathcal{K}(y - z) \right]
\end{aligned}$$

$$\begin{aligned}
& \times U^{ab}(y) U^{ac}(x) \{ \rho^c(x), \rho^b(y) \} \\
& + \frac{1}{2} N_c \mathcal{K}(x-w') \cdot \mathcal{K}(x'-w) \left[\mathcal{K}(z-w') \cdot \mathcal{K}(x-z) + \frac{1}{2} \frac{1}{(x-z)^2} \right] \\
& \times U^{ab}(x') U^{ac}(x) \{ \rho^c(x), \rho^b(x') \} \}
\end{aligned} \tag{8.11}$$

Group III (with c.c.)

$$\begin{aligned}
\frac{d^3 N^{(2\rho)}}{d^3 k} \Big|_{\text{GIII.I}} = \mathcal{N} \int e^{-ik \cdot (w'-w)} \{ & \\
& + \frac{1}{2} \mathcal{K}(x-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(x'-z) + \frac{1}{2} \mathcal{K}(x-z) \cdot \mathcal{K}(x'-z) \right] \\
& \times f^{abc} f^{deh} U^{eb}(z) U^{da}(w) U^{hl}(x') \{ \rho^l(x'), \rho^c(x) \} \\
& + \frac{1}{2} \mathcal{K}(x'-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \mathcal{K}(x'-z) \cdot \mathcal{K}(y-z) \right] \\
& \times f^{abc} f^{dce} U^{hd}(x') U^{lm}(y) U^{lb}(z) U^{ha}(w) \{ \rho^e(x'), \rho^m(y) \} \\
& + \frac{1}{2} \mathcal{K}(y-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \frac{1}{(y-z)^2} \right] \\
& \times f^{abc} f^{dce} U^{hl}(x') U^{md}(y) U^{mb}(z) U^{ha}(w) \{ \rho^e(y), \rho^l(x') \} \\
& - \frac{1}{2} N_c \mathcal{K}(x-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(x'-z) + \frac{1}{2} \mathcal{K}(x-z) \cdot \mathcal{K}(x'-z) \right] \\
& \times U^{ab}(x) U^{ac}(x') \{ \rho^c(x'), \rho^b(x) \} \\
& + \frac{1}{2} N_c \mathcal{K}(x'-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \mathcal{K}(x'-z) \cdot \mathcal{K}(y-z) \right] \\
& \times U^{ab}(y) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \\
& - \frac{1}{2} N_c \mathcal{K}(y-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \frac{1}{(y-z)^2} \right] \\
& \times U^{ab}(x') U^{ac}(y) \{ \rho^c(y), \rho^b(x') \} \\
& - \frac{1}{2} \mathcal{K}(x-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(x'-z) + \frac{1}{2} \mathcal{K}(x-z) \cdot \mathcal{K}(x'-z) \right] \\
& \times f^{abc} f^{deh} U^{bl}(z) U^{le}(x') U^{md}(w') U^{ma}(w) \{ \rho^h(x'), \rho^c(x) \} \\
& - \frac{1}{2} \mathcal{K}(x'-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \mathcal{K}(x'-z) \cdot \mathcal{K}(y-z) \right] \\
& \times f^{abc} f^{dce} U^{bh}(z) U^{hl}(y) U^{md}(w') U^{ma}(w) \{ \rho^e(x'), \rho^l(y) \} \\
& - \frac{1}{2} \mathcal{K}(y-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \frac{1}{(y-z)^2} \right] \\
& \times f^{abc} f^{dce} U^{bh}(z) U^{hd}(y) U^{lm}(w') U^{la}(w) \{ \rho^e(y), \rho^m(x') \} \\
& + \frac{1}{2} \mathcal{K}(x-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(x'-z) + \frac{1}{2} \mathcal{K}(x-z) \cdot \mathcal{K}(x'-z) \right] \\
& \times f^{abc} f^{deh} U^{be}(x') U^{ad}(w') U^{cl}(x) \{ \rho^h(x'), \rho^l(x) \} \\
& + \frac{1}{2} \mathcal{K}(x'-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \mathcal{K}(x'-z) \cdot \mathcal{K}(y-z) \right] \\
& \times f^{abc} f^{deh} U^{bl}(y) U^{ad}(w') U^{ce}(x') \{ \rho^h(x'), \rho^l(y) \} \\
& + \frac{1}{2} N_c \mathcal{K}(y-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \frac{1}{(y-z)^2} \right] \\
& \times U^{ab}(w') U^{ac}(y) \{ \rho^c(y), \rho^b(x') \} \} + (w' \leftrightarrow w)
\end{aligned} \tag{8.12}$$

$$\begin{aligned}
\frac{d^3 N^{(2\rho)}}{d^3 k} \Big|_{\text{GIII.II}} = \mathcal{N} \int e^{-ik \cdot (w'-w)} \{ & \\
& + \frac{1}{2} \mathcal{K}(x-w) \cdot \mathcal{K}(y-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \mathcal{K}(x-z) \cdot \mathcal{K}(y-z) \right] \\
& \times f^{abc} f^{deh} U^{bd}(z) U^{ea}(w) U^{hl}(y) \{ \rho^l(y), \rho^c(x) \} \\
& + \frac{1}{2} \mathcal{K}(y-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \frac{1}{(y-z)^2} \right] \\
& \times f^{abc} f^{dce} U^{hd}(y) U^{lm}(x') U^{bh}(z) U^{la}(w) \{ \rho^e(y), \rho^m(x') \} \\
& + \frac{1}{2} \mathcal{K}(x'-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \mathcal{K}(x'-z) \cdot \mathcal{K}(y-z) \right]
\end{aligned}$$

$$\begin{aligned}
& \times f^{abc} f^{dce} U^{hl}(y) U^{md}(x') U^{bh}(z) U^{ma}(w) \{\rho^e(x'), \rho^l(y)\} \\
& + \frac{1}{2} N_c \mathcal{K}(x-w) \cdot \mathcal{K}(y-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \mathcal{K}(x-z) \cdot \mathcal{K}(y-z) \right] \\
& \times U^{ab}(x) U^{ac}(y) \{\rho^c(y), \rho^b(x)\} \\
& - \frac{1}{2} N_c \mathcal{K}(y-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \frac{1}{(y-z)^2} \right] \\
& \times U^{ab}(x') U^{ac}(y) \{\rho^c(y), \rho^b(x')\} \\
& + \frac{1}{2} N_c \mathcal{K}(x'-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \mathcal{K}(x'-z) \cdot \mathcal{K}(y-z) \right] \\
& \times U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \\
& - \frac{1}{2} \mathcal{K}(x-w) \cdot \mathcal{K}(y-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \mathcal{K}(x-z) \cdot \mathcal{K}(y-z) \right] \\
& \times f^{abc} f^{bde} U^{hd}(y) U^{ha}(w) \{\rho^e(y), \rho^c(x)\} \\
& - \frac{1}{2} N_c \mathcal{K}(y-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \frac{1}{(y-z)^2} \right] \\
& \times U^{ab}(x') U^{ac}(w) \{\rho^c(y), \rho^b(x')\} \\
& - \frac{1}{2} \mathcal{K}(x'-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \mathcal{K}(x'-z) \cdot \mathcal{K}(y-z) \right] \\
& \times f^{abc} f^{dce} U^{hd}(x') U^{ha}(w) \{\rho^e(x'), \rho^b(y)\} \\
& + \frac{1}{2} \mathcal{K}(x-w) \cdot \mathcal{K}(y-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \mathcal{K}(x-z) \cdot \mathcal{K}(y-z) \right] \\
& \times f^{abc} f^{deh} U^{ae}(y) U^{db}(z) U^{cl}(x) \{\rho^h(y), \rho^l(x)\} \\
& + \frac{1}{2} \mathcal{K}(y-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \frac{1}{(y-z)^2} \right] \\
& \times f^{abc} f^{deh} U^{al}(x') U^{db}(z) U^{ce}(y) \{\rho^h(y), \rho^l(x')\} \\
& - \frac{1}{2} N_c \mathcal{K}(x'-w) \cdot \mathcal{K}(x'-w') \left[\mathcal{K}(z-w) \cdot \mathcal{K}(y-z) + \frac{1}{2} \mathcal{K}(x'-z) \cdot \mathcal{K}(y-z) \right] \\
& \times U^{ab}(z) U^{bc}(x') \{\rho^c(x'), \rho^a(y)\} \} + (w' \leftrightarrow w)
\end{aligned} \tag{8.13}$$

$$\begin{aligned}
\left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{\text{GIII.III}} &= \mathcal{N} \int e^{-ik \cdot (y'-w)} \left\{ \right. \\
& + \frac{1}{2} \mathcal{K}(y'-z) \cdot \mathcal{K}(x-z) \mathcal{K}(x'-y') \cdot \mathcal{K}(x'-w) \\
& \times f^{abc} f^{deh} U^{da}(y') U^{bl}(z) U^{ce}(w) \{\rho^h(x'), \rho^l(x)\} \\
& + \frac{1}{2} \mathcal{K}(y'-z) \cdot \mathcal{K}(x'-z) \mathcal{K}(x'-y') \cdot \mathcal{K}(y-w) \\
& \times f^{abc} f^{deh} U^{da}(y') U^{be}(z) U^{cl}(w) \{\rho^h(x'), \rho^l(y)\} \\
& - \frac{1}{2} \mathcal{K}(y'-z) \cdot \mathcal{K}(x-z) \mathcal{K}(x'-y') \cdot \mathcal{K}(x'-w) \\
& \times f^{abc} f^{deh} U^{da}(y') U^{ce}(x') U^{bl}(x) \{\rho^h(x'), \rho^l(x)\} \\
& - \frac{1}{2} \mathcal{K}(y'-z) \cdot \mathcal{K}(x'-z) \mathcal{K}(x'-y') \cdot \mathcal{K}(y-w) \\
& \times f^{abc} f^{deh} U^{da}(y') U^{cl}(y) U^{be}(x') \{\rho^h(x'), \rho^l(y)\} \\
& - \frac{1}{2} \mathcal{K}(y'-z) \cdot \mathcal{K}(x-z) \mathcal{K}(x'-y') \cdot \mathcal{K}(x'-w) \\
& \times f^{abc} f^{deh} U^{lb}(z) U^{ad}(x') U^{ce}(w) \{\rho^h(x'), \rho^l(x)\} \\
& - \frac{1}{2} \mathcal{K}(y'-z) \cdot \mathcal{K}(x'-z) \mathcal{K}(x'-y') \cdot \mathcal{K}(y-w) \\
& \times f^{abc} f^{deh} U^{eb}(z) U^{ad}(x') U^{cl}(w) \{\rho^h(x'), \rho^l(y)\} \\
& - \frac{1}{2} N_c \mathcal{K}(y'-z) \cdot \mathcal{K}(x-z) \mathcal{K}(x'-y') \cdot \mathcal{K}(x'-w) \\
& \times U^{ab}(x) U^{ac}(x') \{\rho^c(x'), \rho^b(x)\} \\
& + \frac{1}{2} N_c \mathcal{K}(y'-z) \cdot \mathcal{K}(x'-z) \mathcal{K}(x'-y') \cdot \mathcal{K}(y-w) \\
& \times U^{ab}(y) U^{ac}(x') \{\rho^c(x'), \rho^b(y)\} \} + (y' \leftrightarrow w)
\end{aligned} \tag{8.14}$$

$$\begin{aligned}
\left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{\text{GIII.V}} &= \mathcal{N} \int e^{-ik \cdot (y'-w)} \left\{ \right. \\
& + \frac{1}{2} \mathcal{K}(y'-z) \cdot \mathcal{K}(x'-z) \mathcal{K}(x'-y') \cdot \mathcal{K}(y-w) \\
& \times f^{abc} f^{deh} U^{da}(y') U^{be}(x') U^{cl}(y) \{\rho^h(x'), \rho^l(y)\}
\end{aligned}$$

$$\begin{aligned}
& + \frac{1}{2} \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x' - w) \\
& \quad \times f^{abc} f^{deh} U^{da}(y') U^{bl}(x) U^{ce}(x') \{ \rho^h(x'), \rho^l(x) \} \\
& + \frac{1}{2} N_c \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x - w) \\
& \quad \times U^{ab}(y') U^{bc}(x) \{ \rho^c(x), \rho^a(x') \} \\
& - \frac{1}{2} \mathcal{K}(y' - z) \cdot \mathcal{K}(x' - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \\
& \quad \times f^{abc} f^{deh} U^{da}(y') U^{cl}(w) U^{be}(x') \{ \rho^h(x'), \rho^l(y) \} \\
& - \frac{1}{2} \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x' - w) \\
& \quad \times f^{abc} f^{deh} U^{da}(y') U^{ce}(w) U^{bl}(x) \{ \rho^h(x'), \rho^l(x) \} \\
& - \frac{1}{2} \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x - w) \\
& \quad \times f^{abc} f^{deh} U^{la}(y') U^{ce}(w) U^{bd}(x) \{ \rho^h(x), \rho^l(x') \} \\
& - \frac{1}{2} N_c \mathcal{K}(y' - z) \cdot \mathcal{K}(x' - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \\
& \quad \times U^{ab}(y) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \\
& + \frac{1}{2} N_c \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x' - w) \\
& \quad \times U^{ab}(x) U^{ac}(x') \{ \rho^c(x'), \rho^b(x) \} \\
& - \frac{1}{2} N_c \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x - w) \\
& \quad \times U^{ab}(x') U^{ac}(x) \{ \rho^c(x), \rho^b(x') \} \\
& + \frac{1}{2} N_c \mathcal{K}(y' - z) \cdot \mathcal{K}(x' - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \\
& \quad \times U^{ab}(w) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \\
& + \frac{1}{2} \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x' - w) \\
& \quad \times f^{abc} f^{deh} U^{ad}(x') U^{ce}(w) U^{bl}(x) \{ \rho^h(x'), \rho^l(x) \} \\
& + \frac{1}{2} \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x - w) \\
& \quad \times f^{abc} f^{deh} U^{al}(x') U^{ce}(w) U^{bd}(x) \{ \rho^h(x), \rho^l(x') \} \} + (y' \leftrightarrow w)
\end{aligned} \tag{8.15}$$

$$\begin{aligned}
\left. \frac{d^3 N^{(2\rho)}}{d^3 k} \right|_{\text{GIII.VI}} &= \mathcal{N} \int e^{-ik \cdot (y' - w)} \{ \\
& + \frac{1}{2} \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x' - w) \\
& \quad \times f^{abc} f^{deh} U^{da}(y') U^{ce}(x') U^{bl}(x) \{ \rho^h(x'), \rho^l(x) \} \\
& + \frac{1}{2} \mathcal{K}(y' - z) \cdot \mathcal{K}(x' - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \\
& \quad \times f^{abc} f^{deh} U^{da}(y') U^{cl}(y) U^{be}(x') \{ \rho^h(x'), \rho^l(y) \} \\
& - \frac{1}{2} N_c \mathcal{K}(y' - z) \cdot \mathcal{K}(y - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \\
& \quad \times U^{ab}(y') U^{bc}(y) \{ \rho^c(y), \rho^a(x') \} \\
& - \frac{1}{2} \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x' - w) \\
& \quad \times f^{abc} f^{deh} U^{da}(y') U^{bl}(z) U^{ce}(x') \{ \rho^h(x'), \rho^l(x) \} \\
& - \frac{1}{2} \mathcal{K}(y' - z) \cdot \mathcal{K}(x' - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \\
& \quad \times f^{abc} f^{deh} U^{da}(y') U^{be}(z) U^{cl}(y) \{ \rho^h(x'), \rho^l(y) \} \\
& - \frac{1}{2} \mathcal{K}(y' - z) \cdot \mathcal{K}(y - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \\
& \quad \times f^{abc} f^{deh} U^{la}(y') U^{be}(z) U^{cd}(y) \{ \rho^h(y), \rho^l(x') \} \\
& + \frac{1}{2} N_c \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x' - w) \\
& \quad \times U^{ab}(x) U^{ac}(x') \{ \rho^c(x'), \rho^b(x) \} \\
& - \frac{1}{2} N_c \mathcal{K}(y' - z) \cdot \mathcal{K}(x' - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \\
& \quad \times U^{ab}(y) U^{ac}(x') \{ \rho^c(x'), \rho^b(y) \} \\
& + \frac{1}{2} N_c \mathcal{K}(y' - z) \cdot \mathcal{K}(y - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \\
& \quad \times U^{ab}(x') U^{ac}(y) \{ \rho^c(y), \rho^b(x') \} \\
& - \frac{1}{2} N_c \mathcal{K}(y' - z) \cdot \mathcal{K}(x - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(x' - w) \\
& \quad \times U^{ab}(z) U^{bc}(x') \{ \rho^c(x'), \rho^a(x) \} \\
& + \frac{1}{2} \mathcal{K}(y' - z) \cdot \mathcal{K}(x' - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \\
& \quad \times f^{abc} f^{deh} U^{eb}(z) U^{ad}(x') U^{cl}(y) \{ \rho^h(x'), \rho^l(y) \}
\end{aligned}$$

$$\begin{aligned}
& + \frac{1}{2} \mathcal{K}(y' - z) \cdot \mathcal{K}(y - z) \mathcal{K}(x' - y') \cdot \mathcal{K}(y - w) \\
& \times f^{abc} f^{deh} U^{eb}(z) U^{al}(x') U^{cd}(y) \{ \rho^h(y), \rho^l(x') \} \} + (y' \leftrightarrow w)
\end{aligned} \tag{8.16}$$

A The kernel of Row II of Group I in coordinate space

We need

$$F^i(w; x, y) \equiv \int d^2 \mathbf{k} d^2 \mathbf{p} e^{i \mathbf{k} \cdot (w-y)} e^{i \mathbf{p} \cdot (y-x)} \frac{B^i(\mathbf{k}, \mathbf{p})}{\mathbf{k}^2 \mathbf{p}^2}, \quad B^i = -\frac{2\mathbf{k}^2 \mathbf{p}^i + (\mathbf{p}^2 - \mathbf{k} \cdot \mathbf{p}) \mathbf{k}^i}{(\mathbf{k} - \mathbf{p})^2} - 2\mathbf{k}^i. \tag{A.1}$$

(B^i is the bracket of the notes: $-(\mathbf{k}^2 \mathbf{p}^i - \mathbf{k} \cdot \mathbf{p} \mathbf{k}^i) - (\mathbf{p}^2 \mathbf{k}^i + \mathbf{k}^2 \mathbf{p}^i) = -[2\mathbf{k}^2 \mathbf{p}^i + (\mathbf{p}^2 - \mathbf{k} \cdot \mathbf{p}) \mathbf{k}^i]$.)

Step 1: partial fractions. With $\mathbf{q} \equiv \mathbf{k} - \mathbf{p}$ and $\mathbf{p}^2 - \mathbf{k} \cdot \mathbf{p} = -\mathbf{p} \cdot \mathbf{q}$,

$$\frac{B^i}{\mathbf{k}^2 \mathbf{p}^2} = -\frac{2\mathbf{p}^i}{\mathbf{q}^2 \mathbf{p}^2} + \frac{(\mathbf{p} \cdot \mathbf{q}) \mathbf{k}^i}{\mathbf{q}^2 \mathbf{k}^2 \mathbf{p}^2} - \frac{2\mathbf{k}^i}{\mathbf{k}^2 \mathbf{p}^2}. \tag{A.2}$$

Step 2: Fourier transforms. We use $\int d^2 \mathbf{k} e^{i \mathbf{k} \cdot X} \mathbf{k}^i / \mathbf{k}^2 = 2\pi i \mathcal{K}^i(X)$; $\int d^2 \mathbf{q} e^{i \mathbf{q} \cdot X} / \mathbf{q}^2 = -\pi \log X^2 + C_0$, where the constant C_0 is infrared divergent; and $\mathbf{k}^i / \mathbf{k}^2 = \frac{i}{2\pi} \int d^2 u e^{-i \mathbf{k} \cdot u} \mathcal{K}^i(u)$. In the first term we change variables $(\mathbf{k}, \mathbf{p}) \rightarrow (\mathbf{q}, \mathbf{p})$, so the phase is $e^{i \mathbf{q} \cdot (w-y)} e^{i \mathbf{p} \cdot (w-x)}$:

$$-2 \cdot 2\pi i \mathcal{K}^i(w-x) (-\pi \log(w-y)^2 + C_0) = -4\pi^2 i \mathcal{K}^i(x-w) \log(w-y)^2 + 4\pi i C_0 \mathcal{K}^i(x-w). \tag{A.3}$$

The third term factorizes directly:

$$-2 \cdot 2\pi i \mathcal{K}^i(w-y) (-\pi \log(x-y)^2 + C_0) = -4\pi^2 i \mathcal{K}^i(y-w) \log(x-y)^2 + 4\pi i C_0 \mathcal{K}^i(y-w). \tag{A.4}$$

In the second term we write $\mathbf{k}^i / \mathbf{k}^2$ as an integral over u , change variables to $(\mathbf{q}, \mathbf{p})$, and do the two Fourier transforms. With $z' = w - u$,

$$\frac{i}{2\pi} (2\pi i)^2 \int_{z'} \mathcal{K}^i(w-z') \mathcal{K}(z'-y) \cdot \mathcal{K}(z'-x) = -2\pi i \int_{z'} K(x, y, z') \mathcal{K}^i(w-z'). \tag{A.5}$$

Step 3: keep the odd part. The colour structures of Row II are odd under $x \leftrightarrow y$, so only the odd part of F^i contributes. The z' term and the C_0 terms are even and drop out. With $F^i = 2\pi i \mathfrak{F}^i$,

$$\mathfrak{F}^i(w; x, y) = -\pi \left[\mathcal{K}^i(x-w) \log \frac{(w-y)^2}{(x-y)^2} - \mathcal{K}^i(y-w) \log \frac{(w-x)^2}{(x-y)^2} \right]. \tag{A.6}$$

Using $\int_{z'} K(a, b, z') = -\pi \log(a-b)^2 + \text{const}$, this is

$$\mathfrak{F}^i(w; x, y) = \mathcal{K}^i(x-w) \int_{z'} [K(w, y, z') - K(x, y, z')] - \mathcal{K}^i(y-w) \int_{z'} [K(w, x, z') - K(x, y, z')]. \tag{A.7}$$

The coordinate-space expression for $\mathbb{A}^{(3)}$ in your `main.tex` (the commented version), after $\log \frac{\Lambda}{k^+} = -\log \frac{\Lambda}{\Lambda} + \log \frac{\Lambda}{k^+}$, has the same structure: $\mathcal{K}^i(x-w) \int_{z'} [K(w, y, z') - K(x, y, z')] - (x \leftrightarrow y)$.

B Checks

The algebra was generated by a short program, which also checked it numerically.

1. *Reordering identity.* Eqs. (3.4)–(3.7), including the explicit one- ρ remainder R_1 , hold to machine precision for random matrices A, B, C .
2. *Colour algebra.* Each of the 144 pieces was evaluated before and after the colour algebra, for $SU(3)$ with random classical charges and random symmetric adjoint Wilson lines (U of $V = W \text{diag}(1, -1, -1) W^\dagger$, W random). The two agree to 2×10^{-15} for every piece. With general, non-symmetric Wilson lines, 14 pieces differ. These are exactly the pieces whose simplification uses $(U1^*)$ or $(U2^*)$, i.e. the convention $U^{bc} = U^{cb}$.
3. *The whole reordering.* For every sub-term we took the charges as explicit operators on three sites (the δ -function becomes a Kronecker delta), with random kernels and random symmetric Wilson lines. We then compared

$$\sum_{\text{singlets}} \langle s | O - S(O) | s \rangle \quad \text{and} \quad \sum_{\text{singlets}} \langle s | (\text{two-}\rho \text{ pieces}) | s \rangle,$$

where O is the three- ρ operator of the sub-term, $S(O)$ its fully symmetrized form, and $|s\rangle$ runs over the colour singlets, in which every one- ρ term has zero expectation value. We used $SU(3)$ with one quark per site (one singlet) and $SU(2)$ with two spin- $\frac{1}{2}$ per site (five singlets), three random draws each. The largest difference was 4×10^{-14} , for values up to 25. This checks the commutators, the signs, the factors $\frac{1}{4}$ and $\frac{1}{2}$, the δ -functions and the colour algebra together.